



CE EMC Test Report

APPLICANT : Magne AI Global tech limited
EQUIPMENT : MAG1
BRAND NAME : MaQ
MODEL NAME : MA1
STANDARD : ETSI EN 301 489-1 V2.2.3 (2019-11)
ETSI EN 301 489-3 V2.3.2 (2023-01)
ETSI EN 301 489-17 V3.3.1 (2024-09)
ETSI EN 301 489-19 V2.2.1 (2022-09)
ETSI EN 301 489-52 V1.3.1 (2024-11)
TEST DATE(S) : Feb. 10, 2026 ~ Mar. 20, 2026

The measurement shown in this test report is tested in accordance with the test procedures given in ETSI EN 301 489-1 V2.2.3 (2019-11), ETSI EN 301 489-3 V2.3.2 (2023-01), ETSI EN 301 489-17 V3.3.1 (2024-09), ETSI EN 301 489-19 V2.2.1 (2022-09), ETSI EN 301 489-52 V1.3.1 (2024-11).

The test results in this report apply exclusively to the tested model / sample. Without written approval of Sporton International Inc. (KunShan), the test report shall not be reproduced except in full.

Jason Jia

Approved by: Jason Jia



Sporton International Inc. (Kunshan)

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REVISION HISTORY

REPORT NO.	VERSION	DESCRIPTION	ISSUED DATE
EW612311	Rev. 01	Initial issue of report	May 25, 2026

SUMMARY OF TEST RESULT

CLAUSE (EN301489-1)	TEST ITEMS	TEST STANDARD	RESULT (PASS/FAIL)	REMARK
EMC Emission Measurements				
8.2	Radiated Emission	EN 55032:2015 Class B	PASS	Under limit 3.12 dB at 93.050 MHz for Quasi-Peak
8.3 / 8.4 / 8.7	Conducted Emission	EN 55032:2015 Class B	PASS	Under limit 4.02 dB at 3.381 MHz
8.5	Harmonic Current Emissions	EN IEC 61000-3-2 : 2019 + A1 : 2021 Class A	PASS	-
8.6	Voltage Fluctuations and Flicker	EN 61000-3-3 : 2013 + A2 : 2021	PASS	-
EMC Immunity Tests				
9.2	RF Electromagnetic Field	EN 61000-4-3:2006+A1:2008 +A2:2010	PASS	-
9.3	Electrostatic Discharge	EN 61000-4-2:2009	PASS	-
9.4	Fast Transients, Common Mode	EN 61000-4-4:2012	PASS	-
9.5	Radio frequency, Common Mode	EN 61000-4-6:2014	PASS	-
9.6	Transients and Surges in the vehicular environment	ISO 7637-2:2004	Not Applicable	-
9.7	Voltage Dips and Interruptions	EN 61000-4-11:2004	PASS	-
9.8	Surges	EN 61000-4-5:2014 + A1:2017	PASS	-
Note: Not required means after assessing, test items are not necessary to carry out.				

Conformity Assessment Condition:

The test results (PASS/FAIL) with all measurement uncertainty excluded are presented against the regulation limits or in accordance with the requirements stipulated by the applicant/manufacture who shall bear all the risks of non-compliance that may potentially occur if measurement uncertainty is taken into account. Please refer to each test results in the section "Measurement Uncertainty".

Disclaimer:

The product specifications of the EUT presented in the test report that may affect the test assessments are declared by the manufacturer who shall take full responsibility for the authenticity.

APPLIED STANDARDS

According to the specifications of the manufacturer, the EUT must comply with the requirements of

- **ETSI EN 301 489-1 V2.2.3 (2019-11)**
- **ETSI EN 301 489-3 V2.3.2 (2023-01)**
- **ETSI EN 301 489-17 V3.3.1 (2024-09)**
- **ETSI EN 301 489-19 V2.2.1 (2022-09)**
- **ETSI EN 301 489-52 V1.3.1 (2024-11)**

Below are basic standards version used for all test items in this report.

Emission Test Items	Basic Standard Version used in this report	Remark
Radiated Emission	EN 55032:2015	There is no additional test requirement for this EUT, when EN 55032: 2015 is updated to 55032: 2015/A11: 2020 versions.
Conducted Emission	EN 55032:2015	
Harmonics Current Emission	EN IEC 61000-3-2 : 2019 + A1 : 2021 Class A	-
Voltage Fluctuation and Flicker	EN 61000-3-3 : 2013 + A2 : 2021	-

Immunity Test Items	Basic Standard Version used in this report	Remark
RF Electromagnetic Field	EN 61000-4-3:2006+A1:2008+A2: 2010	-
Electrostatic Discharge	EN 61000-4-2:2009	-
Fast Transients, Common Mode	EN 61000-4-4:2012	-
Radio frequency, Common Mode	EN 61000-4-6:2014	-
Voltage Dips and Interruptions	EN 61000-4-11:2004	-
Surges	EN 61000-4-5:2014 + A1:2017	-

1. General Description of Equipment under Test

1.1 Applicant

Magne AI Global tech limited

FLAT 1019B,10/F,LIVEN HOUSE,NO.61-63 KING YIP STREET KWUN TONG HK

1.2 Manufacturer

Magne AI Global tech limited

FLAT 1019B,10/F,LIVEN HOUSE,NO.61-63 KING YIP STREET KWUN TONG HK

1.3 Product Feature of Equipment Under Test

Product Feature	
Equipment	MAG1
Brand Name	MaQ
Model Name	MA1
GSM Operating Band(s)	GSM 850/900/1800/1900MHz
WCDMA Operating Band(s)	FDD Band I / II / IV / V / VIII
LTE Operating Band(s)	FDD Band 1 / 2 / 3 / 5 / 7 / 8 / 12 / 13 / 17 / 20 / 25 / 26 / 28 / 66 / 71 TDD Band 38 / 40 / 41 / 42
5G NR	n1 / n3 / n5 / n7 / n8 / n20 / n28 / n66 / n71 / n38 / n40 / n41 / n77 / n78 / n79
Wi-Fi Specification	WLAN 2.4GHz 802.11b/g/n HT20/HT40 WLAN 5GHz 802.11a/n HT20/HT40 WLAN 5GHz 802.11ac VHT20/VHT40/VHT80
Bluetooth Version	Bluetooth BR / EDR / LE
GNSS	BDS/Galileo/GPS
NFC	Type A/B/F/V
WPT	WPT
IMEI Code	Radiation/Conduction: 016813000001873 RS: 016813000002418/016813000002426 CS: 016813000000651/016813000000669 Harmonic/Flicker/Surge/Dips/EFT: 016813000002632/016813000002640 ESD : 016813000001873/016813000001881
EUT Stage	Production Unit

Remark: The above EUT's information was declared by manufacturer. Please refer to the specifications or user's manual for more detailed description.

1.4 Product Specification of Equipment Under Test

Product Specification subjective to this Test Standard	
Transmitter Frequency Range	GSM900: 880 MHz ~ 915 MHz GSM1800: 1710 MHz ~ 1785 MHz WCDMA Band I: 1920 MHz ~ 1980 MHz WCDMA Band V: 824 MHz ~ 849 MHz WCDMA Band VIII: 880 MHz ~ 915 MHz LTE Band 1: 1920 MHz ~ 1980 MHz LTE Band 3: 1710 MHz ~ 1785 MHz LTE Band 5 : 824 MHz ~ 849 MHz LTE Band 7: 2500 MHz ~ 2570 MHz LTE Band 8: 880 MHz ~ 915 MHz LTE Band 20: 832 MHz ~ 862 MHz LTE Band 28: 703 MHz ~ 748 MHz LTE Band 38: 2570 MHz ~ 2620 MHz LTE Band 40: 2300 MHz ~ 2400 MHz LTE Band 41 : 2496 MHz ~ 2690 MHz LTE Band 42 : 3400 MHz ~ 3600 MHz 5G NR n1: 1920 MHz ~ 1980 MHz 5G NR n3: 1710 MHz ~ 1785 MHz 5G NR n5: 824 MHz ~ 849 MHz 5G NR n7: 2500 MHz ~ 2570 MHz 5G NR n8: 880 MHz ~ 915 MHz 5G NR n20: 832 MHz ~ 862 MHz 5G NR n28: 703 MHz ~ 748 MHz 5G NR n38: 2570 MHz ~ 2620 MHz 5G NR n40: 2300 MHz ~ 2400 MHz 5G NR n41: 2496 MHz ~ 2690 MHz 5G NR n77: 3300 MHz ~ 4200 MHz 5G NR n78: 3300 MHz ~ 3800 MHz 5G NR n79: 4400 MHz ~ 5000 MHz WLAN 802.11b/g/n: 2400 MHz ~ 2483.5 MHz WLAN 802.11a/n/ac: 5150 MHz ~ 5250 MHz; 5250 MHz ~ 5350 MHz; 5470 MHz ~ 5725 MHz; 5725 MHz ~ 5850 MHz Bluetooth: 2400 MHz ~ 2483.5 MHz NFC: 13.56 MHz

Receiver Frequency Range	GSM900: 925 MHz ~ 960 MHz GSM1800: 1805 MHz ~ 1880 MHz WCDMA Band I: 2110 MHz ~ 2170MHz WCDMA Band V: 869 MHz ~ 894 MHz WCDMA Band VIII: 925 MHz ~ 960 MHz LTE Band 1: 2110 MHz ~ 2170 MHz LTE Band 3: 1805 MHz ~ 1880 MHz LTE Band 5 : 869MHz ~ 894 MHz LTE Band 7: 2620 MHz ~ 2690 MHz LTE Band 8: 925 MHz ~ 960 MHz LTE Band 20: 791 MHz ~ 821 MHz LTE Band 28: 758 MHz ~ 803 MHz LTE Band 38: 2570 MHz ~ 2620 MHz LTE Band 40: 2300 MHz ~ 2400 MHz LTE Band 41 : 2496 MHz ~ 2690 MHz LTE Band 42 : 3400 MHz ~ 3600 MHz 5G NR n1: 2110 MHz ~ 2170 MHz 5G NR n3: 1805 MHz ~ 1880 MHz 5G NR n5: 869MHz ~ 894 MHz 5G NR n7: 2620 MHz ~ 2690 MHz 5G NR n8: 925 MHz ~ 960 MHz 5G NR n20: 791 MHz ~ 821 MHz 5G NR n28: 758 MHz ~ 803 MHz 5G NR n38: 2570 MHz ~ 2620 MHz 5G NR n40: 2300 MHz ~ 2400 MHz 5G NR n41: 2496 MHz ~ 2690 MHz 5G NR n77: 3300 MHz ~ 4200 MHz 5G NR n78: 3300 MHz ~ 3800 MHz 5G NR n79: 4400 MHz ~ 5000 MHz WLAN 802.11b/g/n: 2400 MHz ~ 2483.5 MHz WLAN 802.11a/n/ac: 5150 MHz ~ 5250 MHz 5250 MHz ~ 5350 MHz 5470 MHz ~ 5725 MHz 5725 MHz ~ 5850 MHz Bluetooth: 2400 MHz ~ 2483.5 MHz GNSS : 1559 MHz ~ 1610 MHz NFC: 13.56 MHz WPT: 100kHz~ 205 kHz
Antenna Type	WWAN : LDS Antenna WLAN : LDS Antenna Bluetooth : LDS Antenna GNSS: LDS Antenna NFC: FPC Antenna WPT: FPC Antenna

Type of Modulation	GSM/GPRS: GMSK WCDMA : BPSK HSPA : QPSK HSPA+ : 16QAM DC-HSDPA : 64QAM LTE: QPSK / 16QAM / 64QAM 5G NR: DFT-s-OFDM (PI/2 BPSK / QPSK / 16QAM / 64QAM / 256QAM) CP-OFDM (QPSK / 16QAM / 64QAM / 256QAM) 802.11b: DSSS (DBPSK / DQPSK / CCK) 802.11a/g/n: OFDM (BPSK / QPSK / 16QAM / 64QAM) 802.11ac: OFDM (BPSK / QPSK / 16QAM / 64QAM / 256QAM) Bluetooth LE : GFSK Bluetooth (1Mbps) : GFSK Bluetooth (2Mbps) : $\pi/4$ -DQPSK Bluetooth (3Mbps) : 8-DPSK GNSS : BPSK NFC: ASK WPT: ASK
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Note: The device supports WPT RX only.

1.5 Modification of EUT

No modifications are made to the EUT during all test items.

2. Test Configuration of Equipment under Test

2.1 Details of EUT Test Modes

Details of Test line Items	
Radiated Emission (Refer to EN301 489-1 Section 8.2)	
Mode 1	: GSM900 Link + Bluetooth Link + WLAN (2.4G) Link + Camera (Rear) + Earphone + Battery + USB Cable (Charging from Adapter) + SIM 1
Mode 2	: GSM1800 Idle + Bluetooth Link + WLAN (5G) Link + Camera (Front) + Earphone + Battery + USB Cable (Charging from Adapter) + ANT 4 + SIM 2
Mode 3	: WCDMA Band I Link + Bluetooth Link + WLAN (2.4G) Link + MPEG4 (Run Color Bar) + Earphone + Battery + USB Cable (Charging from Adapter) + SIM 1
Mode 4	: LTE Band 7 Idle + Bluetooth Link + WLAN (5G) Link + NFC On + Earphone + Battery + USB Cable (Charging from Adapter) + SIM 1
Mode 5	: DC_1A_n78A Idle + Bluetooth Link + WLAN (2.4G) Link + GNSS Rx + Earphone + Battery + USB Cable (EUT (eMMC) USB Data Link to NB) + SIM 1
Mode 6	: WCDMA Band I Link + Bluetooth Link + WLAN (5G) Link + MPEG4 (Run Color Bar) + Earphone + Battery + USB Cable (NB USB Data Link to EUT (eMMC)) + SIM 1
Mode 7	: WCDMA Band I Link + Bluetooth Link + WLAN (2.4G) Link + MPEG4 (Run Color Bar) + Earphone + Battery + USB Cable (Typec to Typec)(EUT Charging to other phones) + SIM 1
Mode 8	: WCDMA Band I Link + Bluetooth Link + WLAN (5G) Link + MPEG4 (Run Color Bar) + Earphone + Battery + USB Cable (Charging from wireless charging cradle) + SIM 1
Mode 9	: WCDMA Band I Link + Bluetooth Idle + WLAN Idle + Earphone + Battery + USB Cable (Charging from Adapter) + SIM 1
Conducted Emission (Refer to EN301 489-1 Section 8.3, 8.4, 8.7)	
Mode 1	: WWAN Link + Bluetooth Link + WLAN (2.4G) Link + Camera (Rear) + Earphone + Battery + USB Cable (Charging from Adapter) + SIM 1
Mode 2	: WWAN Link + Bluetooth Link + WLAN (5G) Link + Camera (Front) + Earphone + Battery + USB Cable (Charging from Adapter) + SIM 2
Mode 3	: WWAN Link + Bluetooth Link + WLAN (5G Band IV) Link + MPEG4 (Run Color Bar) + Earphone + Battery + USB Cable (Charging from Adapter) + SIM 1
Mode 4	: WWAN Link + Bluetooth Link + WLAN (2.4G) Link + GNSS Rx + Earphone + Battery + USB Cable (Charging from Adapter) + SIM 1
Mode 5	: WWAN Idle + Bluetooth Idle + WLAN Idle + NFC On + Earphone + Battery + USB Cable (Charging from Adapter) + SIM 1
Mode 6	: WWAN Link + Bluetooth Link + WLAN (2.4G) Link + Camera(Rear) + Earphone + Battery + USB Cable (Charging from wireless charging cradle) + SIM 1
Harmonic Current Emissions (Refer to EN301 489-1 Section 8.5)	
Mode 1	: GSM900 Link + Bluetooth Link + WLAN (2.4G) Link + Camera(Rear) + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1
Mode 2	: GSM1800 Idle + Bluetooth Link + WLAN (5G) Link + Camera(Front) + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 2
Mode 3	: WCDMA Band I Link + Bluetooth Link + WLAN (5G Band IV) Link + MPEG4(Run Color Bar) + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1
Mode 4	: LTE Band 7 Idle + Bluetooth Link + WLAN (2.4G) Link + GNSS Rx + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1
Mode 5	: DC_1A_n78A Idle + Bluetooth Idle + WLAN Idle + NFC On + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1
Mode 6	: GSM900 Link + Bluetooth Link + WLAN (2.4G) Link + Camera(Rear) + Earphone + Battery + USB Cable (Charging from wireless charging cradle) + SIM 1
Voltage Fluctuations and Flicker (Refer to EN301 489-1 Section 8.6)	
Mode 1	: GSM900 Link + Bluetooth Link + WLAN (2.4G) Link + Camera(Rear) + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1

Mode 2	: GSM1800 Idle + Bluetooth Link + WLAN (5G) Link + Camera(Front) + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 2
Mode 3	: WCDMA Band I Link + Bluetooth Link + WLAN (5G Band IV) Link + MPEG4(Run Color Bar) + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1
Mode 4	: LTE Band 7 Idle + Bluetooth Link + WLAN (2.4G) Link + GNSS Rx + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1
Mode 5	: DC_1A_n78A Idle + Bluetooth Idle + WLAN Idle + NFC On + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1
Mode 6	: GSM900 Link + Bluetooth Link + WLAN (2.4G) Link + Camera(Rear) + Earphone + Battery + USB Cable (Charging from wireless charging cradle) + SIM 1
Radio Frequency Electromagnetic Field (Refer to EN301 489-1 Section 9.2)	
Mode 1	: GSM900 Link + Earphone(L) + SIM 1 + Battery + USB Cable(Charging from Adapter)
Mode 2	: GSM1800 Link + Earphone(R) + SIM 2 + Battery + USB Cable(Charging from Adapter)
Mode 3	: WCDMA Band I Link + Earphone(L) + SIM 1 + Battery + USB Cable(Charging from Adapter)
Mode 4	: LTE Band 5 Link + Earphone(L) + SIM 1 + Battery + USB Cable(Charging from Adapter)
Mode 5	: LTE Band 8 Link + Earphone(L) + SIM 1 + Battery + USB Cable(Charging from Adapter)
Mode 6	: LTE Band 7 Link + Earphone(L) + SIM 1 + Battery + USB Cable(Charging from Adapter)
Mode 7	: LTE Band 40 Link + Earphone(L) + SIM 1 + Battery + USB Cable(Charging from Adapter)
Mode 8	: GSM900 Link + SIM 1 + Battery + USB Cable(Charging from Adapter)
Mode 9	: WCDMA Band I Link (BER) + Bluetooth Link + WLAN (2.4G) Link + Camera(Rear) + SIM 1 + Battery + USB Cable(Charging from Adapter) + Earphone
Mode 10	: LTE Band 5 Link (Throughput) + Bluetooth Link + WLAN (5G) Link + Camera(Front) + SIM 1 + Battery + USB Cable(Charging from Adapter) + Earphone
Mode 11	: DC_1A_n77A Link (Throughput) + Bluetooth Link + WLAN (5G Band IV) Link + MPEG4(Run Color Bar) + SIM 1 + Battery + USB Cable(Charging from Adapter) + Earphone
Mode 12	: DC_8A_n3A Link (Throughput) + Bluetooth Link + WLAN (5G) Link + NFC On + SIM 1 + Battery + USB Cable (EUT (eMMC) USB Data Link to NB) + Earphone
Mode 13	: DC_41A_n78A Link (Throughput) + Bluetooth Link + WLAN (2.4G) Link + GNSS Rx + SIM 1 + Battery + USB Cable (NB USB Data Link to EUT (eMMC)) + Earphone
Mode 14	: GSM900 Idle + Bluetooth Idle + WLAN Idle + Camera(Rear) + SIM 1 + Battery + USB Cable (Charging from wireless charging cradle) + Earphone
Mode 15	: WCDMA Band I Link (BER) + Bluetooth Link + WLAN (2.4G) Link + Camera(Rear) + Battery + USB Cable(Typec to Typec)(EUT Charging to other phones)
Electrostatic Discharge (Refer to EN301 489-1 Section 9.3)	
Mode 1	: GSM900 Link + Bluetooth Link + WLAN (2.4G) Link + Camera(Rear) + Earphone + Battery + USB Cable (Charging from Adapter) + SIM 1
Mode 2	: GSM1800 Idle + Bluetooth Link + WLAN (5G) Link + Camera(Front) + Earphone + Battery + USB Cable (Charging from Adapter) + SIM 2
Mode 3	: WCDMA Band I Link + Bluetooth Link + WLAN (2.4G) Link + MPEG4(Run Color Bar) + Earphone + Battery + USB Cable (Charging from Adapter) + SIM 1
Mode 4	: LTE Band 7 Idle + Bluetooth Link + WLAN (5G) Link + NFC On + Earphone + Battery + USB Cable (Charging from Adapter) + SIM 1
Mode 5	: DC_1A_n78A Idle + Bluetooth Link + WLAN (2.4G) Link + GNSS Rx + Earphone + Battery + USB Cable (EUT (eMMC) USB Data Link to NB) + SIM 1
Mode 6	: GSM900 Link + Bluetooth Link + WLAN (5G) Link + Camera(Rear) + Earphone + Battery + USB Cable (NB USB Data Link to EUT (eMMC)) + SIM 1
Mode 7	: GSM900 Link + Bluetooth Link + WLAN (2.4G) Link + Camera(Rear) + Earphone + Battery + USB Cable(Typec to Typec)(EUT Charging to other phones) + SIM 1
Mode 8	: GSM900 Link + Bluetooth Idle + WLAN Idle + Earphone + Battery + USB Cable (Charging from wireless charging cradle) + SIM 1
Mode 9	: GSM900 Link + Bluetooth Link + WLAN (2.4G) Link + Camera(Rear) + SIM 1
Fast Transient, Common Mode (Refer to EN301 489-1 Section 9.4)	
Mode 1	: GSM900 Link + Bluetooth Link + WLAN (2.4G) Link + Camera(Rear) + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1

Mode 2	: GSM1800 Idle + Bluetooth Link + WLAN (5G) Link + Camera(Front) + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 2
Mode 3	: WCDMA Band I Link + Bluetooth Link + WLAN (5G Band IV) Link + MPEG4(Run Color Bar) + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1
Mode 4	: LTE Band 7 Idle + Bluetooth Link + WLAN (2.4G) Link + GNSS Rx + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1
Mode 5	: DC_1A_n78A Idle + Bluetooth Idle + WLAN Idle + NFC On + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1
Mode 6	: GSM900 Link + Bluetooth Link + WLAN (2.4G) Link + Camera(Rear) + Earphone + Battery + USB Cable (Charging from wireless charging cradle) + SIM 1
Radio Frequency, Common Mode (Refer to EN301 489-1 Section 9.5)	
Mode 1	: GSM900 Link + Earphone(L) + Battery + SIM 1 + USB Cable (Charging from Adapter)
Mode 2	: GSM1800 Link + Earphone(R) + Battery + SIM 2 + USB Cable (Charging from Adapter)
Mode 3	: WCDMA Band I Link + Earphone(L) + Battery + SIM 1 + USB Cable (Charging from Adapter)
Mode 4	: LTE Band 5 Link + Earphone(L) + Battery + SIM 1 + USB Cable (Charging from Adapter)
Mode 5	: LTE Band 8 Link + Earphone(L) + Battery + SIM 1 + USB Cable (Charging from Adapter)
Mode 6	: LTE Band 7 Link + Earphone(L) + Battery + SIM 1 + USB Cable (Charging from Adapter)
Mode 7	: LTE Band 40 Link + Earphone(L) + Battery + SIM 1 + USB Cable (Charging from Adapter)
Mode 8	: GSM900 Link + Battery + SIM 1 + USB Cable(Charging from Adapter)
Mode 9	: WCDMA Band I Link (BER) + Bluetooth Link + WLAN (2.4G) Link + Camera(Rear) + SIM 1 + Battery + USB Cable (Charging from Adapter) + Earphone
Mode 10	: LTE Band 5 Link (Throughput) + Bluetooth Link + WLAN (5G) Link + Camera(Front) + SIM 1 + Battery + USB Cable (Charging from Adapter) + Earphone
Mode 11	: DC_1A_n77A Link (Throughput) + Bluetooth Link + WLAN (5G Band IV) Link + MPEG4 (Run Color Bar) + SIM 1 + Battery + USB Cable (Charging from Adapter) + Earphone
Mode 12	: DC_8A_n3A Link (Throughput) + Bluetooth Link + WLAN (5G) Link + NFC On + SIM 1 + Battery + USB Cable (Charging from Adapter) + Earphone
Mode 13	: DC_41A_n78A Link (Throughput) + Bluetooth Link + WLAN (2.4G) Link + GNSS Rx + SIM 1 + Battery + USB Cable (Charging from Adapter) + Earphone
Mode 14	: GSM900 Idle + Bluetooth Idle + WLAN Idle + SIM 1 + Battery + USB Cable (Charging from wireless charging cradle)
Voltage Dips and Interruptions (Refer to EN301 489-1 Section 9.7)	
Mode 1	: GSM900 Link + Bluetooth Link + WLAN (2.4G) Link + Camera(Rear) + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1
Mode 2	: GSM1800 Idle + Bluetooth Link + WLAN (5G) Link + Camera(Front) + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 2
Mode 3	: WCDMA Band I Link + Bluetooth Link + WLAN (5G Band IV) Link + MPEG4(Run Color Bar) + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1
Mode 4	: LTE Band 7 Idle + Bluetooth Link + WLAN (2.4G) Link + GNSS Rx + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1
Mode 5	: DC_1A_n78A Idle + Bluetooth Idle + WLAN Idle + NFC On + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1
Mode 6	: GSM900 Link + Bluetooth Link + WLAN (2.4G) Link + Camera(Rear) + Earphone + Battery + USB Cable (Charging from wireless charging cradle) + SIM 1
Surges (Refer to EN301 489-1 Section 9.8)	
Mode 1	: GSM900 Link + Bluetooth Link + WLAN (2.4G) Link + Camera(Rear) + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1
Mode 2	: GSM1800 Idle + Bluetooth Link + WLAN (5G) Link + Camera(Front) + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 2
Mode 3	: WCDMA Band I Link + Bluetooth Link + WLAN (5G Band IV) Link + MPEG4(Run Color Bar) + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1
Mode 4	: LTE Band 7 Idle + Bluetooth Link + WLAN (2.4G) Link + GNSS Rx + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1

Mode 5	: DC_1A_n78A Idle + Bluetooth Idle + WLAN Idle + NFC On + Earphone + Battery + USB Cable(Charging from Adapter) + SIM 1
Mode 6	: GSM900 Link + Bluetooth Link + WLAN (2.4G) Link + Camera(Rear) + Earphone + Battery + USB Cable (Charging from wireless charging cradle) + SIM 1

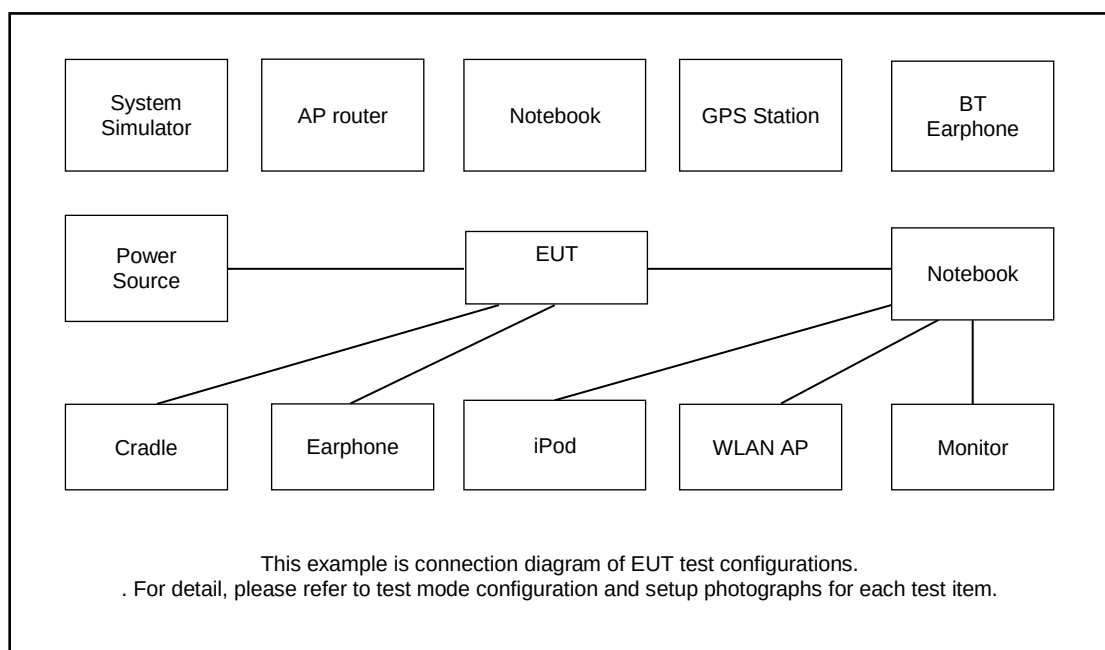
Remark: Data Linking with Notebook means data application transferred mode between EUT and Notebook

Worst mode of all test items listed in section 2.1

Test items	Worst mode
Radiated Emission	8
Conducted Emission	6
Harmonic Current Emissions	1
Voltage Fluctuations and Flicker	1
Radio Frequency Electromagnetic Field	8
Electrostatic Discharge	-
Fast Transient, Common Mode	-
Radio Frequency, Common Mode	8
Voltage Dips and Interruptions	-
Surges	-

Remark: Only data of worst mode (if test item has) was reported in test result.

2.2 Connection Diagram of Test System



2.3 Supported Unit Used in Test Configuration and System

Item	Equipment	Trade Name	Model Name	FCC ID	Data Cable	Power Cord
1.	System Simulator	Anritsu	MT8821C	N/A	N/A	Unshielded,1.8m
2.	5G NR Base Station	Anritsu	MT8000A	N/A	N/A	Unshielded,1.8m
3.	LABSAT GPS Simulator	RACELOGIC	RLLS03-2RP	N/A	N/A	Unshielded,1.8m
4.	GNSS Station	R&S	SMBV100A	N/A	N/A	Unshielded, 1.8m
5.	WLAN AP	D-Link	DIR-655	KA21R655B1	N/A	Unshielded,1.8m
6.	WLAN AP	D-Link	DIR-855	KA2DIR855A2	N/A	Unshielded,1.8m
7.	WLAN AP	D-Link	DIR-815	KA21R815A1	Unshielded,1.8m	Unshielded,1.8m
8.	Notebook	Lenovo	V130-15IKB005	N/A	N/A	AC I/P: Unshielded, 1.8 m DC O/P: Shielded, 1.8 m
9.	Notebook	Lenovo	G480	N/A	N/A	AC I/P: Unshielded, 1.8 m DC O/P: Shielded, 1.8 m
10.	Notebook	Lenovo	G410	N/A	N/A	AC I/P: Unshielded, 0.9 m DC O/P: Shielded, 1.8 m
11.	Notebook	acer	N20C5	N/A	N/A	AC I/P: Unshielded, 0.9 m DC O/P: Shielded, 1.8 m
12.	Earphone	Lenovo	P121	N/A	N/A	Unshielded,1.2m
13.	Bluetooth Earphone	Lenovo	thinkplus-BH3	N/A	N/A	N/A
14.	Bluetooth Earphone	Lenovo	LBH505	N/A	N/A	N/A
15.	SD Card	Kingston	4/8GB	N/A	N/A	N/A
16.	NFC Card	N/A	N/A	N/A	N/A	N/A

2.4 EUT Operation Test Setup

The EUT was set in below conditions during EMI and EMS testing.

Read/Write and Storage of Data

1. Let EUT be connected with notebook or PC.
2. Execute the program for files transferring between EUT and storage device.
3. Monitor the data transmission by checking whether there some error or abnormal action occurred.

NFC

1. Turn on the NFC function of EUT.
2. NFC signal is kept transmitting, and monitoring it by using RF receiver.

GNSS

1. Execute program to make the EUT continuously receive signals from GNSS simulator.
2. The GNSS receiver performance is kept monitoring.

WLAN

1. Enable WLAN function of the EUT.
2. The EUT links with supported units
3. Execute "PING IP" function under the "cmd" of Window system to transfer packet bi-directionally between the EUT and supported units.
4. Monitor the packet loss and WLAN radio performance.

Bluetooth (include Bluetooth Data Link and Bluetooth Headset)

1. Link with supported unit via Bluetooth radio function.
2. Monitor the status of connection by checking the Bluetooth link performance without radio link drop.

GSM Link

1. The GSM radio function is linked with system simulator, and set GSM900 PCL=5, and DCS1800 PCL=0 for maximum output power.
2. The Uplink/Downlink radio quality and audio quality are monitored via the system simulator.

WCDMA Link

1. The WCDMA (RMC12.2Kbps) radio function is linked with system simulator, and set all up bits for maximum output power, and the DTX is disabled.
2. The Uplink/Downlink radio quality, and audio quality are monitored via the system simulator.
3. For call established mode, check from the phone display that the call is maintained throughout the test.
4. For data transmitting, the BER shall be monitored and not over limit.

LTE Link

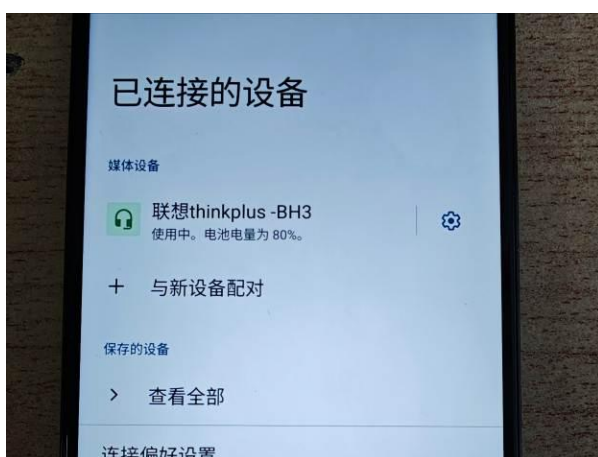
1. The LTE radio function is linked with system simulator, and set all up bits for maximum output power, and the DTX is disabled.
2. For data transmitting, the throughput shall be monitored via the system simulator and not over limit.

5G NR Mode

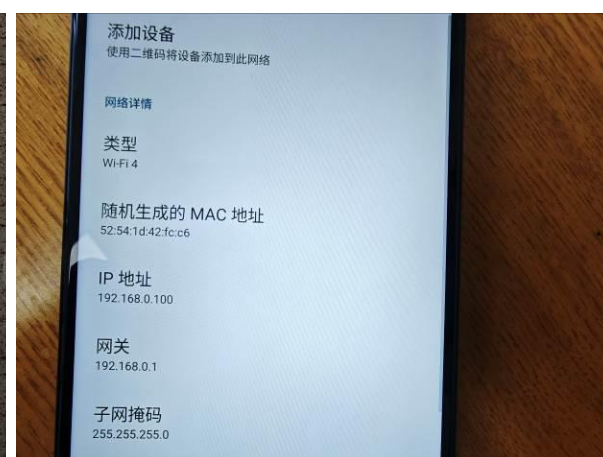
For data transmitting, the throughput shall be monitored via the system simulator.

GSM/WCDMA/LTE/5G NR Idle

For Idle mode, the connection between EUT and GSM/WCDMA/LTE/5G NR system simulator is established.



Monitor the Bluetooth function



Monitor the WLAN function



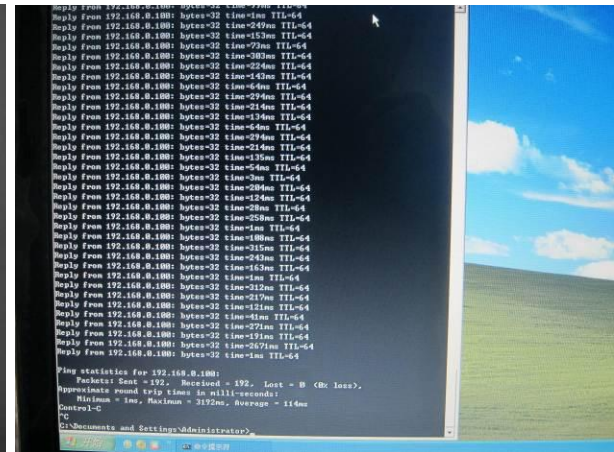
EUT with GNSS function linkage, and the GNSS signal is receipt continuously



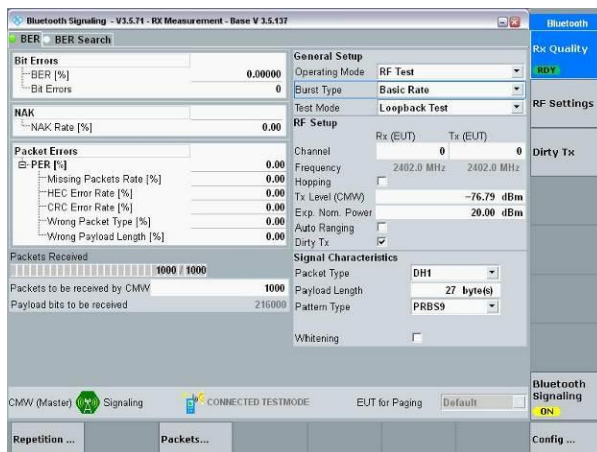
Monitor the NFC function



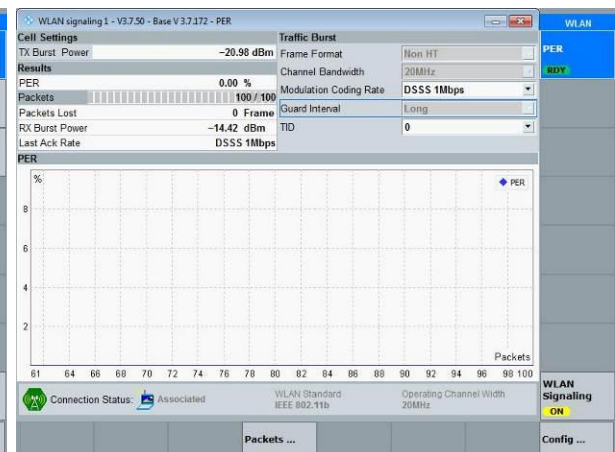
Monitor the MPEG4 function



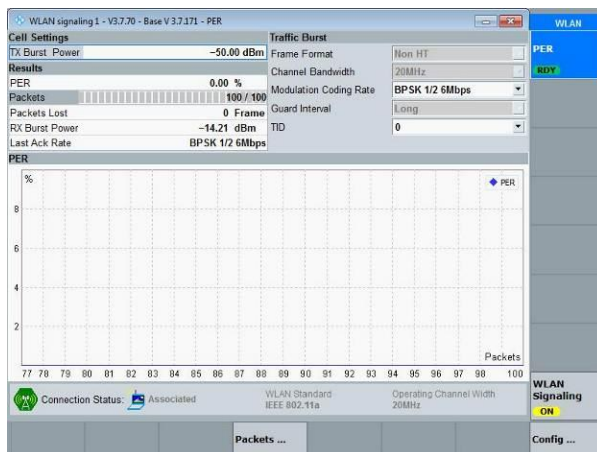
Monitor the WLAN packet lost (0%)



Monitor the Bluetooth PER function



Monitor the WLAN(2.4GHz) PER function



Monitor the WLAN(5GHz) PER function

2.5 Summary of Environment Condition, Test Date and Test Engineer for all Test Items

Test items	Ambient Temperature (°C)	Relative Humidity (%)	Atmospheric Pressure (kPa)	Test Date	Test Engineer
Radiated Emission	23.7~24.3	47~48	98	Mar. 20, 2026	Yoke Si
Conducted Emission	24.2~25.6	37~39	98	Feb. 28, 2026	Buck
Harmonic Current Emissions	22~26	33~36	98	Feb. 12, 2026	Eko Guan
Voltage Fluctuations and Flicker	22~26	33~36	98	Feb. 12, 2026	Eko Guan
Electrostatic Discharge (ESD)	23.5~24	43~45	98	Feb. 10, 2026	sun
Continuous Radiated Disturbances (RS)	18~19	45~46	98	Mar. 08, 2026	DAHE
Electrical Fast Transients (EFT/BURST)	22~26	33~36	98	Feb. 12, 2026	Eko Guan
Continuous Conducted Disturbances (CS)	21~23	43~45	98	Mar. 10, 2026	LUYU
Surges	22~26	33~36	98	Feb. 12, 2026	Eko Guan
Voltage Dips and Interruptions	22~26	33~36	98	Feb. 12, 2026	Eko Guan

2.6 Test Software

Item	Site	Manufacturer	Name	Version
1.	03CH07-KS	AUDIX	E3	210616
2.	10CH01-KS	AUDIX	E3	210616
3.	CO01-KS	AUDIX	E3	6.2009-8-24
4.	RS01-KS	R&S	EMC32	10.50.00
5.	CS01-KS	R&S	EMC32	10.50.00
6.	EX01-KS (surge Dips EFT)	TQSEQ	win3000	1.3.2
7.	EX01-KS (H&F)	TQSEQ	Win2100V4	4.30.0

3. Test Conditions of 301489 Series Standards

3.1 Special Conditions of Applied Standards for EUT

Below each section is special condition applied for each application of EUT.

3.1.1 Emission

EN301 489-3

Reference to clauses in EN 301 489-1	Special product-related conditions, additional to or modifying the test conditions in EN 301 489-1, clause 8
8.3 DC power input/output	The requirements of ETSI EN 301 489-1 [1] clause 8.3 shall be applied where the cable length exceeds 3 m or when connected to a vehicle power supply.

EN301 489-17

No special conditions shall apply to UE in the scope of the present document.

EN301 489-19

Reference to clauses in EN 301 489-1	Special product-related conditions, additional to or modifying the test conditions in EN 301 489-1, clause 8
8.3 DC power input/output	The requirements of ETSI EN 301 489-1 [1] clause 8.3 shall be applied where the cable length exceeds 3 m or when connected to a vehicle power supply.

EN301 489-52

Reference to clauses in EN 301 489-1	Special product-related conditions, additional to or modifying the test conditions in EN 301 489-1, clause 8
8.2 Enclosure port of ancillary equipment	According to typical of intended use of radio equipment, ancillary equipment can also be measured in combination with the radio equipment under test. When the ancillary equipment is measured in combination with the radio equipment, radiated emissions from the transmitter/transceiver shall be ignored, but recorded in the test report.

3.1.2 Immunity
EN301 489-3

Reference to clauses in EN 301 489-1	Special product-related conditions, additional to or modifying the test conditions in EN 301 489-1, clause 9
9.2.2: Test method; Radiofrequency electromagnetic field	Where the EUT is subject to EMC Immunity testing under a Harmonised Standard of a Directive other than the Directive 2014/53/EU then the modulating signal frequency specified in that Harmonised Standard may be used. If this alternative modulating frequency is used, then the applicable Directive, Harmonised Standard & modulating frequency shall be noted in the test report.
9.4: Fast transients common mode	The requirements of ETSI EN 301 489-1 [1], clauses 9.4.1 and 9.4.2 shall be applied with the exception of clause 7.4 of EN 61000-4-4 [4].
9.5.2: Test method; Radiofrequency, common mode	Where the EUT is subject to EMC Immunity testing under a Harmonised Standard of a Directive other than the Directive 2014/53/EU then the modulating signal frequency specified in that Harmonised Standard may be used. If this alternative modulating frequency is used, then the applicable Directive, Harmonised Standard & modulating frequency shall be noted in the test report.

EN301489-17

Reference to clauses in EN 301 489-1	Special product-related conditions, additional to or modifying the test conditions in EN 301 489-1, clause 9
9.8.2 Surges	9.8.2.0 General Where the surge coupling network for the 10/700 (5/320) μ s waveform affects the functioning of high-speed data ports, the test shall be carried out using a 1,2/50 (8/20) μ s waveform and appropriate coupling network.

EN301489-19

No special conditions are relevant for products covered in the present document.

EN301489-52

Reference to clauses in EN 301 489-1	Special product-related conditions, additional to or modifying the test conditions in EN 301 489-1, clause 9
9.3 Electrostatic discharge	When applying direct discharge to a portable or handheld battery-powered EUT with a display screen, it may not be possible to observe the screen for a given EUT orientation. If observation of the screen is necessary during the test, the EUT may be mounted vertically using non-metallic supports as specified in the last paragraph of clause 4.2.1 of EN 55035.

3.2 RF Exclusion Band of Radio Equipment

3.2.1 EN301489-1

3.2.1.1 Exclusion band for transmitters or the transmitter part of transceivers

a. General

Exclusion bands shall not be applied when measuring transmitters in standby mode.

b. Channelised Equipment

For channelised equipment the exclusion band shall extend 250% of the channel width either side of the transmitter centre frequency.

NOTE: Exclusion band of 250 % is based on the definition from ITU Radio Regulations [i.8] 1.146, 1.146A and 1.146B

c. Non-Channelised Equipment

For non-channelised equipment the exclusion band shall extend 250% of the occupied bandwidth either side of the transmitter centre frequency.

NOTE: Exclusion band of 250 % is based on the definition from ITU Radio Regulations [i.8] 1.146, 1.146A and 1.146B.

3.2.1.2 Exclusion band for receivers or the receiver part of transceivers

a. Applicability

Exclusion bands are not applied when testing emissions of receivers or receiver part of transceivers.

b. Channelised Equipment

For channelised equipment the exclusion band shall be calculated by using the following formulae:

For the lower edge for the exclusion band:-

$$\text{EXband(lower)} = \text{Band}_{\text{RX}}(\text{lower}) - n\text{ChW}_{\text{RX}}$$

and for the upper edge of the exclusion band:-

$$\text{EXband(upper)} = \text{Band}_{\text{RX}}(\text{upper}) + n\text{ChW}_{\text{RX}}$$

Where n = number of channel widths required for exclusion band

NOTE: For equipment that support multiple channel widths the Channel Width used should be the widest support by the EUT.

Where the present document is being used in a stand-alone basis, the value of n shall be 1.

c. Non-Channelised Equipment

For non-channelized equipment the exclusion band shall be calculated by using the following

formula:

For the lower edge for the exclusion band:-

$$\text{EXband(lower)} = \text{Band}_{\text{RX}}(\text{lower}) - n\text{BW}_{\text{RX}}$$

and for the upper edge of the exclusion band:-

$$\text{EXband(upper)} = \text{Band}_{\text{RX}}(\text{upper}) + n\text{BW}_{\text{RX}}$$

Where n = multiple of whole bandwidths required to define exclusion band

Bandwidth of Receiver is the occupied bandwidth of the corresponding transmitter signal.

Where the present document is being used in a stand-alone basis, the value of n shall be 1.

3.2.2 EN301489-3

3.2.2.1 Exclusion bands for emissions testing

a. Transmitters

Where the applicable harmonised radio standard specifies an Operating Channel and an Out of Band domain, then these together shall form the exclusion band.

Where this is not so specified the exclusions bands shall be as below:

For transmitters operating, or intended to operate, in a channelized frequency band, the exclusion band is five times (i.e., +/-250%) the maximum operating channel width (OCW) allowed for that service, centred around the operating frequency.

For all other transmitters, the exclusion band is twice the intended operating frequency band centred around the centre frequency of the intended operating frequency band.

The exclusion band shall only apply when the EUT is in transmit mode of operation.

b. Receivers

No exclusion band applies.

c. Duplex and multi-mode equipment

In the case of EUT tested with a simultaneous transmit and receive mode, the exclusion band used shall be the one for the transmitter. I.e. only one exclusion band shall be applied.

In the case of transmitters capable of operating on more than one frequency band, testing shall be carried out on each band separately.

3.2.2.2 Exclusion bands for immunity testing

a. Transmitters

The exclusion band shall be as specified for emissions testing in clause 4.3.2.1.

b. Receivers

The exclusion band is based on the lower edge (FOC_{low}) and the upper edge (FOC_{high}) and the centre (f_0) of the Operating Channel (OC).

Where the OC is defined in the applicable harmonised radio standard, the values for that shall be used and f_0 shall be the centre of the OC. In all other cases, f_0 shall be the receiver operating frequency and the OC shall be $\pm 0,05\%$ around f_0 .

Tables 2 and 3 show the lower and upper limits of the exclusion band.

Table 2: Lower limit of exclusion band

Lower edge of OC, FOC_{low}	Lower limit of exclusion band
< 3 MHz	0
3 MHz to < 30 MHz	$FOC_{low} - 3\text{ MHz}$
30 MHz to < 42 MHz	27 MHz
42 MHz to < 1 GHz	$FOC_{low} - 15\text{ MHz}$, or $FOC_{low} - 0,05 \times f_0$, whichever is lower
1 GHz to < 1,05 GHz	950 MHz
1,05 GHz to < 6 GHz	$FOC_{low} - 100\text{ MHz}$
6 GHz to < 6,3 GHz	5,9 GHz
$\geq 6,3\text{ GHz}$	$FOC_{low} - 0,05 \times f_0$

Table 3: Upper limit of exclusion band

Upper edge of OC, FOC_{high}	Upper limit of exclusion band
< 300 kHz	$FOC_{high} + 300\text{ kHz}$
300 kHz to < 30 MHz	$FOC_{high} + 3\text{ MHz}$
30 MHz to < 42 MHz	$FOC_{high} + 5\text{ MHz}$
42 MHz to < 1 GHz	$FOC_{high} + 15\text{ MHz}$, or $FOC_{high} + 0,05 \times f_0$, whichever is higher
1 GHz to < 6 GHz	$FOC_{high} + 100\text{ MHz}$
$\geq 6\text{ GHz}$	$FOC_{high} + 0,05 \times f_0$

c. Duplex and multi-mode equipment

In the case of EUT tested with a simultaneous transmit and receive mode, the exclusion band used shall be the combination of the exclusion band for the transmitter and the exclusion band for the receiver. I.e. both exclusion bands shall be applied.

In the case of transmitters capable of operating on more than one frequency band, testing shall be carried out on each band separately.

In the case of receivers operating on more than one frequency, the exclusion band used shall be the combination of the exclusion bands for each operating frequency, i.e. an exclusion band for each operating frequency shall be applied

NOTE: Where the operating frequencies are in the same operational frequency band, the result will usually be an enlarged single exclusion band. Where the frequencies are widely spaced, e.g. in different bands, the result will be to create multiple separate exclusion bands.

3.2.3 EN301489-17

The frequencies on which the transmitter part of the EUT is intended to operate shall be excluded from radiated emission measurements when performed in transmit mode of operation.

There shall be no frequency exclusion band applied to emission measurements of the receiver part of transceivers or the stand alone receiver under test, and/or associated ancillary equipment. For EUT that operate above 6 GHz there is no exclusion band specified as test ranges stop at 6 GHz.

The exclusion band for immunity testing of equipment operating in the 2.4GHz band shall be:

- 'lower limit of exclusion band = lowest allocated band edge frequency -120MHz, i.e. 2 280MHz;
- 'upper limit of exclusion band = highest allocated band edge frequency +120MHz, i.e. 2 603.5MHz.

Note: This is based upon a channel size of 40 MHz and a value of $n = 3$ from ETSI EN 301 489-1 [1], clause 4.3.3. These values also take into account the blocking test frequencies from ETSI EN 300 328 [i.8].

The exclusion band for immunity testing of equipment operating in the 5GHz WiFi band shall be:

- 'lower limit of exclusion band = lowest allocated band edge frequency -320MHz, i.e. 4 830MHz;
- 'as the immunity requirements have an upper frequency range of 6 GHz and any upper edge exclusion band would be greater than this for both the 5 470 MHz - 5 725 MHz and 5 725 MHz - 5 850 MHz bands. Therefore the test stops at the lower limit of exclusion band (i.e. 4 830 MHz).

Note: This is based upon a channel size of 80 MHz and a value of $n = 4$ from ETSI EN 301 489-1 [1], clause 4.3.3. These values also take into account the blocking test frequencies from ETSI EN 301 893 [i.3]

The exclusion band for immunity testing of equipment operating in the 5.8GHz band shall be:

- 'lower limit of exclusion band = lowest allocated band edge frequency -440MHz, i.e. 5 285MHz;
- 'as the immunity requirements have an upper frequency range of 6 GHz and any upper edge exclusion band would be greater than this for the 5,8 GHz band. Therefore the test stops at the lower limit of exclusion band (i.e. 5 285 MHz).

Note: This is based upon a channel size of 40 MHz and a value of $n = 11$ from ETSI EN 301 489-1 [1], clause 4.3.3. These values also take into account the blocking test frequencies from ETSI EN 302 502 [i.4]

The exclusion band for immunity testing of equipment operating in 6 GHz WLAN band shall be:

- 'lower limit of exclusion band = lowest allocated band edge frequency -480 MHz, i.e. 5 465 MHz;
- 'as the immunity requirements have an upper frequency range of 6 GHz and any upper edge exclusion band would be greater than this for the 5 945 MHz to 6 425 MHz band. Therefore, the test stops at the lower limit of exclusion band (i.e. 5 465 MHz).

NOTE: This is based upon a channel size of 160 MHz and a value of $n = 3$ from ETSI EN 301

489-1 [1], clause 4.3.3. These values also take into account the blocking test frequencies from ETSI EN 303 687 [i.2]

3.2.4 EN301489-19

The provision of ETSI EN 301 489-1, clause 4.3.1 shall apply with the following modifications:

- There shall be no exclusion bands for the ancillary equipment;
- The receiver exclusion band as defined below shall apply.

3.2.4.1 Receiver exclusion band

The receiver exclusion band is the band of frequencies over which no tests of radiated immunity of a receiver are made.

a. GNSS exclusion bands

The exclusion band for immunity testing of equipment operating in the 1 559 MHz to 1 610 MHz band shall be:

· lower limit of exclusion band = 1 492 MHz (-67 MHz of the lowest band edge frequency).

· upper limit of exclusion band = 1 706 MHz (+96 MHz of the highest band edge frequency).

The exclusion band for immunity testing of equipment operating in the 1 164 MHz to 1 300 MHz band shall be:

· lower limit of exclusion band = 1 100 MHz (-64 MHz of the lowest band edge frequency).

· upper limit of exclusion band = 1 364 MHz (+64 MHz of the highest band edge frequency).

b. ROMES Exclusion Bands

The receiver exclusion band is the band of frequencies over which no tests of radiated immunity are conducted.

The exclusion band for immunity testing of equipment operating in the 1 518 MHz to 1 559 MHz band shall be:

· lower limit of exclusion band = 1 492 MHz (-26 MHz of the lowest band edge frequency).

· upper limit of exclusion band = 1 627 MHz (+68 MHz of the highest band edge frequency).

c. Combined Equipment GNSS & ROMES

When equipment is designed to perform both GNSS and ROMES receive functions with shared hardware, the exclusion band shall be the union of the GNSS and the ROMES exclusion bands:

· lower limit of exclusion band = 1 492 MHz (-26 MHz of the lowest band edge frequency for the ROMES component).

· upper limit of exclusion band = 1 706 MHz (+96 MHz of the highest band edge frequency for the GNSS component).

3.2.5 EN301489-52**3.2.5.1 GSM Transmitter exclusion band**

The exclusion band for transmitters and transmitter sections of transceivers is the band of frequencies over which no immunity tests with radiated RF are made.

The exclusion band for transmitters is as defined in clause 4.3.2.2 of ETSI EN 301 489-1 [1], 250 % of the channel width either side of the nominal operating frequency of the transmitter

3.2.5.2 GSM Receiver exclusion band

The exclusion band for receivers and receiver sections of transceivers is the band of frequencies over which no immunity tests with radiated RF are made as defined in clause 4.3.3 of ETSI EN 301 489-1 [1] where $n = 1$ and Channel Width is 200 kHz.

3.2.5.3 UTRA and E-UTRA Transmitter exclusion band**UTRA**

The frequency bands including in band emissions and out of band emissions are covered by the RF spectral mask specification and need no further consideration

For the purpose of EMC specifications the transmitter exclusion band this shall be as defined in clause 4.3.2.2 of ETSI EN 301 489-1 [1].

E-UTRA

For the purpose of EMC specifications there shall be a transmitter exclusion band as defined in clause 4.3.2.2 of ETSI EN 301 489-1 where BW Channel is the channel bandwidth as defined in clause 5.6 ETSI TS 136 101.

3.2.5.4 UTRA and E-UTRA Receiver exclusion band

As defined in clause 4.3.3 of ETSI EN 301 489-1 where $n=1$ and Channel Width is as follows:

- UTRA Channel Width 5MHz.
- E-UTRA Channel Width 20MHz (see note).

NOTE: For systems that support multiple channel widths the Channel Width used should be the widest support by the EUT.

3.2.5.5 NR SA and NSA Transmitter exclusion band for FR1

For the purpose of EMC specifications there shall be a transmitter exclusion band as defined in clause 4.3.2.2 of ETSI EN 301 489-1 [1] where BW Channel is the channel bandwidth as defined in ETSI TS 138 101-1 [14]

3.2.5.6 NR SA and NSA Receiver exclusion band for FR1

As defined in clause 4.3.3 of ETSI EN 301 489-1 [1] where $n = 1$ and Channel Width is as follows:

- NR Channel Width 100 MHz.
- E-UTRA Channel Width 20 MHz (see note at clause 4.3.4).

3.2.5.7 NR SA and NSA Transmitter exclusion band for FR2

For EUTs that operate above 6 GHz there is no exclusion band specified as test ranges stop at 6 GHz.

3.2.5.8 NR SA and NSA Receiver exclusion band for FR2

For EUTs that operate above 6 GHz there is no exclusion band specified as test ranges stop at 6 GHz.

4. EMC Emission Measurements

4.1 Radiated Emission Test (Refer to EN301 489-1 Section 8.2)

4.1.1 Limits for Radiated Emission Test

<Class B limit>

Frequency Range (MHz)	Measurement		Class B limits dB (μV/m)
	Distance (m)	Detector Type/ Bandwidth	OATS/SAC
30 ~ 230	10	Quasi Peak / 120 kHz	30
230 ~ 1000			37
30 ~ 230	3		40
230 ~ 1000			47

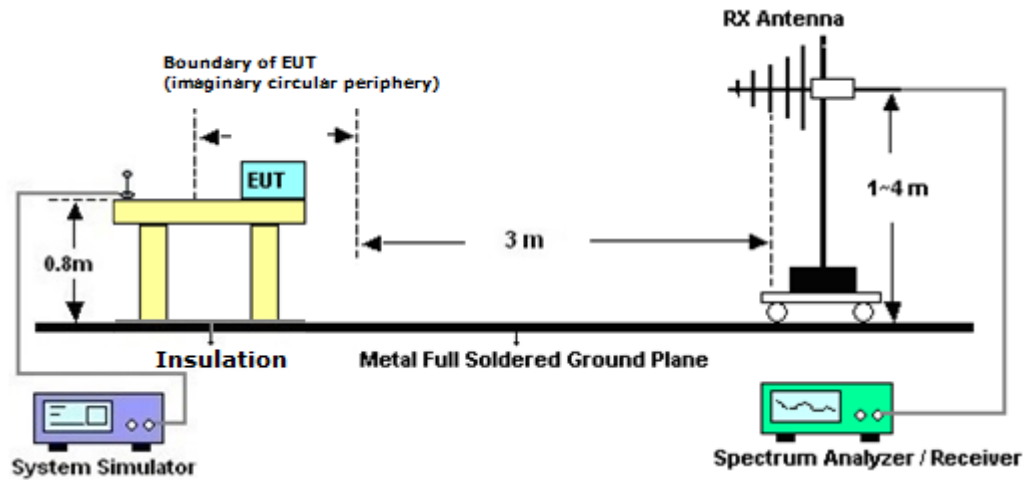
Frequency Range (MHz)	Measurement		Class B limits dB(μV/m)
	Distance (m)	Detector Type/ Bandwidth	FSOATS
1000 ~ 3000	3	Average / 1 MHz	50
3000 ~ 6000			54
1000 ~ 3000		Peak / 1 MHz	70
3000 ~ 6000			74

Conditional testing frequency:

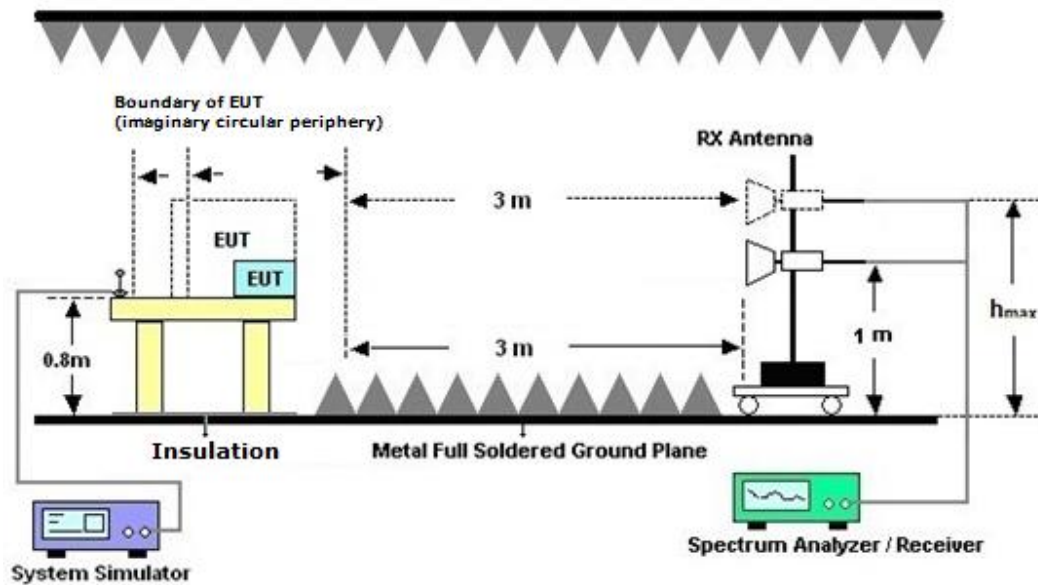
Highest measured frequency	Highest measured frequency
$F_x \leq 108 \text{ MHz}$	1 GHz
$108 \text{ MHz} < F_x \leq 500 \text{ MHz}$	2 GHz
$500 \text{ MHz} < F_x \leq 1 \text{ GHz}$	5 GHz
$F_x > 1 \text{ GHz}$	5 x F_x up to a maximum of 6 GHz

4.1.2 Test Setup

<Radiated Emissions Frequency: 30 MHz to 1000 MHz>

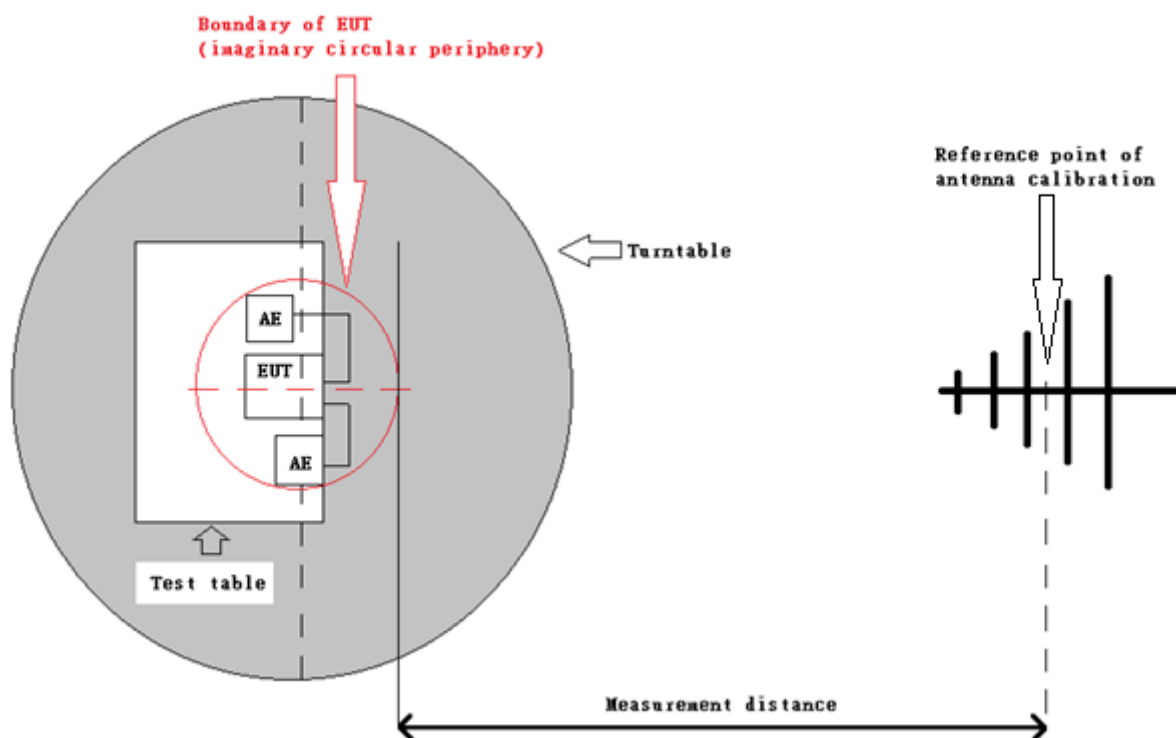


<Radiated Emissions Frequency: 1000 MHz to 6000 MHz>



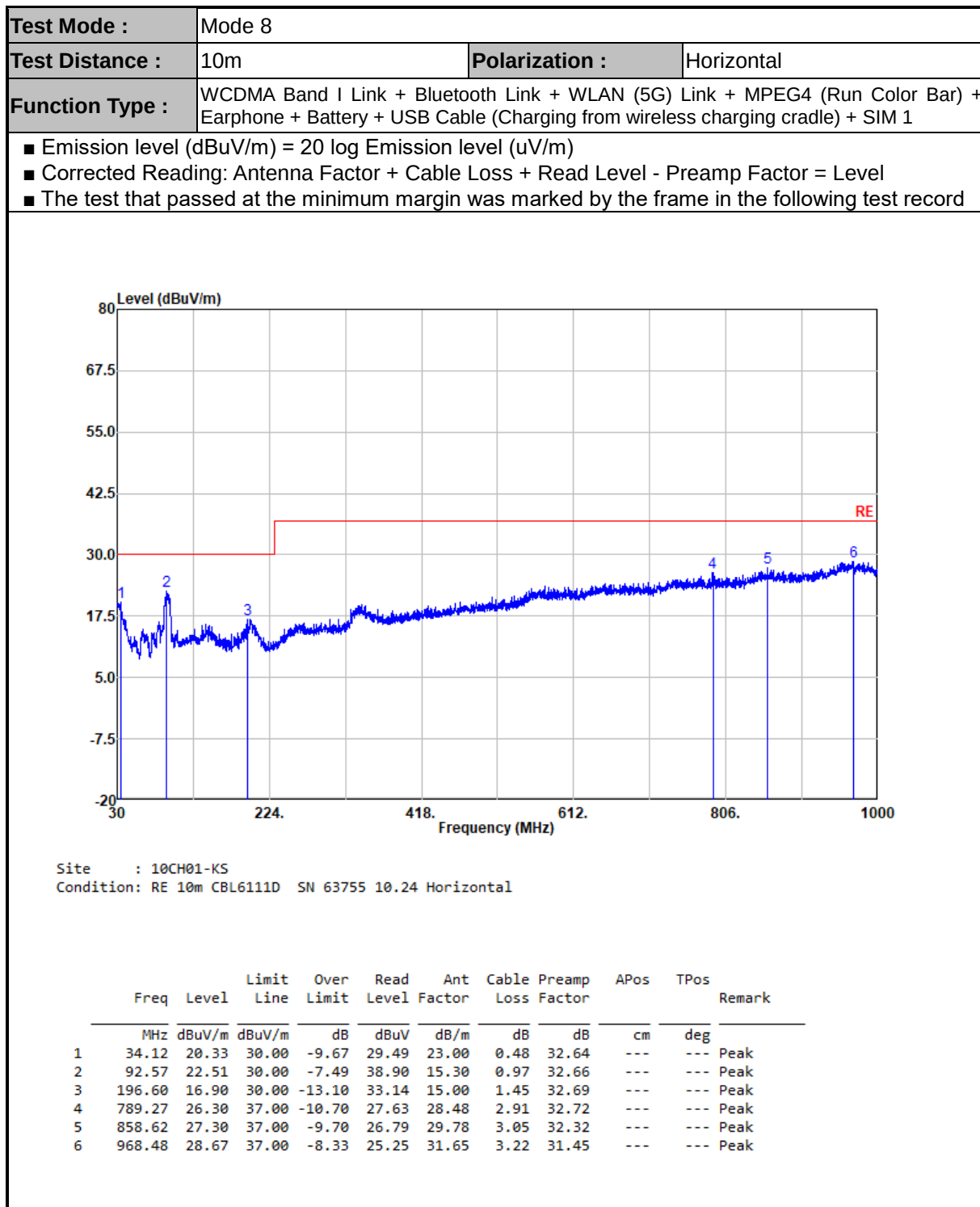
Remark: When EUT's height is over 172cm, h_{max} = top of EUT

<Radiated Emissions Setup Configuration>



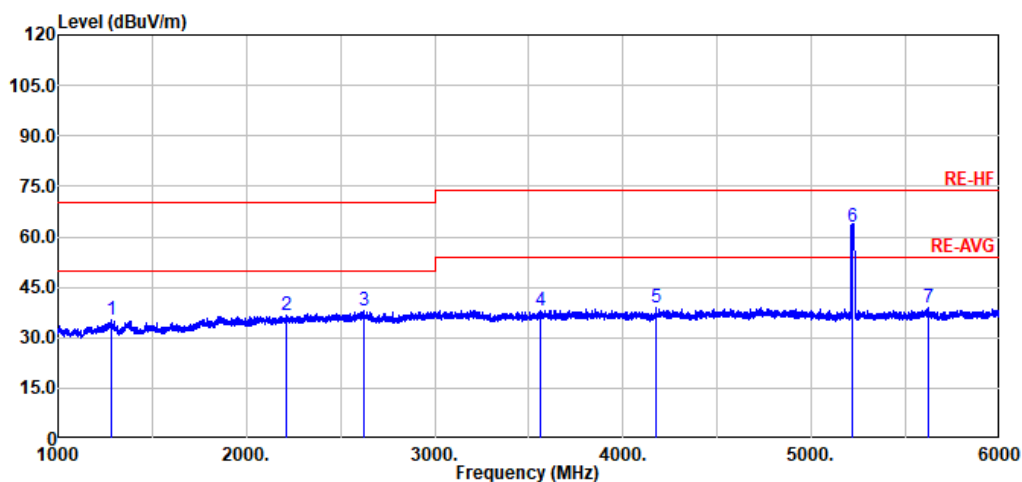
4.1.3 Test Procedures

- a. The EUT shall be placed upon a non-conductive table 0,8 m above the horizontal ground reference plane of the test site.
- b. The boundary of EUT was set 10 meters(30 MHz~1G) and 3 meters(1G~6G) from the receiving antenna which was mounted on the top of a variable height antenna tower. Cables connecting to outside area is directly dropped to, but with an insulation holder less than 150mm height, the reference ground plane.
- c. The table was rotated 360 degrees to determine the position of the highest radiation.
- d. The height of the antenna is varied between 1 m and 4 m above ground to find the maximum value of the field strength for both horizontal polarization and vertical polarization of the antenna.
- e. For each suspected emission, the EUT was arranged to its worst case and then tune the antenna tower (from 1 m to 4 m) and turntable (from 0 degree to 360 degrees) to find the maximum reading values.
- f. Ideally, the central point of the arrangement shall be positioned at the centre of the turntable and the rear of the arrangement shall be flush with the back of the supporting tabletop unless that would not be possible or typical of normal use.
- g. All units of equipment forming the system under test (includes the EUT as well as connected peripherals and associated equipment or devices) shall be arranged such that a nominal 0,1 m separation is achieved between the neighboring units.
- h. Where the mains cable supplied by the manufacturer is longer than 1 m, the excess should be folded at the centre into a bundle no longer than 0,4 m, so that its length is shortened to 1 m.
- i. If the emission level of the EUT in peak mode was 3 dB lower than the limit specified, peak values of EUT will be reported. Otherwise, the emission will be repeated by using the quasi-peak method and reported for frequency range below 1GHz.
- j. If emission level of the EUT in Peak measurement mode is 20dB lower than Peak limit line (that means the emission level in Peak measurement mode complies with both Peak and Average limit lines), then only Peak measurement result is reported. Otherwise, emissions in Average measurement mode shall be measured, and reported.
- k. Test Results Explanation:
$$\text{Level(dB}\mu\text{V/m)} = \text{Read Level(dB}\mu\text{V)} + \text{Antenna Factor(dB/m)} + \text{Cable Loss(dB)} - \text{Preamp Factor(dB)}$$
$$\text{Over Limit(dB)} = \text{Level(dB}\mu\text{V/m)} - \text{Limit Line(dB}\mu\text{V/m)}$$

4.1.4 Test Result


Test Mode :	Mode 8		
Test Distance :	3m	Polarization :	Horizontal
Function Type :	WCDMA Band I Link + Bluetooth Link + WLAN (5G) Link + MPEG4 (Run Color Bar) + Earphone + Battery + USB Cable (Charging from wireless charging cradle) + SIM 1		
Remark :	#6 is RF signal which comes from WLAN Access Point used to connect the EUT, and which can be ignored.		

- Emission level (dBuV/m) = 20 log Emission level (uV/m)
- Corrected Reading: Antenna Factor + Cable Loss + Read Level - Preamp Factor = Level
- The test that passed at the minimum margin was marked by the frame in the following test record

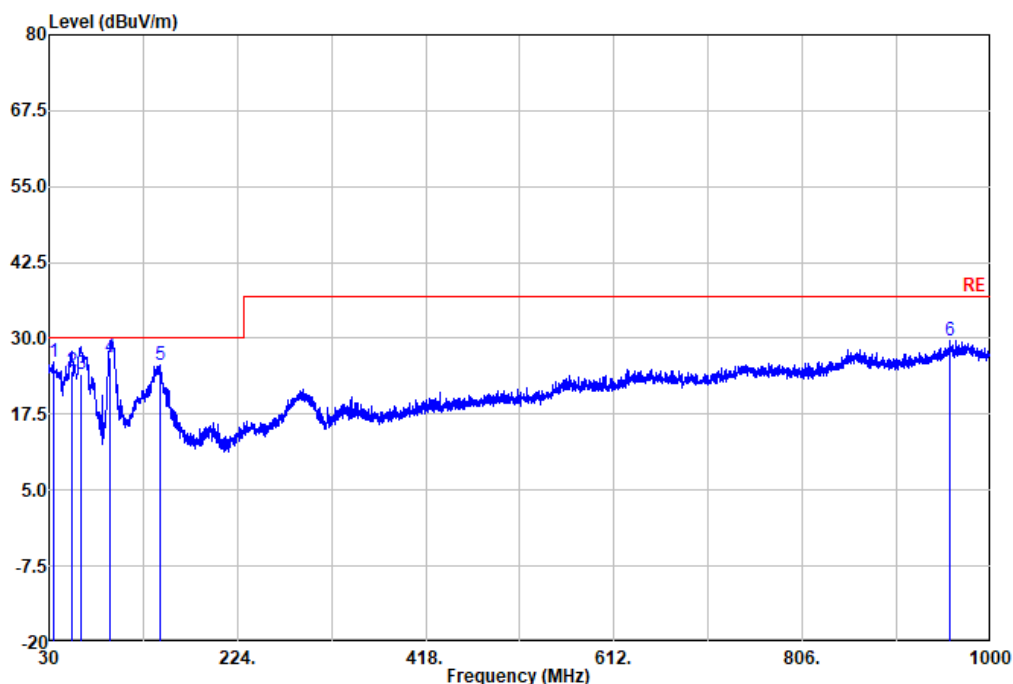


Site : 03CH07-KS
Condition: RE-HF 3m 3117-00240132 Horizontal

	Freq	Level	Limit	Over	Read	Ant	Cable	Preamp	APos	TPos	Remark
	MHz	dBuV/m	dBuV/m	dB	dBuV	dB/m	dB	dB	cm	deg	
1	1281.00	35.15	70.00	-34.85	64.25	29.15	3.87	62.12	---	---	Peak
2	2212.50	36.56	70.00	-33.44	62.72	32.05	5.08	63.29	---	---	Peak
3	2626.00	37.87	70.00	-32.13	62.36	32.90	5.55	62.94	---	---	Peak
4	3562.50	38.14	74.00	-35.86	61.66	33.20	6.48	63.20	---	---	Peak
5	4176.50	38.83	74.00	-35.17	61.34	34.48	7.06	64.05	---	---	Peak
6	5217.00	62.86			84.61	34.80	7.95	64.50	---	---	Peak
7	5618.00	39.01	74.00	-34.99	60.44	34.90	8.29	64.62	---	---	Peak

Test Mode :	Mode 8		
Test Distance :	10m	Polarization :	Vertical
Function Type :	WCDMA Band I Link + Bluetooth Link + WLAN (5G) Link + MPEG4 (Run Color Bar) + Earphone + Battery + USB Cable (Charging from wireless charging cradle) + SIM 1		

- Emission level (dBuV/m) = 20 log Emission level (uV/m)
- Corrected Reading: Antenna Factor + Cable Loss + Read Level - Preamp Factor = Level
- The test that passed at the minimum margin was marked by the frame in the following test record

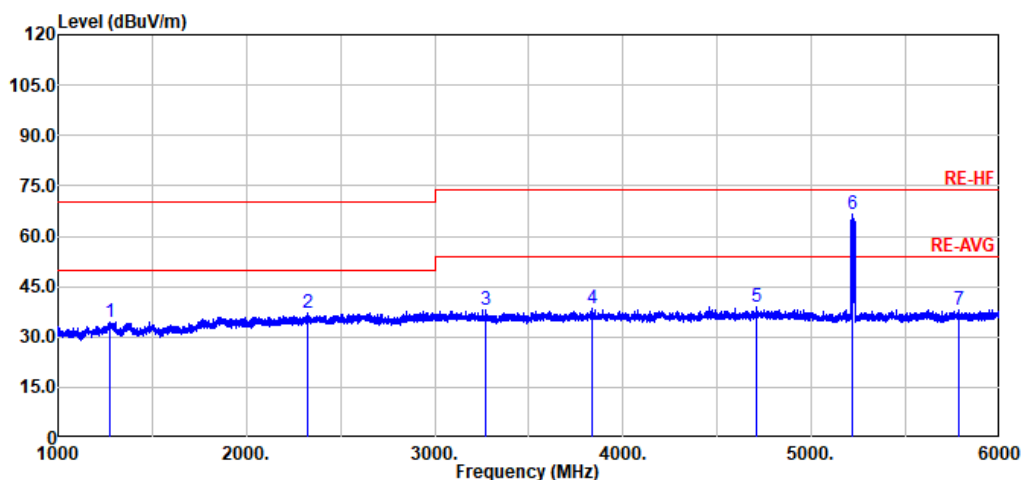


Site : 10CH01-KS
Condition: RE 10m CBL6111D SN 63161 10.24 Vertical

	Freq	Level	Limit	Over	Read	Ant	Cable	Preamp	APos	TPos	Remark
	MHz	dBuV/m	dBuV/m	dB	dBuV	dB/m	dB	dB	cm	deg	
1	34.12	26.02	30.00	-3.98	35.20	22.96	0.52	32.66	---	---	Peak
2	53.28	24.68	30.00	-5.32	43.34	13.29	0.71	32.66	100	20	QP
3	62.74	24.22	30.00	-5.78	44.13	11.96	0.80	32.67	300	360	QP
4	93.05	26.88	30.00	-3.12	43.26	15.27	1.03	32.68	100	345	QP
5	144.46	25.53	30.00	-4.47	39.36	17.50	1.32	32.65	---	---	Peak
6	957.81	29.66	37.00	-7.34	26.50	31.28	3.36	31.48	---	---	Peak

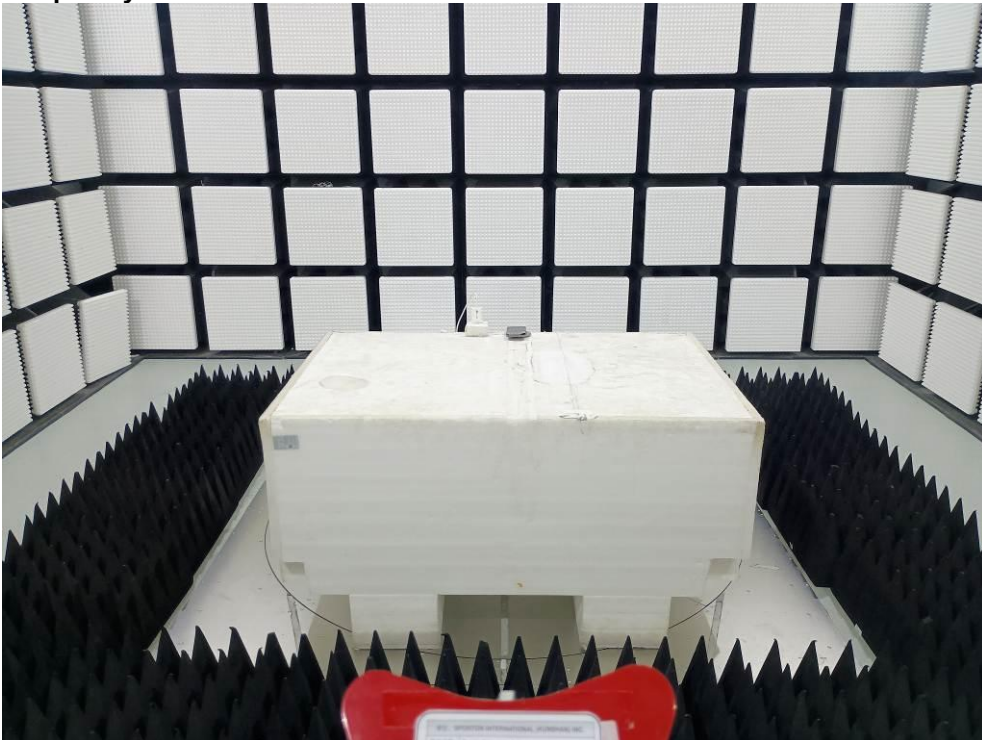
Test Mode :	Mode 8		
Test Distance :	3m	Polarization :	Vertical
Function Type :	WCDMA Band I Link + Bluetooth Link + WLAN (5G) Link + MPEG4 (Run Color Bar) + Earphone + Battery + USB Cable (Charging from wireless charging cradle) + SIM 1		
Remark :	#6 is RF signal which comes from WLAN Access Point used to connect the EUT, and which can be ignored.		

- Emission level (dBuV/m) = 20 log Emission level (uV/m)
- Corrected Reading: Antenna Factor + Cable Loss + Read Level - Preamp Factor = Level
- The test that passed at the minimum margin was marked by the frame in the following test record



Site : 03CH07-KS
Condition: RE-HF 3m 3117-00240132 Vertical

	Freq	Level	Limit	Over	Read	Ant	Cable	Preamp	APos	TPos	Remark
	MHz	dBuV/m	dBuV/m	dB	dBuV	dB/m	dB	dB	cm	deg	
1	1273.00	34.53	70.00	-35.47	63.72	29.08	3.86	62.13	---	---	Peak
2	2324.00	37.12	70.00	-32.88	62.96	32.19	5.21	63.24	---	---	Peak
3	3271.50	38.19	74.00	-35.81	62.12	32.99	6.22	63.14	---	---	Peak
4	3832.50	38.46	74.00	-35.54	61.48	33.67	6.76	63.45	---	---	Peak
5	4708.50	38.75	74.00	-35.25	60.68	34.72	7.55	64.20	---	---	Peak
6	5217.00	66.40			88.15	34.80	7.95	64.50	---	---	Peak
7	5780.50	38.17	74.00	-35.83	59.44	35.06	8.45	64.78	---	---	Peak

4.1.5 Setup Photographs**Mode 8****Frequency: 30 MHz to 1000 MHz****Frequency: 1000 MHz to 6000 MHz**

4.2 Conducted Emission Test (Refer to EN301 489-1 Section 8.3, 8.4 and 8.7)

4.2.1 Limits for Conducted Emissions

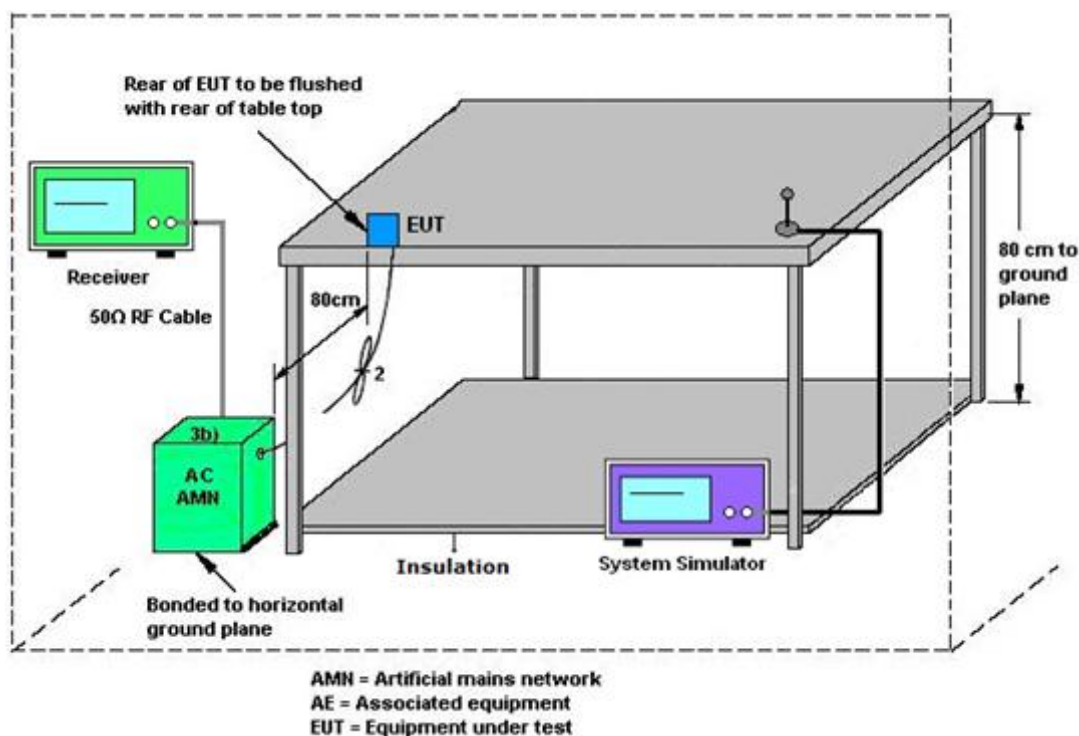
<Class B>

Applicable to AC mains power ports			
Frequency Range (MHz)	Coupling Device	Detector Type / Bandwidth	Class B limits dB(μV)
0.15~0.5	AMN	Quasi Peak / 9 kHz	66~56
0.5~5			56
5~30			60
0.15~0.5	AMN	Average / 9 kHz	56~46
0.5~5			46
5~30			50

The limit decreases linearly with the logarithm of the frequency in the range 0,15 MHz to 0,50 MHz.

4.2.2 Test Setup

<AC power port>

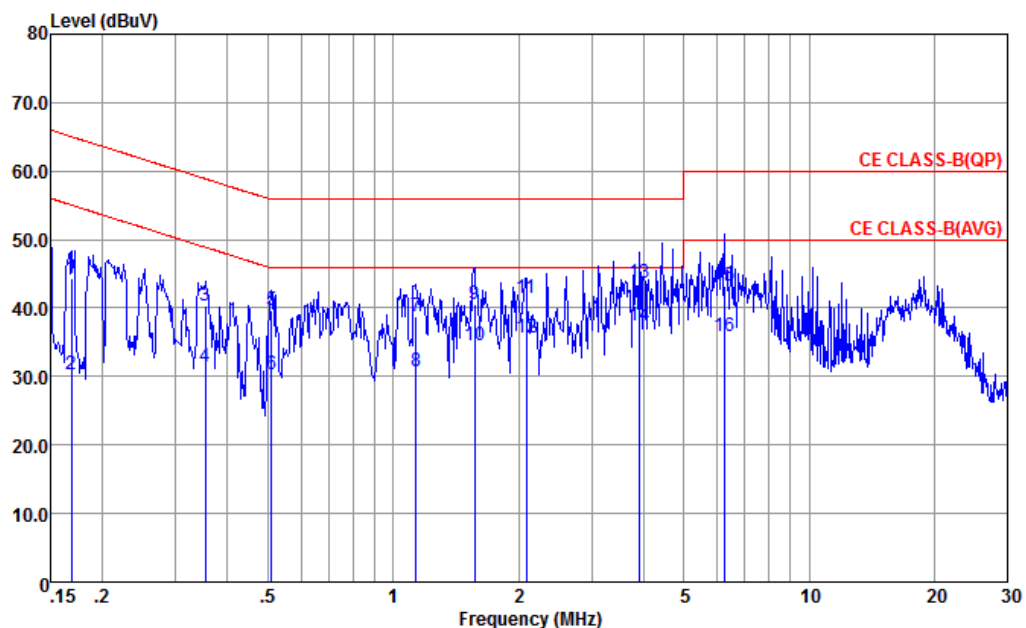


4.2.3 Test Procedure

- a. Finish the requirements of test setup for EUT.
- b. The EUT was placed on a desk 0.8 m height from the metal ground plane and 0.4 m from the conducting wall of the shielding room and it was kept at least 0.8 m from any other grounded conducting surface. The size of the table will nominally be 1,5 m × 1,0 m.
- c. The rear of the arrangement shall be flush with the back of the supporting tabletop unless that would not be possible or typical of normal use.
- d. All units of equipment forming the system under test (includes the EUT as well as connected peripherals and associated equipment or devices) shall be arranged such that a nominal 0,1 m separation is achieved between the neighboring units.
- e. Connect EUT to the power mains through an artificial mains network (AMN). Where the mains cable supplied by the manufacturer is longer than 1 m, the excess should be folded at the centre into a bundle no longer than 0,4 m, so that its length is shortened to 1 m.
- f. All the support units are connecting to the other AMN.
- g. The AMN provides 50 ohm coupling impedance for the measuring instrument.
- h. The CISPR states that the AMN with 50 ohm and 50 microhenry should be used.
- i. The frequency range from 150 kHz to 30 MHz was sweep.
- j. Set the test-receiver system to Quasi Peak detect function and Average detect function, and to measure the conducted emissions values.
- k. Test Results Explanation:
$$\text{Level(dB}\mu\text{V)} = \text{Read Level(dB}\mu\text{V)} + \text{LISN Factor(dB)} + \text{Cable Loss(dB)}$$
$$\text{Over Limit(dB)} = \text{Level(dB}\mu\text{V)} - \text{Limit Line(dB}\mu\text{V)}$$

4.2.4 Test Results

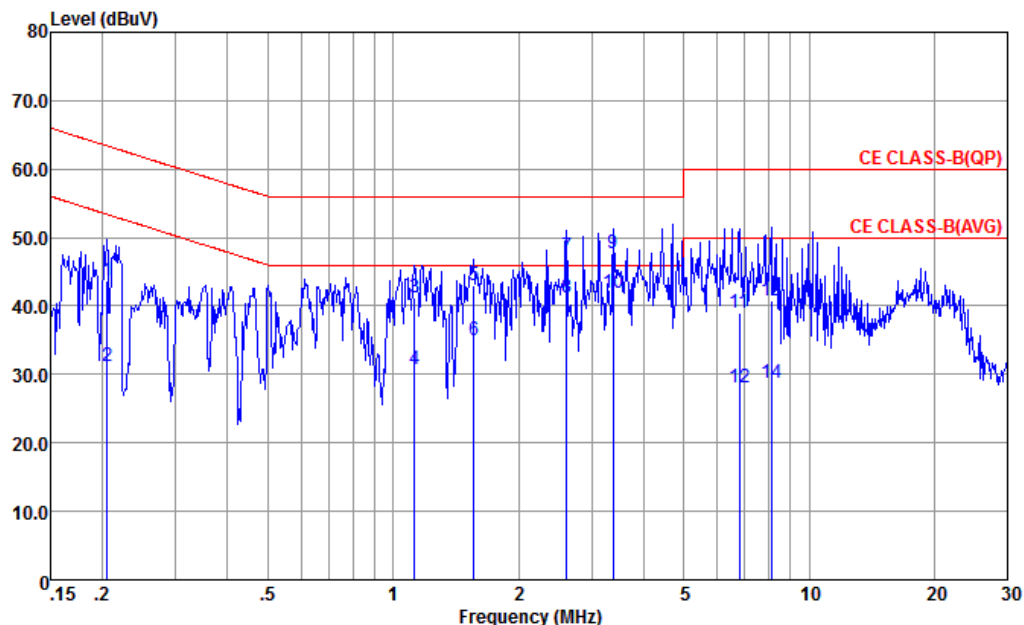
Test Mode :	Mode 6		
Test Voltage :	230Vac / 50Hz	Phase :	Line
Function Type :	WWAN Link + Bluetooth Link + WLAN (2.4G) Link + Camera(Rear) + Earphone + Battery + USB Cable (Charging from wireless charging cradle) + SIM 1		



Site : CO01-KS
Condition : CE CLASS-B(QP) LISN-100334-L 2025 LINE

	Freq	Level	Over	Limit	Read	LISN	Cable	
	MHz	dBuV	Limit	Line	Level	Factor	Loss	Remark
			dB	dBuV	dBuV	dB	dB	
1	0.169	44.32	-20.71	65.03	24.30	9.55	10.47	QP
2	0.169	30.32	-24.71	55.03	10.30	9.55	10.47	Average
3	0.354	40.33	-18.54	58.87	20.20	9.64	10.49	QP
4	0.354	31.53	-17.34	48.87	11.40	9.64	10.49	Average
5	0.810	39.60	-16.40	56.00	19.50	9.59	10.51	QP
6	0.810	30.20	-15.80	46.00	10.10	9.59	10.51	Average
7	1.135	38.86	-17.14	56.00	18.71	9.59	10.56	QP
8	1.135	30.66	-15.34	46.00	10.51	9.59	10.56	Average
9	1.568	40.65	-15.35	56.00	20.50	9.60	10.55	QP
10	1.568	34.65	-11.35	46.00	14.50	9.60	10.55	Average
11	2.088	41.34	-14.66	56.00	21.20	9.60	10.54	QP
12	2.088	35.64	-10.36	46.00	15.50	9.60	10.54	Average
13	3.922	43.62	-12.38	56.00	23.50	9.61	10.51	QP
14 *	3.922	37.42	-8.58	46.00	17.30	9.61	10.51	Average
15	6.252	43.18	-16.82	60.00	22.89	9.71	10.58	QP
16	6.252	35.98	-14.02	50.00	15.69	9.71	10.58	Average

Test Mode :	Mode 6		
Test Voltage :	230Vac / 50Hz	Phase :	Neutral
Function Type :	WWAN Link + Bluetooth Link + WLAN (2.4G) Link + Camera(Rear) + Earphone + Battery + USB Cable (Charging from wireless charging cradle) + SIM 1		



Site : CO01-KS
 Condition : CE CLASS-B(QP) LISN-100334-N 2025 NEUTRAL

	Freq	Level	Over	Limit	Read	LISN	Cable	
	MHz	dBuV	Limit	Line	Level	Factor	Loss	Remark
	MHz	dBuV	dB	dBuV	dBuV	dB	dB	
1	0.205	43.69	-19.71	63.40	23.60	9.61	10.48	QP
2	0.205	31.19	-22.21	53.40	11.10	9.61	10.48	Average
3	1.123	41.31	-14.69	56.00	20.80	9.95	10.56	QP
4	1.123	30.81	-15.19	46.00	10.30	9.95	10.56	Average
5	1.560	42.92	-13.08	56.00	22.50	9.87	10.55	QP
6	1.560	35.02	-10.98	46.00	14.60	9.87	10.55	Average
7	2.608	47.17	-8.83	56.00	26.90	9.74	10.53	QP
8	2.608	41.17	-4.83	46.00	20.90	9.74	10.53	Average
9	3.381	47.78	-8.22	56.00	27.60	9.66	10.52	QP
10 *	3.381	41.98	-4.02	46.00	21.80	9.66	10.52	Average
11	6.805	39.02	-20.98	60.00	18.80	9.65	10.57	QP
12	6.805	28.02	-21.98	50.00	7.80	9.65	10.57	Average
13	8.105	40.72	-19.28	60.00	20.50	9.67	10.55	QP
14	8.105	28.72	-21.28	50.00	8.50	9.67	10.55	Average

4.2.5 Setup Photographs**Mode 6**

Remote View



Rear View



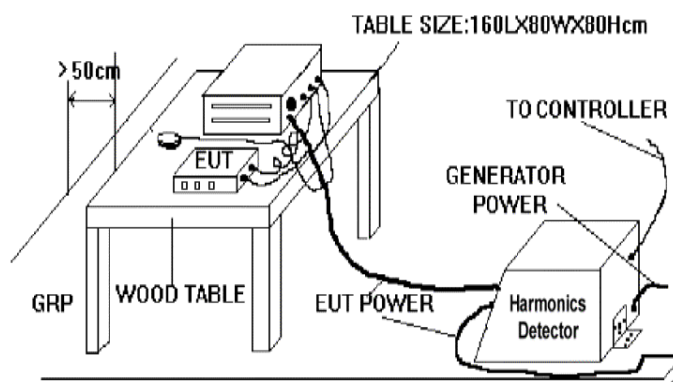
4.3 Harmonics Current Emission Test (Refer to EN301 489-1 Section 8.5)

4.3.1 Limit of Harmonics Current Emission Test

Class A Limits Table:

Harmonic Order	Max. permissible harmonic current (A)
Odd harmonics	
3	2.30
5	1.14
7	0.77
9	0.40
11	0.33
13	0.21
$15 \leq n \leq 39$	$0.15 \cdot 15/n$
Even harmonics	
2	1.08
4	0.43
6	0.30
$8 \leq n \leq 40$	$0.23 \cdot 8/n$

4.3.2 Test Setup



4.3.3 Test Instrument Setting

Harmonic Distortion Tester	Setting
Test Voltage	230 Vac
Test Frequency	50 Hz
Test Duration	10 minutes

4.3.4 Test Procedure

1. Finish the requirements of test setup for EUT.
2. The measurement of harmonic currents shall be performed as follows:
 - i. For each harmonic order, measure the 1.5 s smoothed r.m.s. harmonic current in each DFT time window as defined in EN / IEC 61000-4-7: 2009.
 - ii. Calculate the arithmetic average of the measured values from the DFT time windows, over the entire observation period Short cyclic ($T_{\text{cycle}} \leq 2.5 \text{ min}$). Because of synchronisation to meet the requirements for repeatability in 5%.

4.3.5 Test Result

Pass.

Following is detailed test data

<Mode 1>

Harmonics – Class-A

Test category: Class-A (European limits)

Test Margin: 100

Test date: 2026/2/12

Start time: 8:50:01

End time: 9:00:12

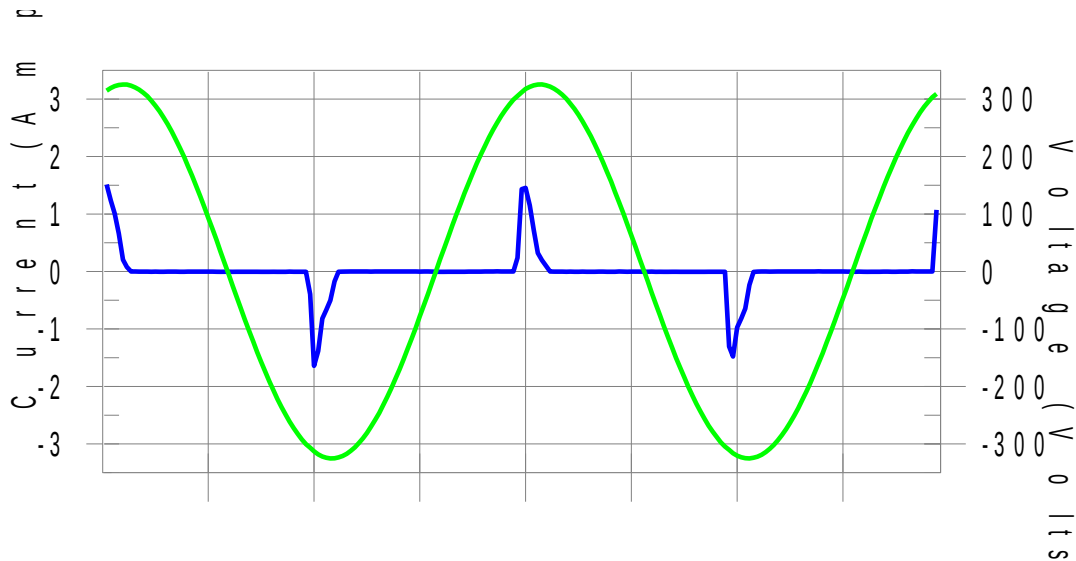
Test duration (min): 10

Data file name: H-000310.cts_data

Test Result: Pass

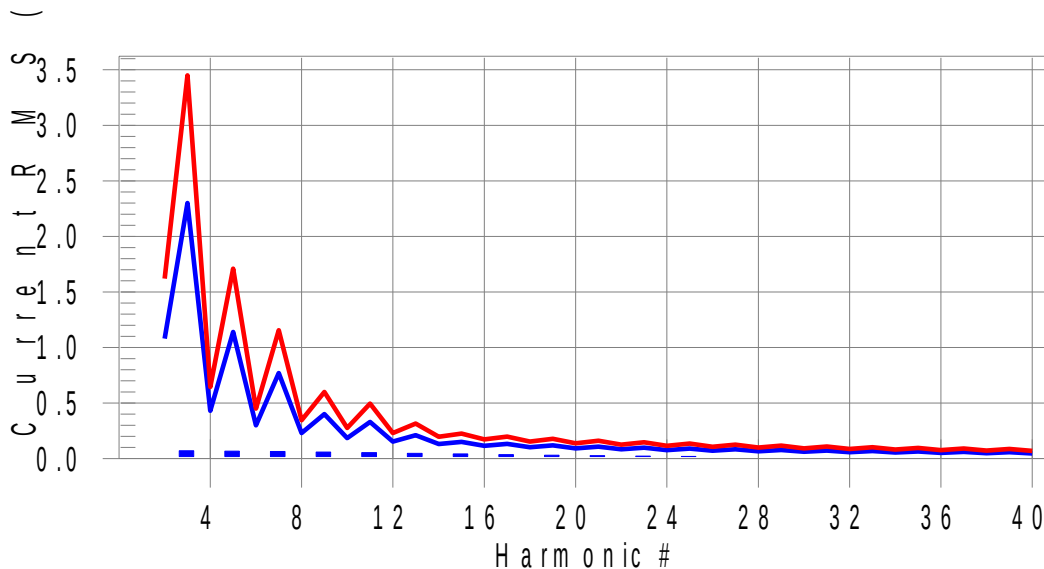
Source qualification: Normal

Current & voltage waveforms



Harmonics and Class A limit line

European Limits



**Current Test Result Summary (Run time)**

Test category: Class-A (European limits) Test Margin: 100
Test date: 2026/2/12 Start time: 8:50:01 End time: 9:00:12
Test duration (min): 10 Data file name: H-000310.cts_data

Test Result: Pass Source qualification: Normal
THC(A): 0.170 I-THD(%): 110.9 POHC(A): 0.051 POHC Limit(A): 0.251

Highest parameter values during test:

V_RMS (Volts):	229.95	Frequency(Hz):	50.00
I_Peak (Amps):	1.812	I_RMS (Amps):	0.348
I_Fund (Amps):	0.153	Crest Factor:	9.851
Power (Watts):	34.5	Power Factor:	0.435

Harm#	Harms(avg)	100%Limit	%of Limit	Harms(max)	150%Limit	%of Limit	Status
2	0.004	1.080	N/A	0.005	1.620	N/A	Pass
3	0.070	2.300	3.1	0.148	3.450	4.3	Pass
4	0.004	0.430	N/A	0.005	0.645	N/A	Pass
5	0.068	1.140	5.9	0.138	1.710	8.1	Pass
6	0.004	0.300	N/A	0.005	0.450	N/A	Pass
7	0.064	0.770	8.3	0.125	1.155	10.8	Pass
8	0.004	0.230	N/A	0.005	0.345	N/A	Pass
9	0.059	0.400	14.7	0.109	0.600	18.1	Pass
10	0.004	0.184	N/A	0.005	0.276	N/A	Pass
11	0.053	0.330	16.1	0.092	0.495	18.5	Pass
12	0.004	0.153	N/A	0.005	0.230	N/A	Pass
13	0.047	0.210	22.4	0.074	0.315	23.6	Pass
14	0.004	0.131	N/A	0.005	0.197	N/A	Pass
15	0.041	0.150	27.4	0.059	0.225	26.2	Pass
16	0.004	0.115	N/A	0.005	0.173	N/A	Pass
17	0.035	0.132	26.9	0.047	0.198	23.6	Pass
18	0.004	0.102	N/A	0.005	0.153	N/A	Pass
19	0.030	0.118	25.7	0.038	0.178	21.7	Pass
20	0.004	0.092	N/A	0.005	0.138	N/A	Pass
21	0.026	0.107	24.5	0.034	0.161	21.4	Pass
22	0.004	0.084	N/A	0.005	0.125	N/A	Pass
23	0.023	0.098	23.1	0.033	0.147	22.2	Pass
24	0.004	0.077	N/A	0.005	0.115	N/A	Pass
25	0.020	0.090	21.7	0.031	0.135	23.2	Pass
26	0.004	0.071	N/A	0.005	0.107	N/A	Pass
27	0.017	0.083	20.2	0.030	0.125	23.6	Pass
28	0.004	0.066	N/A	0.005	0.099	N/A	Pass
29	0.015	0.078	18.7	0.027	0.116	23.0	Pass
30	0.003	0.061	N/A	0.004	0.092	N/A	Pass
31	0.013	0.073	17.3	0.023	0.109	21.3	Pass
32	0.003	0.058	N/A	0.004	0.086	N/A	Pass
33	0.011	0.068	16.0	0.020	0.102	19.1	Pass
34	0.003	0.054	N/A	0.004	0.081	N/A	Pass
35	0.010	0.064	14.9	0.016	0.096	16.8	Pass
36	0.003	0.051	N/A	0.004	0.077	N/A	Pass
37	0.008	0.061	13.9	0.014	0.091	15.1	Pass
38	0.003	0.048	N/A	0.004	0.073	N/A	Pass
39	0.008	0.058	13.2	0.012	0.087	14.2	Pass
40	0.002	0.046	N/A	0.003	0.069	N/A	Pass

4.3.6 Setup Photographs**Mode 1**

4.4 Voltage Fluctuation and Flicker Measurement (Refer to EN301 489-1 Section 8.6)

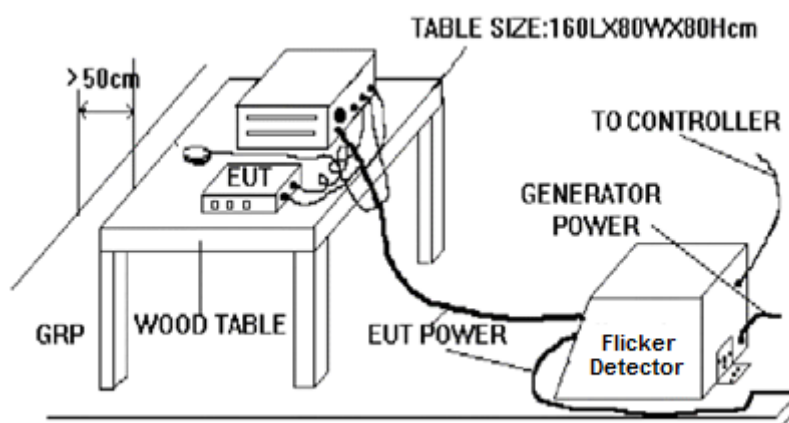
4.4.1 Limit of Voltage Fluctuation and Flicker Test

Voltage Fluctuation and Flicker Limits:

- the value of Pst shall not be greater than 1.0;
- the value of Plt shall not be greater than 0.65;
- the value of $d(t)$ during a voltage change shall not exceed 3.3 % for more than 500 ms.
- the relative steady-state voltage change, d_c , shall not exceed 3.3 %;
- the maximum relative voltage change, d_{max} , shall not exceed 4.0 %;

Note: Tests need not be made on equipment which is unlikely to produce significant voltage fluctuations or flicker. Test results are for reference only.

4.4.2 Test Setup



4.4.3 Test Instrument Setting

Voltage Fluctuation and Flicker Tester	Setting
Test Voltage	230 Vac
Test Frequency	50 Hz
Pst Integration Time	10 minutes
Pst Integration Periods	1
Measurement Delay	10.0 seconds
Test Duration	10 minutes

4.4.4 Test Procedure

1. Finish the requirements of test setup for EUT.
2. The total impedance of the test circuit, excluding the appliance under test, but including the internal impedance of the supply source, shall be equal to the reference impedance. The stability and tolerance of the reference impedance shall be adequate to ensure that the overall accuracy of $\pm 8\%$ is achieved during the whole assessment procedure.

4.4.5 Test Result

Pass.

Following is test data



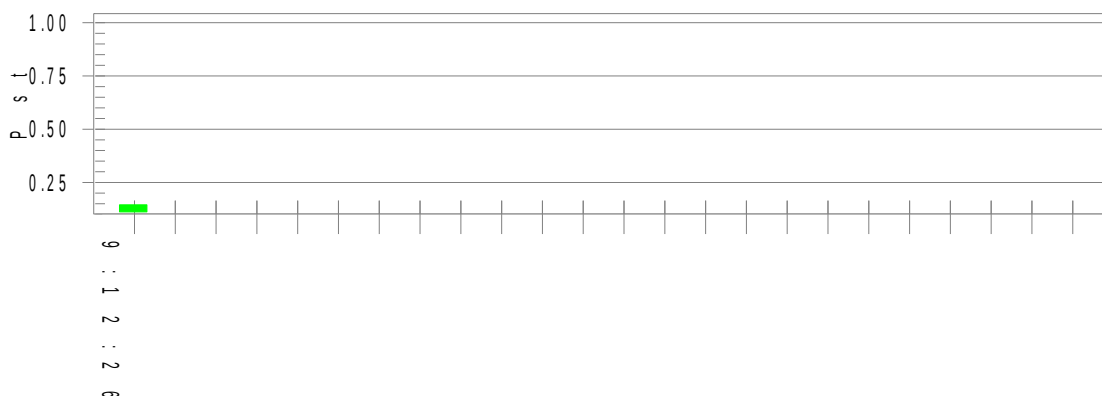
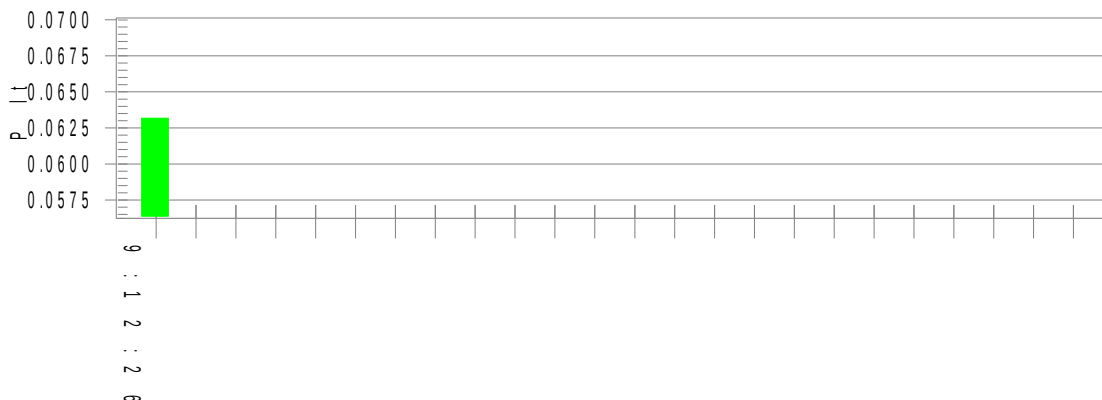
<Mode 1>

Flicker

Test category: dt,dmax,dc and Pst (European limits) Test Margin: 100
Test date: 2026/2/12 Start time: 9:02:05 End time: 9:12:32
Test duration (min): 10 Data file name: F-000311.cts_data

Test Result: Pass

Status: Test Completed

Pst_i and limit lineEuropean LimitsPlt and limit line

Parameter values recorded during the test:

Vrms at the end of test (Volt): 229.86

Highest dt (%):

T-max (mS): 0

Highest dc (%): 0.00

Highest dmax (%): 0.00

Highest Pst (10 min. period): 0.145

Test limit (%):

Test limit (mS): 500.0 Pass

Test limit (%): 3.30 Pass

Test limit (%): 4.00 Pass

Test limit: 1.000 Pass

4.4.6 Setup Photographs**Mode 1**

5. Immunity Tests

5.1 Requirements of Limit and EUT Performance Criteria for all Immunity Test Items

Test limit including test level, test frequency range, pulse type, test duration...etc. requirements.

This section is intended to integrate requirements of limit, and required performance criteria for all immunity test items.

In subsection 5.1.1, includes two parts:

1. Subsection 5.1.1.1 : Support ports list of EUT, accessory, and cable record, where EUT intended to use in. These information will be used for decide test items and test limit
 - (1) Supported ports list of EUT: Because test limit are based on supported ports of EUT, this is necessary information.
 - (2) Accessory : include adapter type and remark EUT has battery or not.
 - (3) Cable record : includes cable type, cable length, indoor/outdoor. These parameters will decide tests shall be carrying out or not.
2. Subsection 5.1.1.2 : tables of immunity test level specified in EN301489 series standards and immunity test level specified by manufacturer.

If immunity test level specified by manufacturer are higher/stronger than level specified in EN301489 series standards, they will be also record in this table. Therefore anyone could distinguish requirements specified by standard or manufacturer from these tables.

In subsection 5.1.2, required performance criteria of EUT per EN301489 series standards.

Integrated required performance criteria of EN301489 series standards, they are used for all immunity test of this report.

5.1.1 Test Limit
5.1.1.1 Information of supported ports of EUT, accessory, cable record where EUT intended to use in.

Supported ports of EUT are listed as below (symbol ☒ means supported port):

☒	Enclosure Port
☒	Input AC power port
☐	Input DC power port
☐	Telecommunication port

Accessory (symbol ☒ means have used with EUT during test)

☒	☒ AC Adapter ☐ DC Adapter ☐ Car charger ☐ PoE adapter	Pins : ☒ 2pins ☐ 3pins Cable Length : ☐ >3m ☒ <3m
☒	Battery	

As per above information, corresponded test limit (including test level, test frequency range, pulse type, test duration...etc. requirements) specified in below table 1~4 have been selected to carry out test in this report.

5.1.1.2 Tables of Immunity Test Level Specified in EN301489 series standards and Immunity Test Level Specified by Manufacturer

When immunity test level specified by manufacturer are higher (stronger) than level specified in EN301489 series standards, they will be also record in this table. But if manufacturer doesn't specify immunity test level, "N/A" is filling in table and test level of all immunity test items are following requirements of EN301489 series standards.

Table 1 - Enclosure Port

Test item	Immunity test level specified in EN301489-1	Immunity test level specified by manufacturer
Electrostatic discharge (ESD)	± 2 kV, ± 4 kV contact	N/A
	± 2 kV, ± 4 kV, ± 8 kV air	N/A
Radio frequency electromagnetic field (RS)	3 V/m	N/A
	Frequency range : 80 MHz – 6 GHz	N/A
	Modulation: 80 % AM at 1 kHz	N/A

Table 2 – Input AC Power Port

Test Item	Immunity test level specified in EN301489-1	Immunity Test level specified by manufacturer
Fast transients, common mode (EFT)	± 1 kV 5 kHz repetition frequency	N/A
Surges Line-to-line	± 0,5 kV, ± 1 kV	N/A
Surges Line-to-ground	± 0,5 kV, ± 1 kV, ± 2 kV	N/A
Radio frequency, common mode (CS)	3 Vrms	N/A
	Frequency range: 0,15 MHz – 80 MHz	N/A
	Modulation: 80 % AM at 1 kHz	N/A
Voltage dips	0 % residual; 0,5 cycle , 50Hz Phase At 0°, 45°, 90°, 135°, 180°, 225°, 270° and 315°	N/A
	0 % residual; 1 cycle , 50Hz Phase At 0°, 45°, 90°, 135°, 180°, 225°, 270° and 315°	
	70 % residual; 25 cycles , 50Hz Phase At 0°, 45°, 90°, 135°, 180°, 225°, 270° and 315°	
Voltage interruptions	0 % residual; 250cycle , 50Hz	N/A

Table 3 – Input DC Power Port (not necessary performed on EUT of this report)

Test Item	Immunity test level specified in EN301489-1	Immunity Test level specified by manufacturer
Fast transients, common mode (EFT)	± 0.5 kV 5 kHz repetition frequency When cable length >3m	N/A
Radio frequency, common mode (CS)	3 Vrms When cable length >3m	N/A
	Frequency range: 0,15 MHz – 80 MHz	N/A
	Modulation: 80 % AM at 1 kHz	N/A
Transients and surges in the vehicular environment	As specified in ISO 7637-2 : Pulse type : 1, 2a, 2b, 3a, 3b, 4 Level : III	N/A

Table 4 –Telecommunication Port (not necessary performed on EUT of this report)

Test Item	Immunity test level specified in EN301489-1	Immunity Test level specified by manufacturer
Fast transients, common mode (EFT)	± 0.5 kV 5 kHz repetition frequency When cable length >3m	N/A
	± 0.5 kV 100 kHz repetition frequency for xDSL When cable length >3m	N/A
Surges Line-to-ground	± 0.5 kV for indoor cable When cable length >30m	N/A
	± 1 kV for outdoor cable	N/A
Radio frequency, common mode (CS)	3 Vrms When cable length >3m	N/A
	Frequency range: 0,15 MHz – 80 MHz	N/A
	Modulation: 80 % AM at 1 kHz	N/A

5.1.2 Required Performance Criteria of EUT per EN301489 series standards
5.1.2.1 Performance requirements table of 301489-1

CLAUSE 6.1 to 6.2 of EN301489-1	
Criteria	Performance criteria
Continuous Phenomena	During the test, the equipment shall: • continue to operate as intended; • not unintentionally transmit; • not unintentionally change its operating state; • not unintentionally change critical stored data.
Transient Phenomena	For all ports and transient phenomena with the exception described below, the following applies: • The application of the transient phenomena shall not result in a change of the mode of operation (e.g. unintended transmission) or the loss of critical stored data. • After application of the transient phenomena, the equipment shall operate as intended. For surges applied to symmetrically operated wired network ports intended to be connected directly to outdoor lines the following criteria applies: • For products with only one symmetrical port intended for connection to outdoor lines, loss of function is allowed, provided the function is self-recoverable, or can be otherwise restored. Information stored in non-volatile memory, or protected by a battery backup, shall not be lost. • For products with more than one symmetrical port intended for connection to outdoor lines, loss of function on the port under test is allowed, provided the function is self-recoverable. Information stored in non-volatile memory, or protected by a battery backup, shall not be lost.

5.1.2.2 Performance requirements table of 301489-3

The provisions of ETSI EN 301 489-1 clause 6 shall apply.

5.1.2.3 Performance requirements table of 301489-17

CLAUSE 6.2 of EN301489-17		
Criteria	During test	After test
A	Shall operate as intended. (see note). Shall be no loss of function. Shall be no unintentional transmissions.	Shall operate as intended. Shall be no degradation of performance Shall be no loss of function. Shall be no loss of critical stored data
B	May be loss of function.	Functions shall be self-recoverable. Shall operate as intended after recovering. Shall be no loss of critical stored data
C	May be loss of function.	Functions shall be recoverable by the operator. Shall operate as intended after recovering. Shall be no loss of critical stored data
Note : Operate as intended during the test allows a level of degradation in accordance with EN301489-17 Clause 6.2.2 Minimum performance level as following: For equipment that supports a PER or FER, the minimum performance level shall be a PER or FER less than or equal to 10 %. For equipment that does not support a PER or a FER, the minimum performance level shall be no loss of the wireless transmission function needed for the intended use of the equipment.		

CLAUSE 6.3 to 6.4 of EN301489-17	
Criteria	Performance criteria
Continuous phenomena	The performance criteria A shall apply. Where the EUT is a transmitter in standby mode, unintentional transmission shall not occur during the test. Where the EUT is a transceiver in receive mode, unintentional transmission shall not occur during the test.
Transient phenomena	The performance criteria B shall apply for transient phenomena, except for voltage dips greater than or equal to 100 ms, voltage interruptions of 5 000 ms duration, and surges of 10/700 µs for which performance criteria C shall apply. Where the EUT is a transmitter in standby mode, unintentional transmission shall not occur as a result of the application of the test. Where the EUT is a transceiver in receive mode, unintentional transmission shall not occur as a result of the application of the test.

5.1.2.4 Performance requirements table of 301489-19

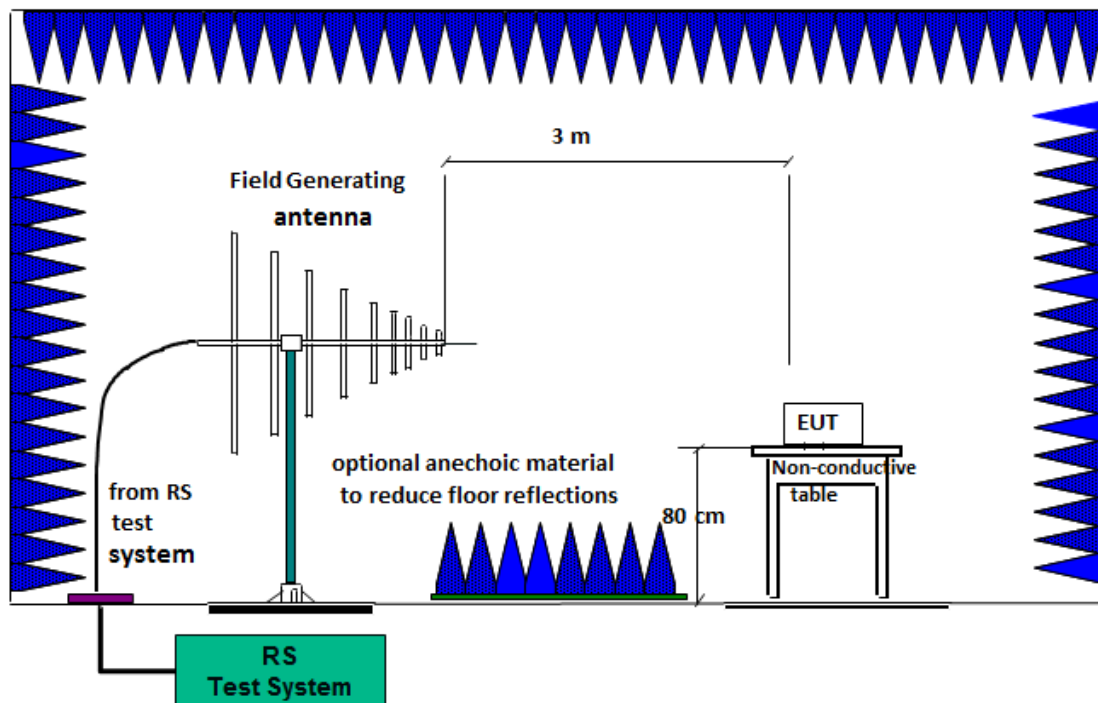
CLAUSE 6.2 to 6.3 of EN301489-19	
Criteria	Performance criteria
CR	For the EUT, excluding spot frequency tests as part of the immunity test with radiated RF electromagnetic fields (see ETSI EN 301 489-1 [1], clause 9.2): • the general performance criteria set out in clause 6.1; • during the test no false calls shall occur; • at the conclusion of the test comprising the series of individual exposures the EUT shall operate as intended with no loss of functions or stored data (messages), as declared by the manufacturer. For the spot frequency test as part of the immunity test with radiated RF electromagnetic fields (see ETSI EN 301 489-1 [1], clause 9.2) the EUT shall be assessed by monitoring the accuracy of the call received alert signal.
TR	For the EUT: • the general performance criteria set out in clause 6.1; • during the test no false calls shall occur; • at the conclusion of the test comprising the series of individual exposures, the EUT shall operate as intended with no loss of function and/or stored data (messages), as declared by the manufacturer.

5.1.2.5 Performance requirements table of 301489-52

Criteria	Performance criteria
CT/CR	GSM and voice call: During the test, the uplink speech output level shall be at least 35 dB (± 3 dB) less than the previously recorded reference levels, when measured through an audio band pass filter of width 200 Hz, centred on 1 kHz (audio breakthrough check) NOTE: When there is a high-level background noise present, the filter bandwidth can be reduced down to a minimum of 40 Hz At the conclusion of the test, the EUT shall operate as intended with no loss of user control functions or critical stored data, and the communication link shall have been maintained. In addition to confirming the above performance during a call, the test shall also be performed in idle mode, and the transmitter shall not unintentionally operate. During the test, the RXQUAL of the downlink shall not exceed the value of three, measured during each individual exposure in the test sequence. During the test, the downlink speech output level shall be at least 35 dB (± 3 dB) less than the previously recorded reference levels, when measured through an audio band pass filter of width 200 Hz, centred on 1 kHz (audio breakthrough check). NOTE: When there is a high-level background noise present, the filter bandwidth can be reduced down to a minimum of 40 Hz. At the conclusion of the test, the EUT shall operate as intended with no loss of user control functions or critical stored data, and the communication link shall have been maintained. UTRA In the data transfer mode, the performance criteria can be one of the following: • if the BER (as referred in clause 5.3.1 of ETSI TS 134 109 [4]) is used, it shall not exceed 0,001 during the test sequence; • if the BLER (as referred in ETSI TS 134 109 [4]) is used, it shall not exceed 0,01 during the test sequence. The BLER calculation shall be based on evaluating the CRC on each transport block. Details are specified in annex C. In the case of narrow band responses, the procedure in clause 4.4.2.1 shall be followed. When testing a voice call, the performance criteria in clause 6.1.1 shall apply. E-UTRA In data transfer mode, the data throughput of the EUT shall not fall below 95 % of the maximum data throughput. Details are specified in annex C. In the case of narrow band responses, the procedure in clause 4.4.2.2 shall be followed. When testing a voice call, the performance criteria in clause 6.1.1 shall apply. NR In data transfer mode, the data throughput of the EUT shall not fall below 95 % of the maximum data throughput. Details are specified in annex C In the case of narrow band responses, the procedure in clause 4.4.3 shall be followed. When testing a voice call, the performance criteria in clause 6.1.1 shall apply.
TT/TR	At the conclusion of each exposure of the transient phenomena, the EUT shall operate without loss of the communication link. At the conclusion of the total test comprising the series of individual exposures, the EUT shall operate as intended without loss of user control functions or critical stored data. In addition where the EUT supports idle mode it should be verified that the transmitter shall not unintentionally operate when transient phenomena are applied.

5.2 Radio frequency electromagnetic field (RS) Test (Refer to EN301 489-1 Section 9.2)

5.2.1 Test Setup



5.2.2 Test Instrument Setting

Frequency Step Size	1% increment ;
Modulation	80% AM (1kHz)
Dwell Time	3 seconds
Tested Antenna Height	1.55m

5.2.3 Test Procedures

The antenna is placed 3m away from the equipment. The required field strength is pre-calibrated and complies with the uniform field area requirement lay down in the IEC/EN 61000-4-3.

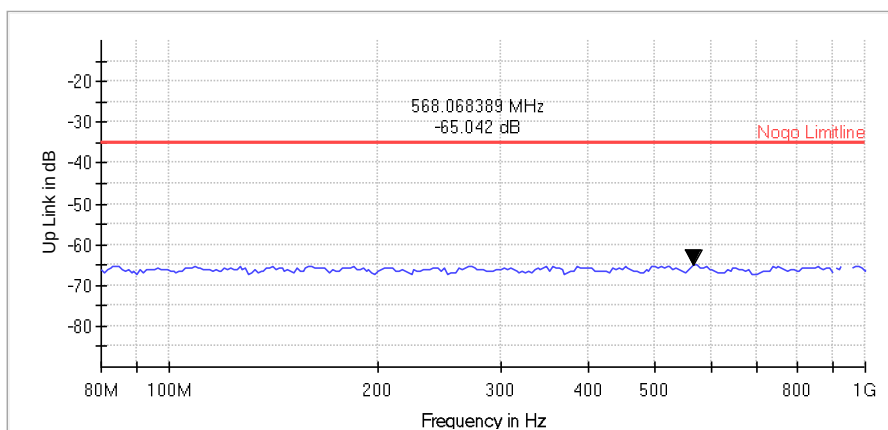
5.2.4 Test Result

Test Standard	EN 61000-4-3:2006+A1:2008+A2:2010
Product Standard	EN 301 489-1, EN 301 489-3, EN 301 489-17, EN 301 489-19, EN 301 489-52
EUT operated voltage during test	230Vac, 50Hz
Test Frequency Range	80 MHz ~ 6 GHz
Test Level	3 V/m
Polarity	Horizontal and Vertical
Azimuth	0°, 180°
Required Performance Criteria	CT/CR
Performance Criteria of EUT	CT/CR
Test Result	PASS

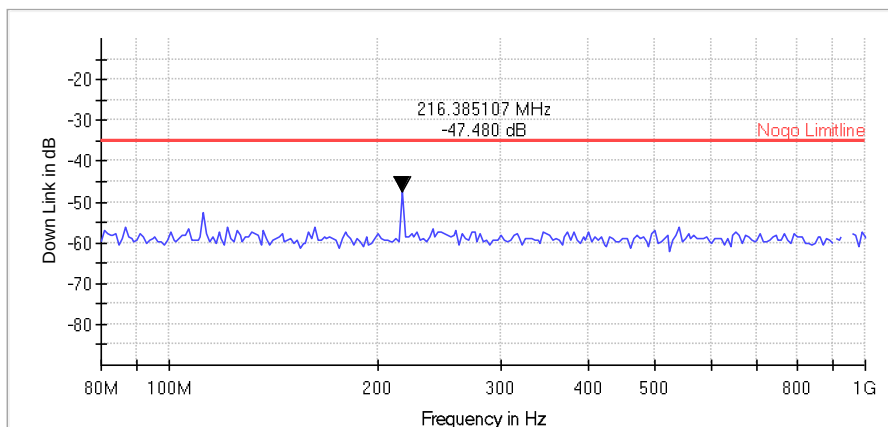
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Test Mode :	Mode 8
Function Type :	GSM900 Link + SIM 1 + Battery + USB Cable(Charging from Adapter)
Audio Calibration Data :	Uplink Ref = 86.00 dBμV, Downlink Ref = 100.20 dBμV
Azimuth	0°
Frequency Range :	80MHz~1GHz Horizontal

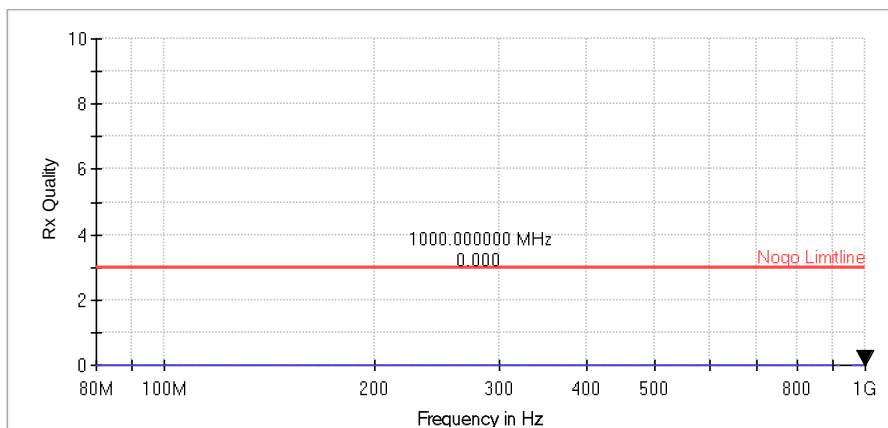
Up Link



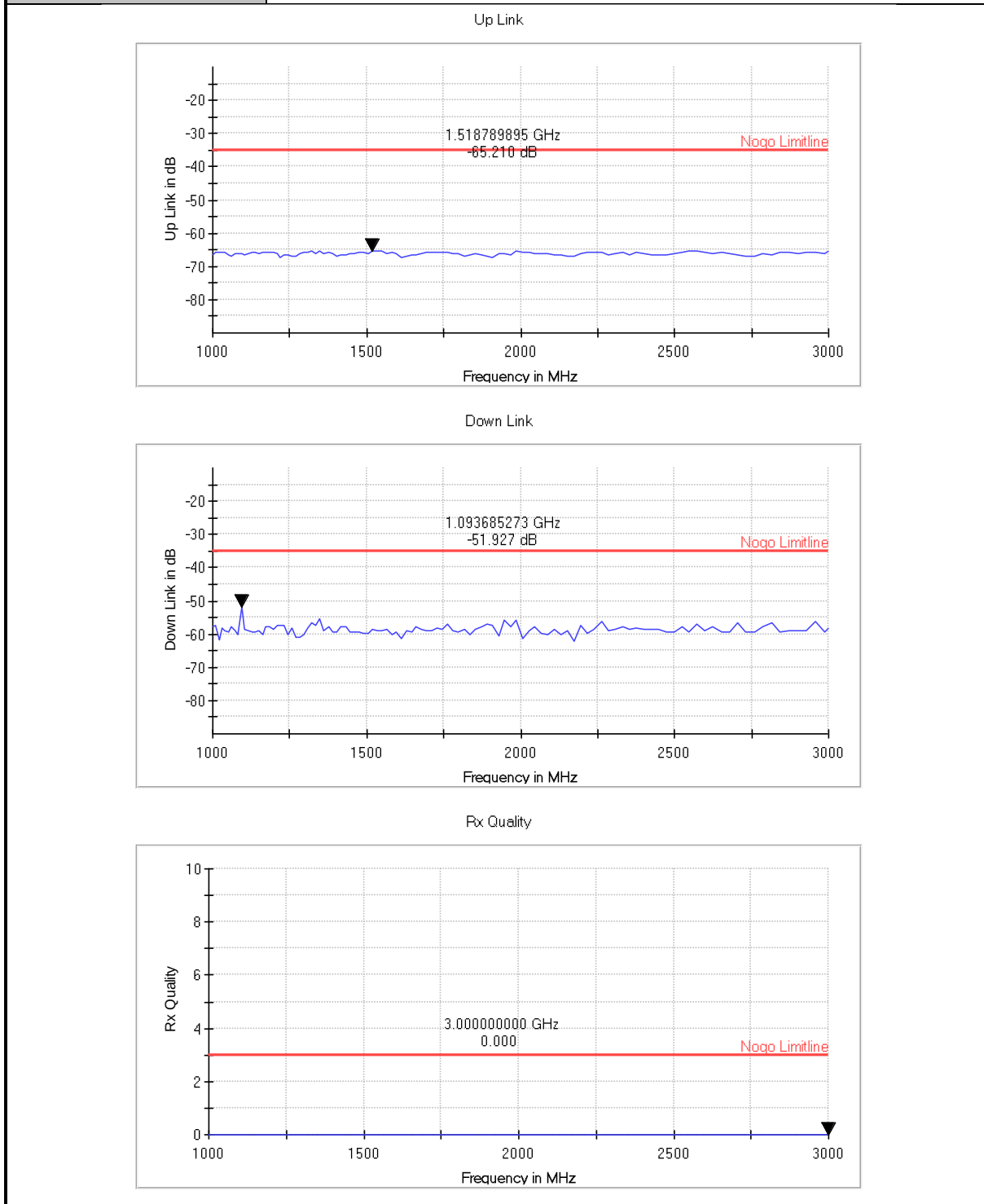
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Rx Quality

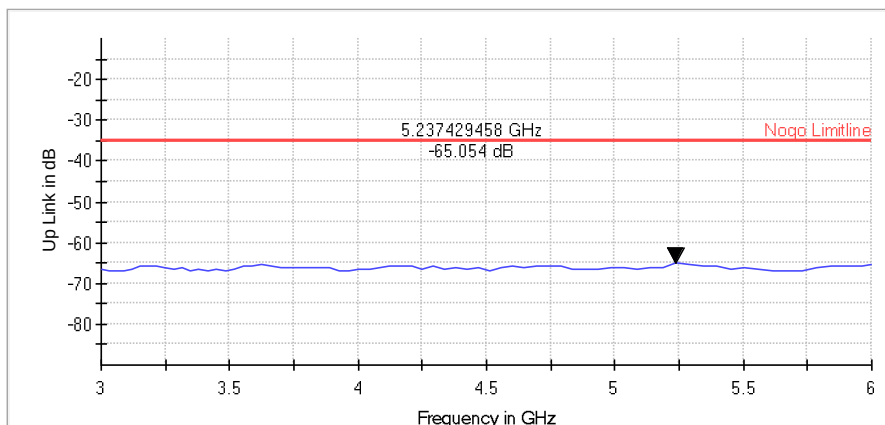


Test Mode :	Mode 8
Function Type :	GSM900 Link + SIM 1 + Battery + USB Cable(Charging from Adapter)
Audio Calibration Data :	Uplink Ref = 86.00 dBμV, Downlink Ref = 100.20 dBμV
Azimuth	0°
Frequency Range :	1GHz~3GHz Horizontal

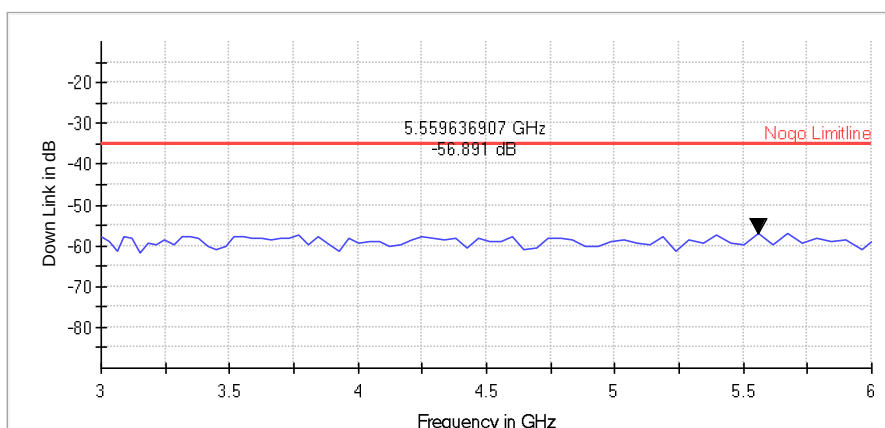


Test Mode :	Mode 8
Function Type :	GSM900 Link + SIM 1 + Battery + USB Cable(Charging from Adapter)
Audio Calibration Data :	Uplink Ref = 86.00 dBμV, Downlink Ref = 100.20 dBμV
Azimuth	0°
Frequency Range :	3GHz~6GHz Horizontal

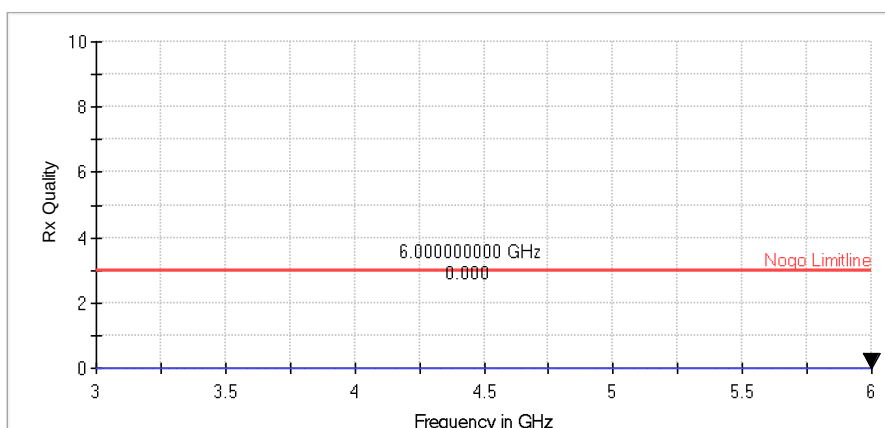
Up Link



Down Link

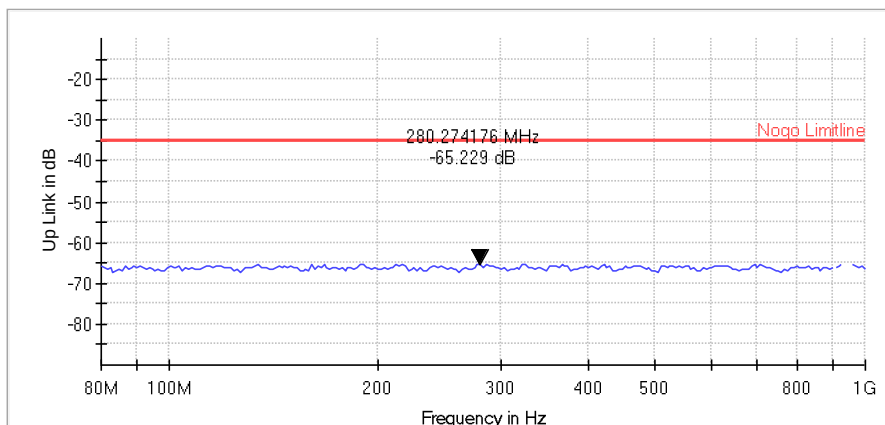


Rx Quality

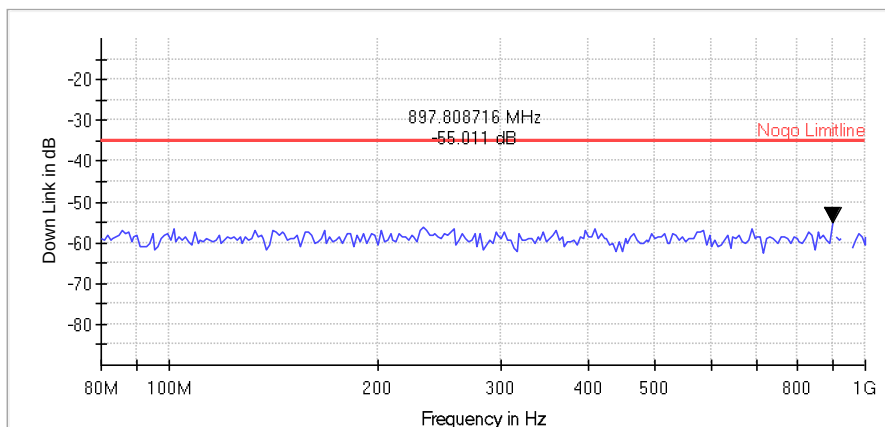


Test Mode :	Mode 8
Function Type :	GSM900 Link + SIM 1 + Battery + USB Cable(Charging from Adapter)
Audio Calibration Data :	Uplink Ref = 86.00 dBμV, Downlink Ref = 100.20 dBμV
Azimuth	0°
Frequency Range :	80MHz~1GHz Vertical

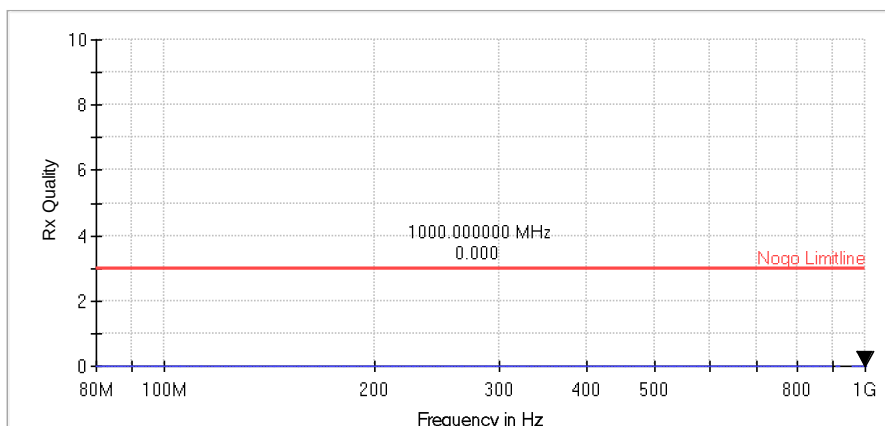
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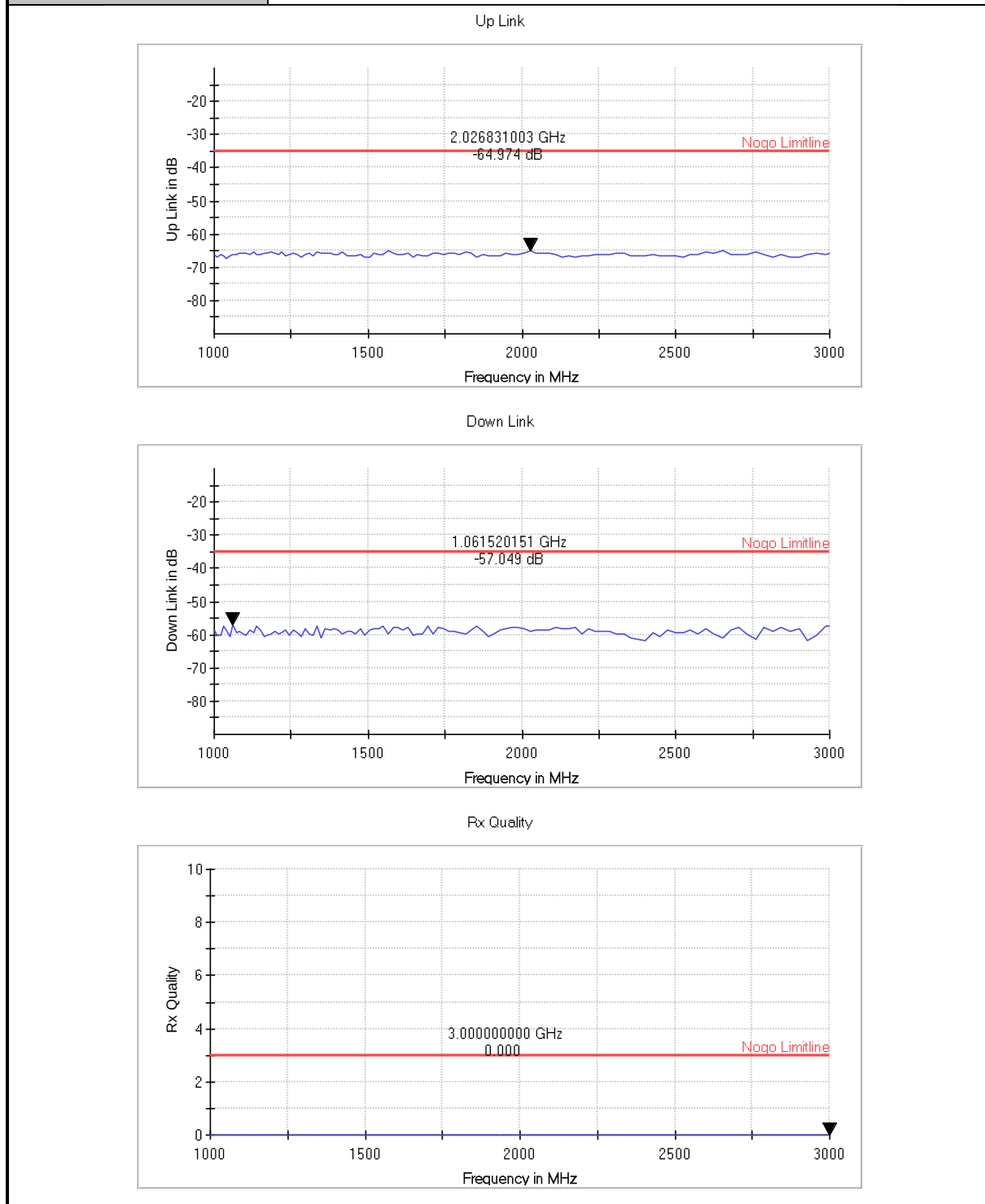
Down Link



Rx Quality

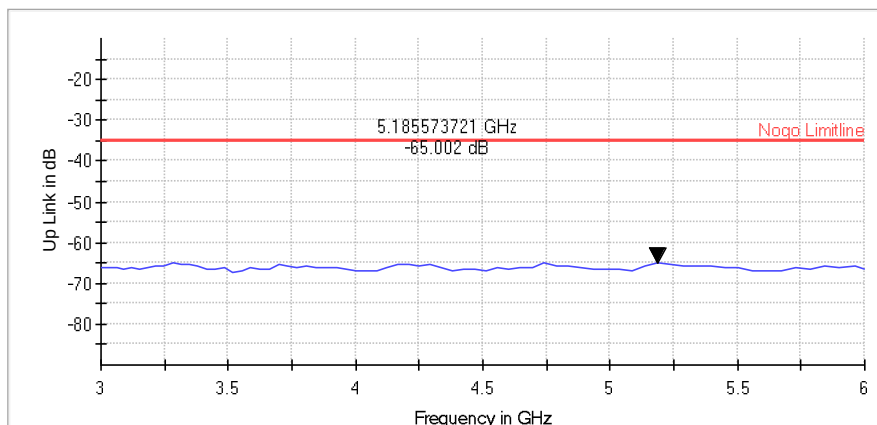


Test Mode :	Mode 8
Function Type :	GSM900 Link + SIM 1 + Battery + USB Cable(Charging from Adapter)
Audio Calibration Data :	Uplink Ref = 86.00 dBμV, Downlink Ref = 100.20 dBμV
Azimuth	0°
Frequency Range :	1GHz~3GHz Vertical

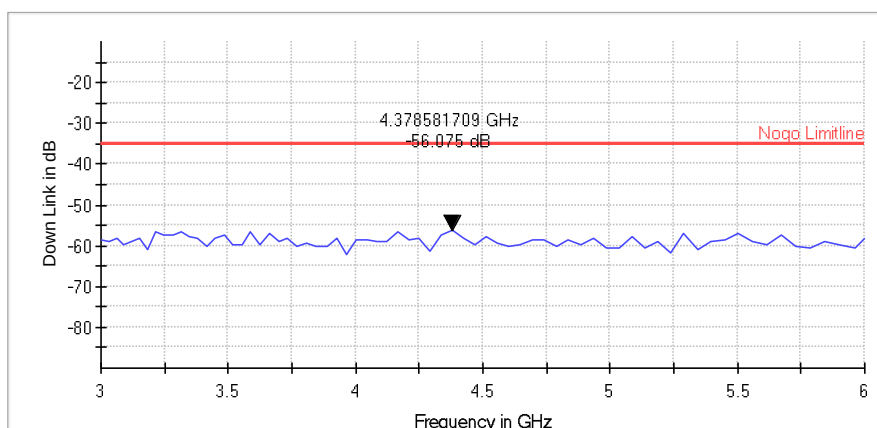


Test Mode :	Mode 8
Function Type :	GSM900 Link + SIM 1 + Battery + USB Cable(Charging from Adapter)
Audio Calibration Data :	Uplink Ref = 86.00 dBμV, Downlink Ref = 100.20 dBμV
Azimuth	0°
Frequency Range :	3GHz~6GHz Vertical

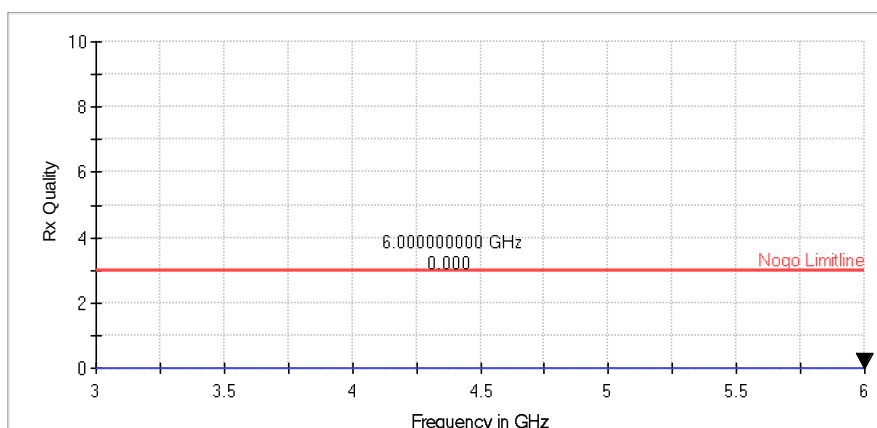
Up Link



Down Link

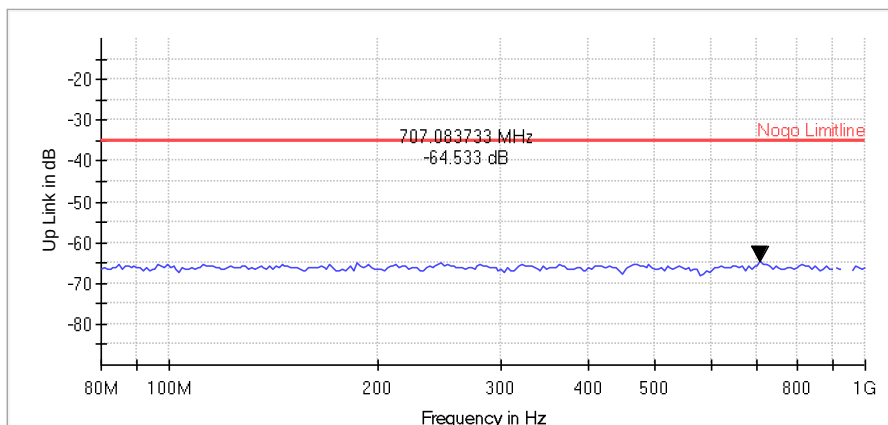


Rx Quality

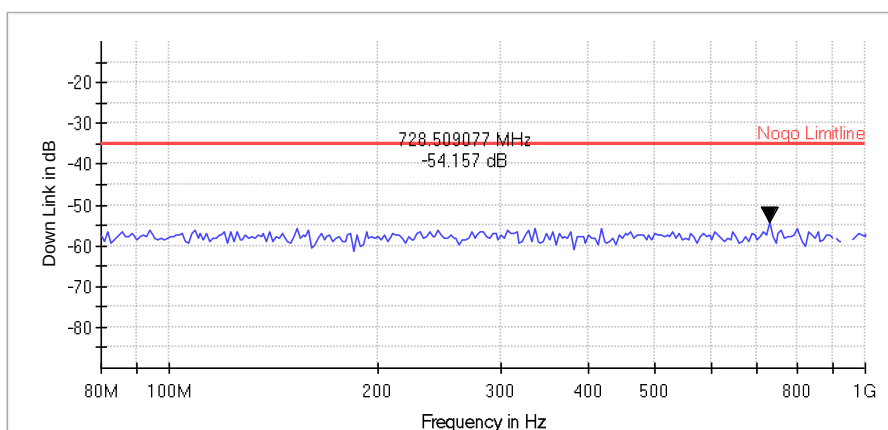


Test Mode :	Mode 8
Function Type :	GSM900 Link + SIM 1 + Battery + USB Cable(Charging from Adapter)
Audio Calibration Data :	Uplink Ref = 86.00 dBμV, Downlink Ref = 100.20 dBμV
Azimuth	180°
Frequency Range :	80MHz~1GHz Horizontal

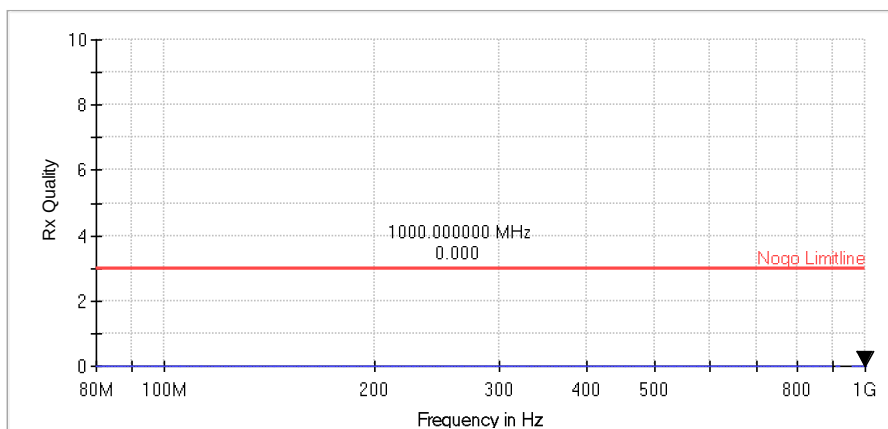
Up Link



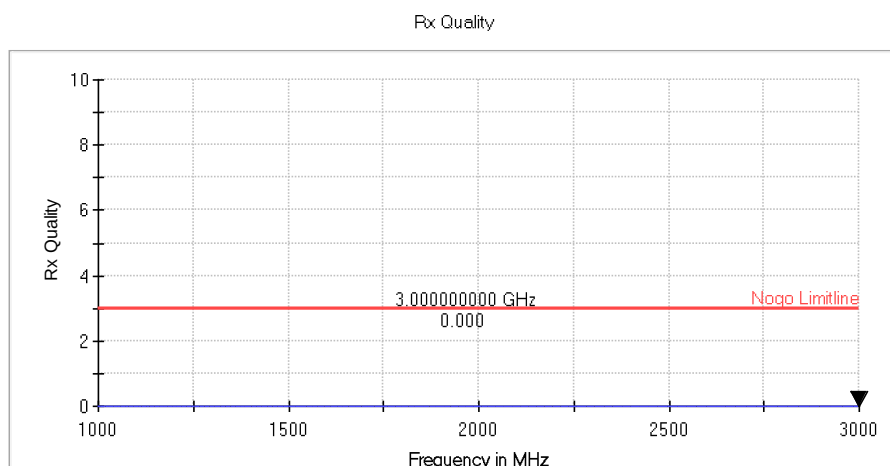
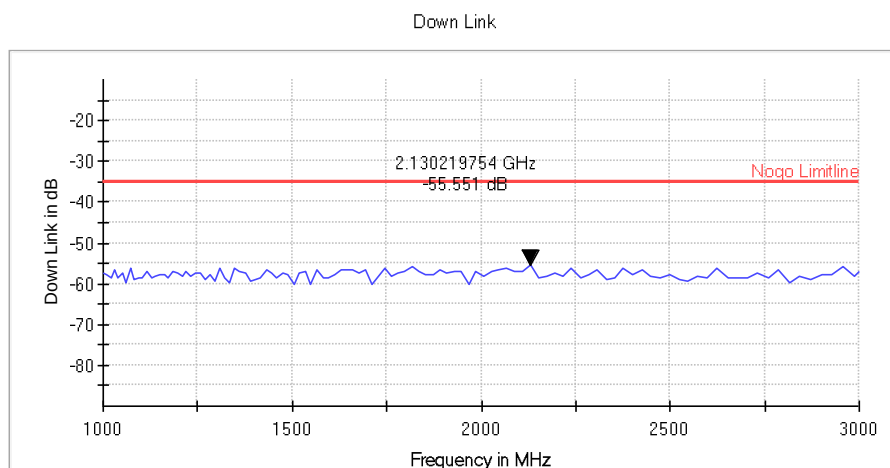
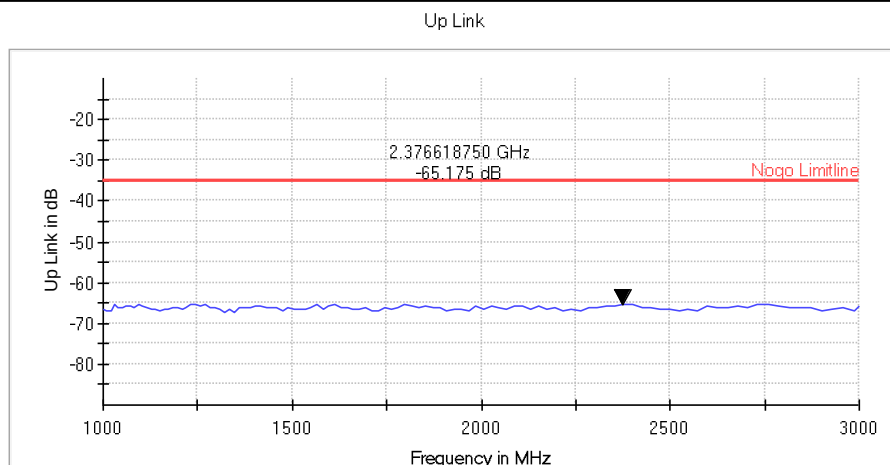
Down Link



Rx Quality

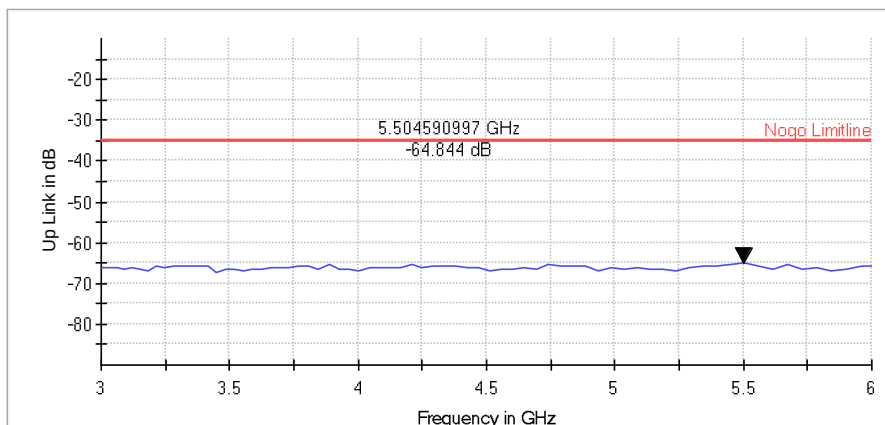


Test Mode :	Mode 8
Function Type :	GSM900 Link + SIM 1 + Battery + USB Cable(Charging from Adapter)
Audio Calibration Data :	Uplink Ref = 86.00 dBμV, Downlink Ref = 100.20 dBμV
Azimuth	180°
Frequency Range :	1GHz~3GHz Horizontal

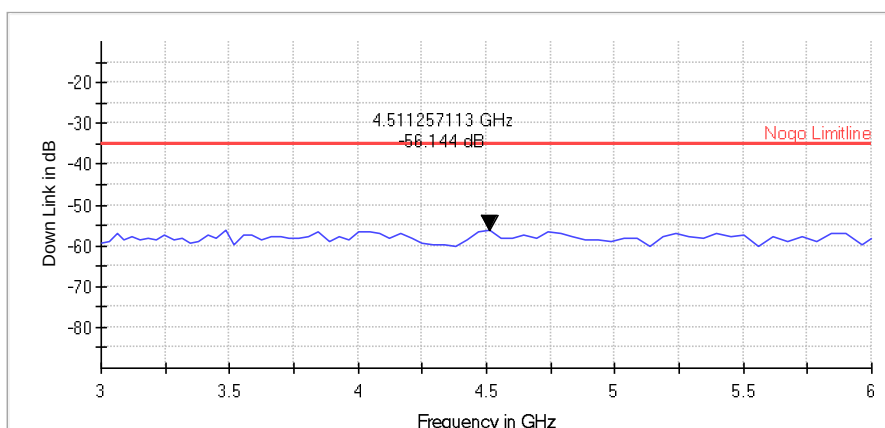


Test Mode :	Mode 8
Function Type :	GSM900 Link + SIM 1 + Battery + USB Cable(Charging from Adapter)
Audio Calibration Data :	Uplink Ref = 86.00 dBμV, Downlink Ref = 100.20 dBμV
Azimuth	180°
Frequency Range :	3GHz~6GHz Horizontal

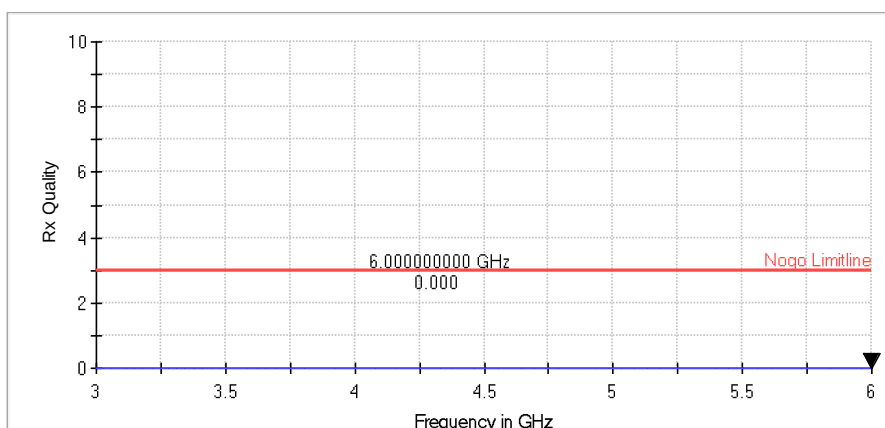
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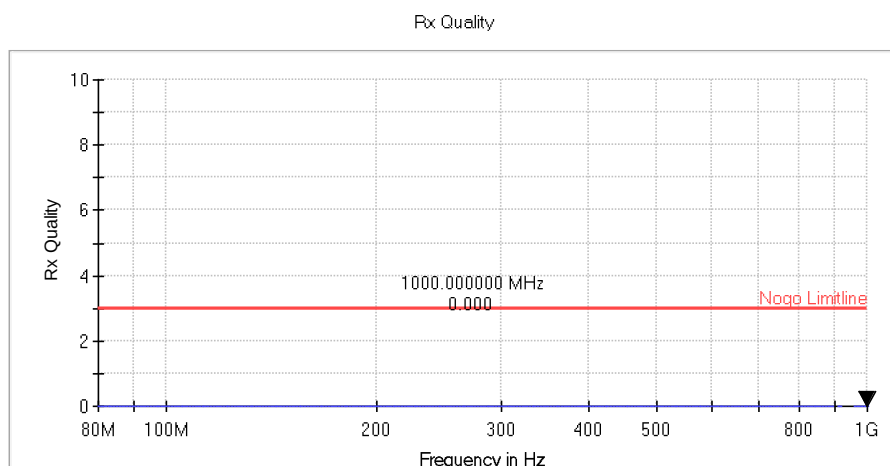
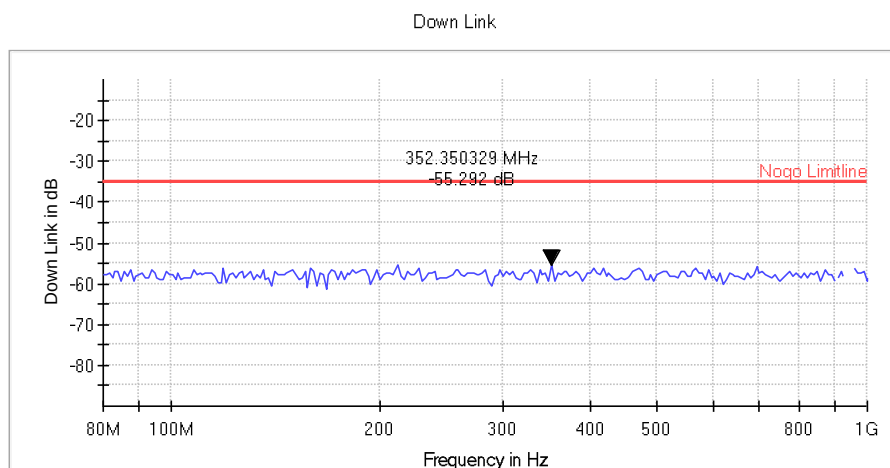
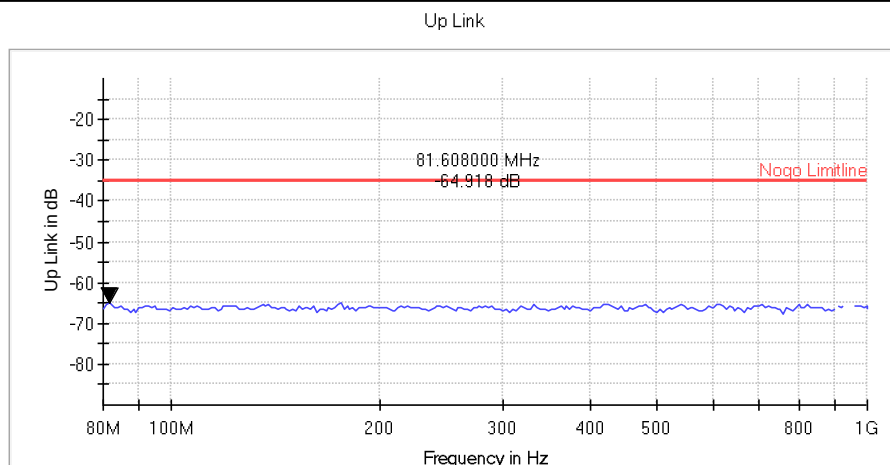
Down Link



Rx Quality

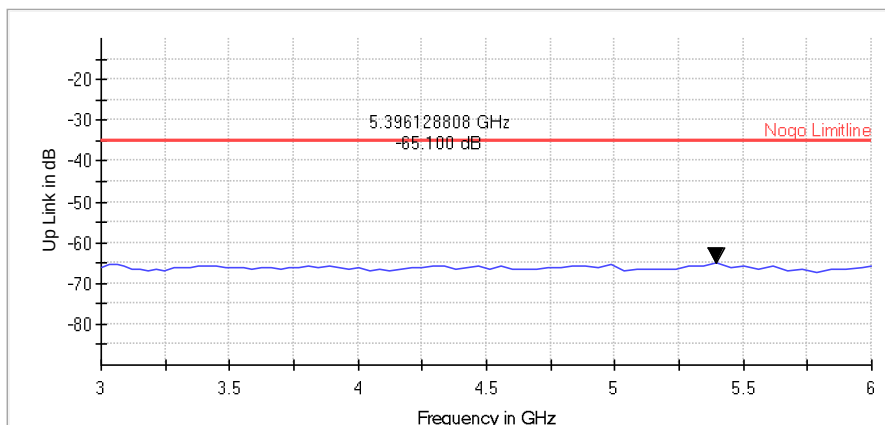


Test Mode :	Mode 8
Function Type :	GSM900 Link + SIM 1 + Battery + USB Cable(Charging from Adapter)
Audio Calibration Data :	Uplink Ref = 86.00 dBμV, Downlink Ref = 100.20 dBμV
Azimuth	180°
Frequency Range :	80MHz~1GHz Vertical

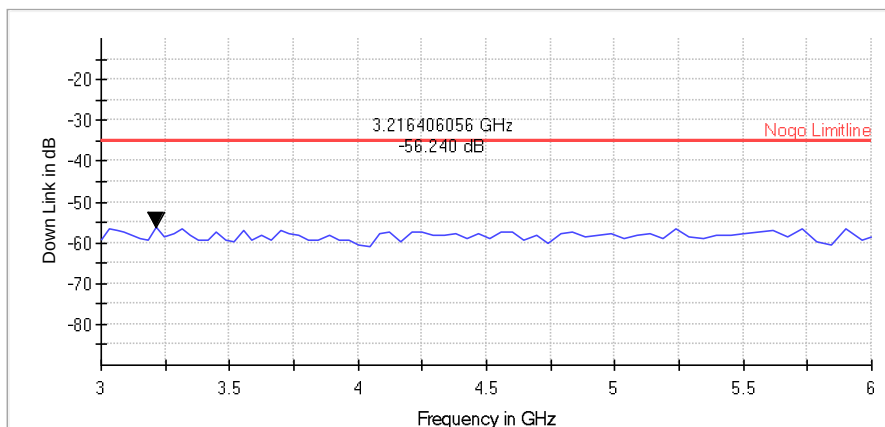


Test Mode :	Mode 8
Function Type :	GSM900 Link + SIM 1 + Battery + USB Cable(Charging from Adapter)
Audio Calibration Data :	Uplink Ref = 86.00 dBμV, Downlink Ref = 100.20 dBμV
Azimuth	180°
Frequency Range :	1GHz~3GHz Vertical

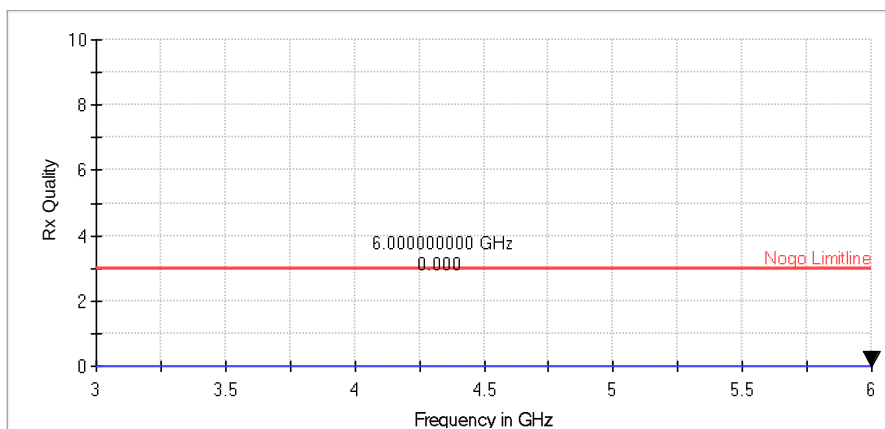
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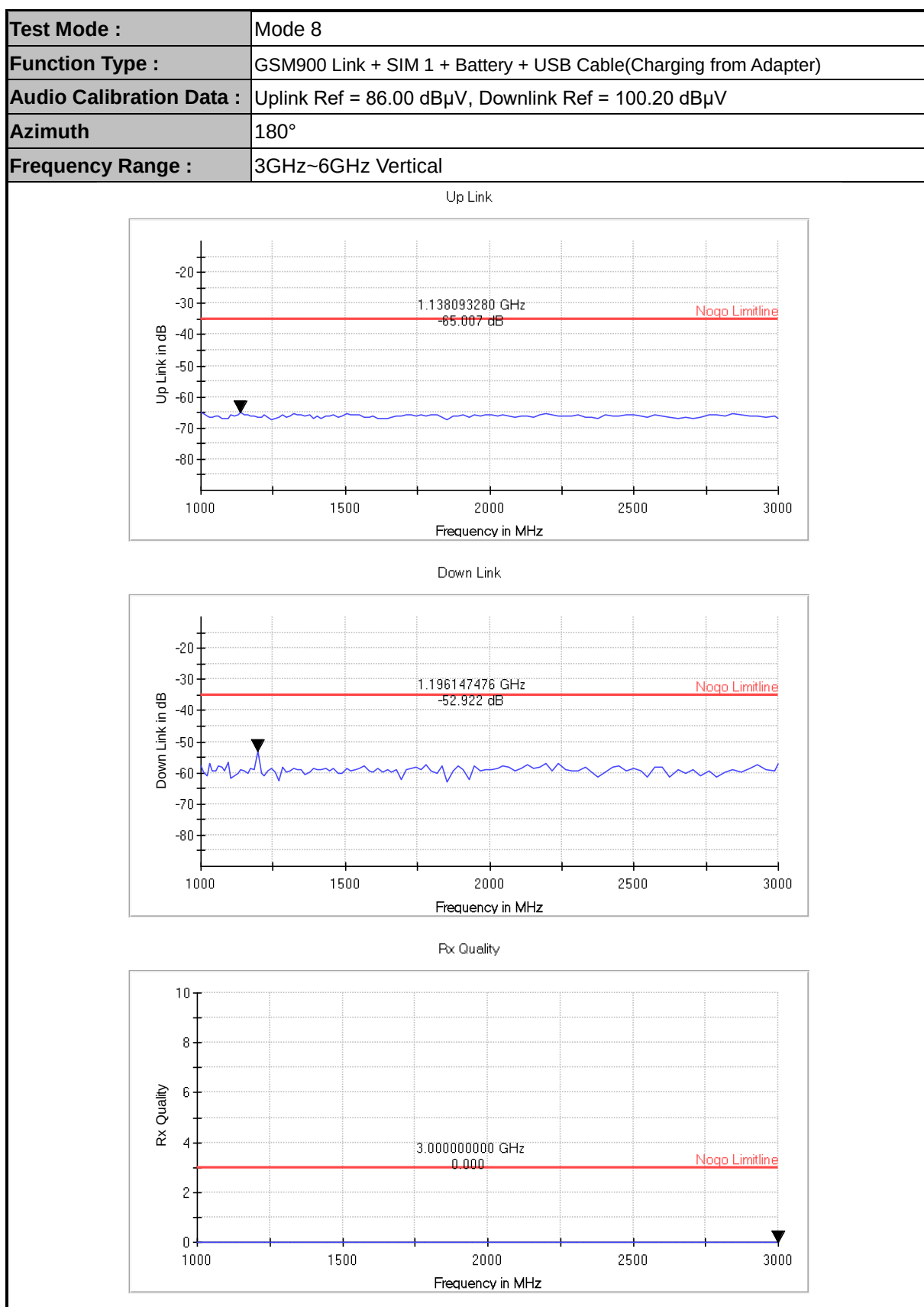


Down Link



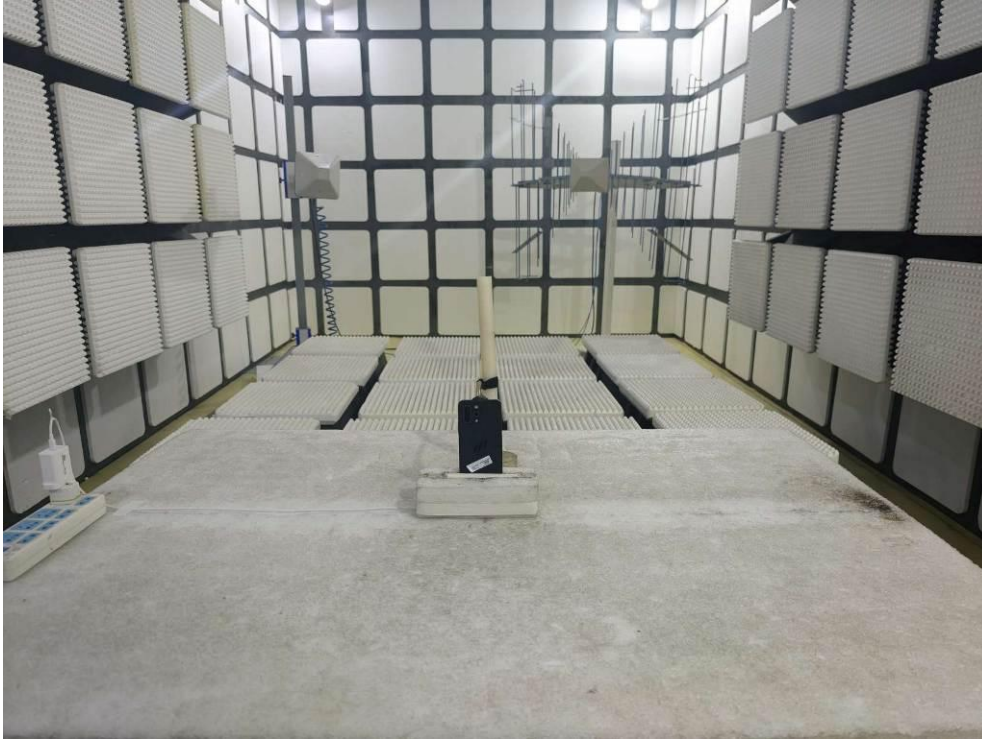
Rx Quality





5.2.5 Setup Photographs

**Mode 8
Position 0°**

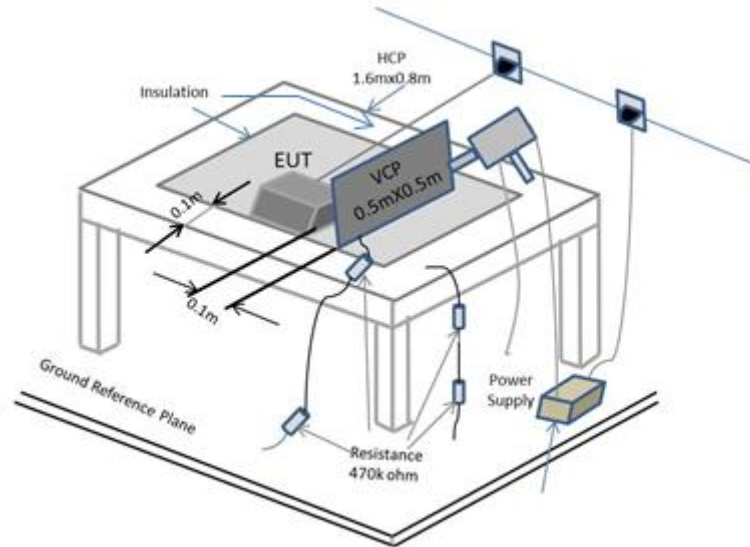


Position 180°



5.3 Electrostatic Discharge (ESD) Test (Refer to EN301 489-1 Section 9.3)

5.3.1 Test Setup



A distance of 1m minimum was provided between the EUT and the wall or any other metallic structure. In cases where this length exceeds the length necessary to apply the discharges to the selected points, the excess length shall, where possible, be placed non-inductively off the ground reference plane and shall not be less than 0.2m to other conductive parts in the test setup.

The coupling plane is placed parallel to, and positioned at a distance of 0,1 m from the EUT.

5.3.2 Test Instrument Setting

1. Tested number of air discharge is at least 10 discharges (in the most sensitive polarity). For contact discharge, tested number is at least 10 discharges.
2. For the time interval between successive single discharges an initial value of 1 s is recommended.
3. Sweeping of the EUT with a grounded carbon fibre brush with bleeder resistors (for example, 2 470 kΩ) in the grounding cable.
4. Ensure parameters of current waveform of an ESD generator is within specifications before test.

5.3.3 Test Procedure

EUT and auxiliary instrument necessary to perform DIRECT and INDIRECT application of discharges to the EUT, in the following manner:

- CONTACT DISCHARGE to the conductive surfaces and to the coupling plane;
- AIR DISCHARGE at insulating surfaces.

a. Contact Discharges to the conductive surfaces and to coupling planes:

In the case of contact discharges, the tip of the discharge electrode shall touch the EUT before the discharge switch is operated.

In the case of painted surface covering a conducting substrate, the following procedure shall be adopted :

- If the coating is not declared to be an insulating coating by the equipment manufacturer, then the pointed tip of the generator shall penetrate the coating so as to make contact with the conducting substrate.
- Coating declared as insulating by the manufacturer shall only be submitted to the air discharge.
- The contact discharge test shall not be applied to such surfaces.

b. Air Discharge to apertures and insulation surfaces:

In the case of air discharges, the round discharge tip of the discharge electrode shall be approached as fast as possible (without causing mechanical damage) to touch the EUT. After each discharge, the ESD generator (discharge electrode) shall be removed from the EUT. The generator is then retriggered for a new single discharge. This procedure shall be repeated until the discharges are completed. In the case of an air discharge test, the discharge switch, which is used for contact discharge, shall be closed.

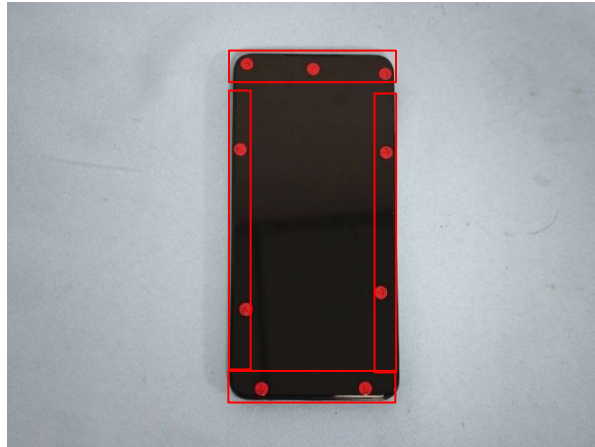
c. Ensure that the applied charge on the EUT has been dis-charged before next ESD pulse.

5.3.4 Test Result

Test Standard	EN 61000-4-2:2009
Product Standard	EN 301 489-1, EN 301 489-3, EN 301 489-17, EN 301 489-19, EN 301 489-52
EUT operated voltage during test	230Vac, 50Hz
Tested Level	$\pm 2 / \pm 4 / \pm 8$ kV for air discharge
	$\pm 2 / \pm 4$ kV for contact discharge
Required Performance Criteria	TT/TR
Performance Criteria of EUT	TT/TR
Test Result	PASS

5.3.5 TT/TR situation during Electrostatic Discharge test

Here are only shown ESD test points which TT/TR situation have happened, detailed ESD test points please refer to next section of this report.



Picture 1. Discharge point (Air Discharge)

Remark : When test air discharge with level $\pm 8\text{KV}$ at the discharge points remarked above, brightness reduced of the panel, but can automatically recover to normal operation as its intention.

5.3.6 Photos for Identification of ESD Test Points**Remark:**

1. Air Discharge points refer to the red arrows on the photo.
2. There is no any contact discharge points on the EUT, only testing for HCP/VCP.

<Mode 1~9>

<Adapter>**<USB Cable>**

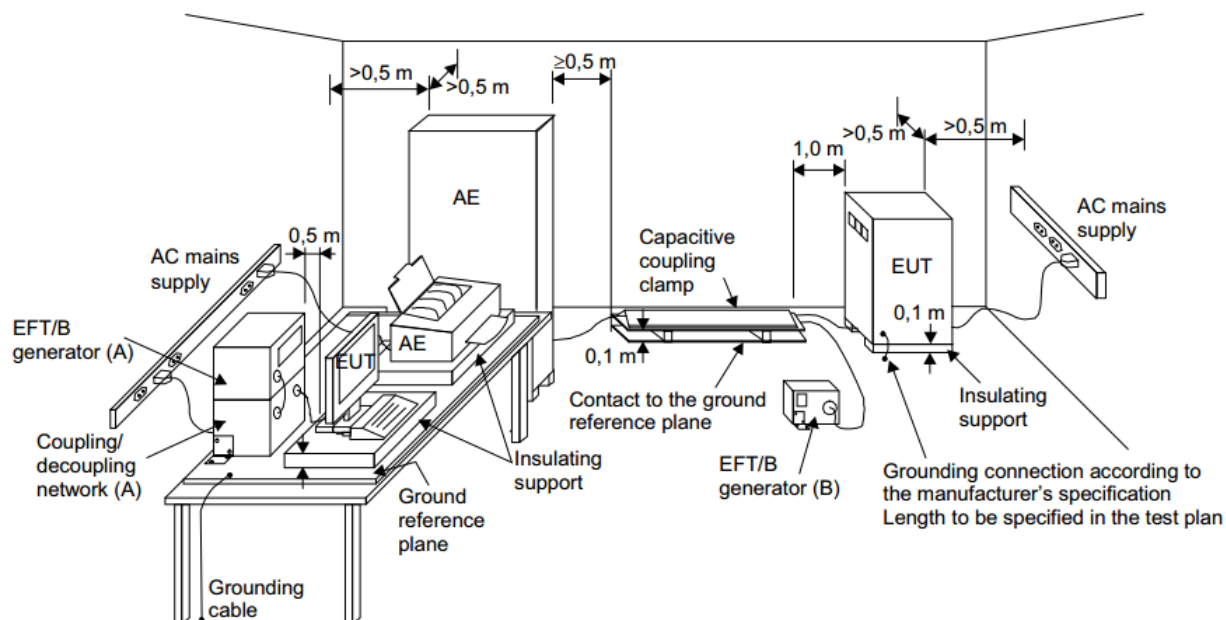
5.3.7 Setup Photographs**Mode 1~4****Mode 5~6**

Mode 7**Mode 8**

Mode 9

5.4 Fast Transients, Common Mode (EFT/BURST) Test (Refer to EN301 489-1 Section 9.4)

5.4.1 Test Setup



The EUT was placed on a ground reference plane and was insulated from it by an insulating support about 0.1m thick. If the EUT is table-top equipment, it was located approximately 0.8m above the GRP. The GRP was a metallic sheet (copper or aluminum) of 0.25 mm, minimum thickness; other metallic may be used but they shall have at least 0.65 mm thickness.

5.4.2 Test Instrument Setting

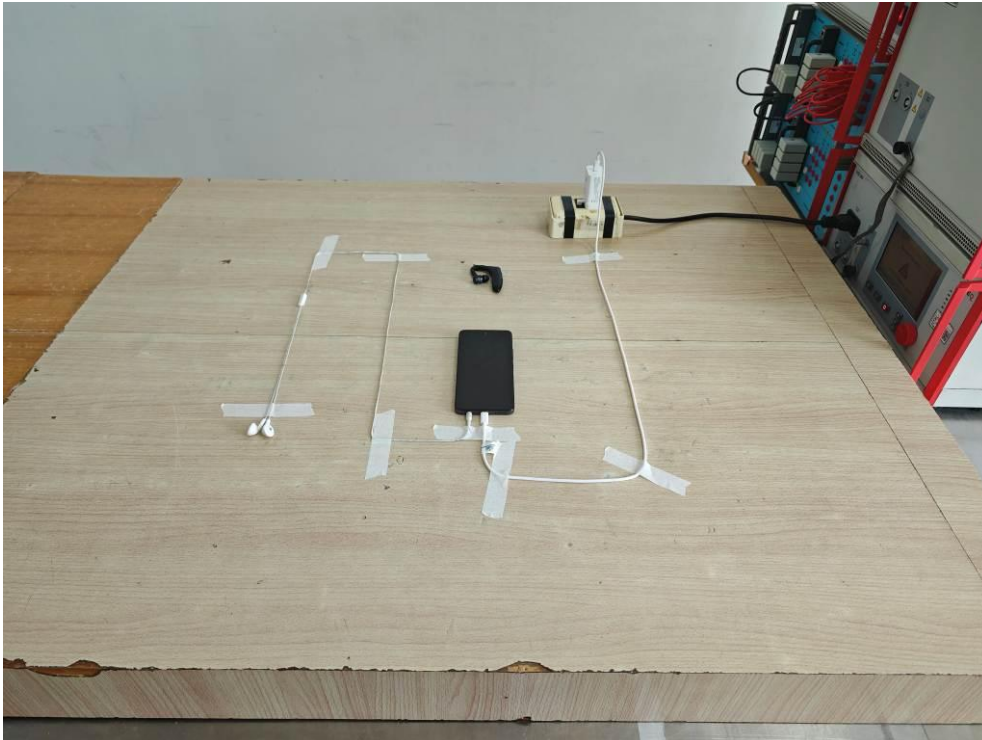
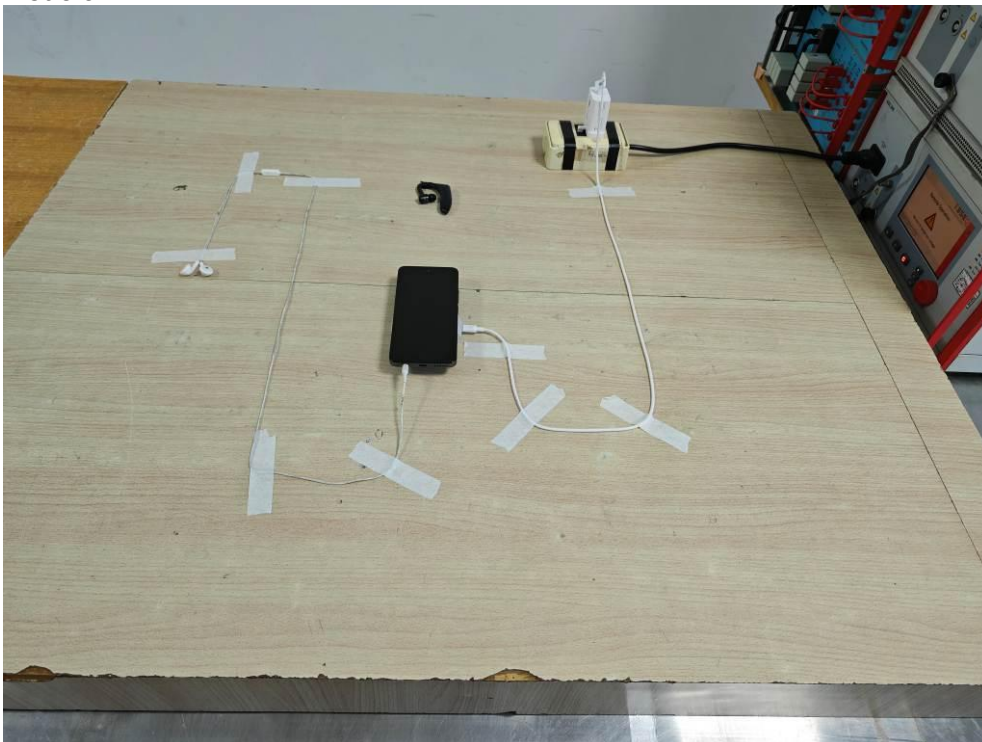
Port under Test	Pulse repetition rate	Burst duration/ Burst period	Duration of test per port
AC power port DC power port wired network ports (excluding xDSL)	5kHz	15 ms / 300 ms	1 minute
xDSL	100kHz	0.75 ms / 300 ms	1 minute

5.4.3 Test Procedure

- a. In order to minimize the effect of environmental parameters on test results, the electromagnetic environment of the laboratory shall not influence the test results.
- b. Duration of test per port shall be longer than 1 minute.
- c. Testing on power port is followed below requirements:
 - i. The EFT/B-generator was located on the GRP. The length from the EFT/B-generator to the EUT does not exceed 1 m.
 - ii. The EFT/B-generator provides the ability to apply the test voltage in a non-symmetrical condition to the power supply input terminals of the EUT.
 - iii. For equipment having a power port with no earth terminal, the test voltage is only applied to L and N lines.
- d. Testing on signal input/output parts port is followed below requirements:
 - i. The coupling clamp is composed of a clamp unit for cable length more than 3 m, and it was placed on the GRP.
 - ii. Telecommunication port whose maximum cable length is less than 3 m in length are excluded Electrical Fast Transients / BURST test

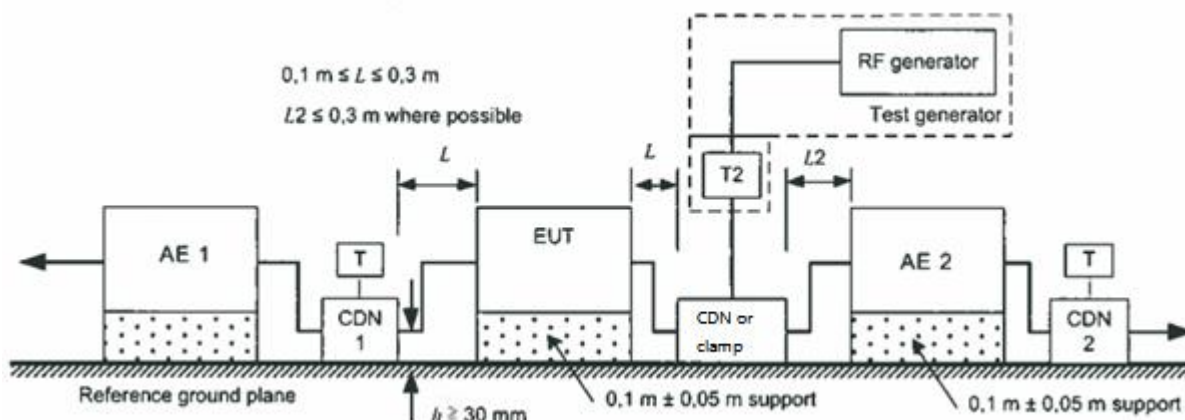
5.4.4 Test Result

Test Standard	EN 61000-4-4:2012
Product Standard	EN 301 489-1, EN 301 489-3, EN 301 489-17, EN 301 489-19, EN 301 489-52
EUT operated voltage during test	230Vac, 50Hz
Test Level	on AC Power Port : ± 1 kV
Required Performance Criteria	TT/TR
Performance Criteria of EUT	CT/CR
Test Result	PASS

5.4.5 Setup Photographs**Mode 1~5****Mode 6**

5.5 Radio frequency, Common mode (CS) Test (Refer to EN301 489-1 Section 9.5)

5.5.1 Test Setup



5.5.2 Test Instrument Setting

Frequency Step Size	1% increment ; For GSM and DCS Applicability, The stepped frequency increments may be 50 kHz increment of the momentary frequency in the frequency range 150 kHz to 5 MHz.
Modulation	80% AM (1kHz)
Dwell Time	3 seconds

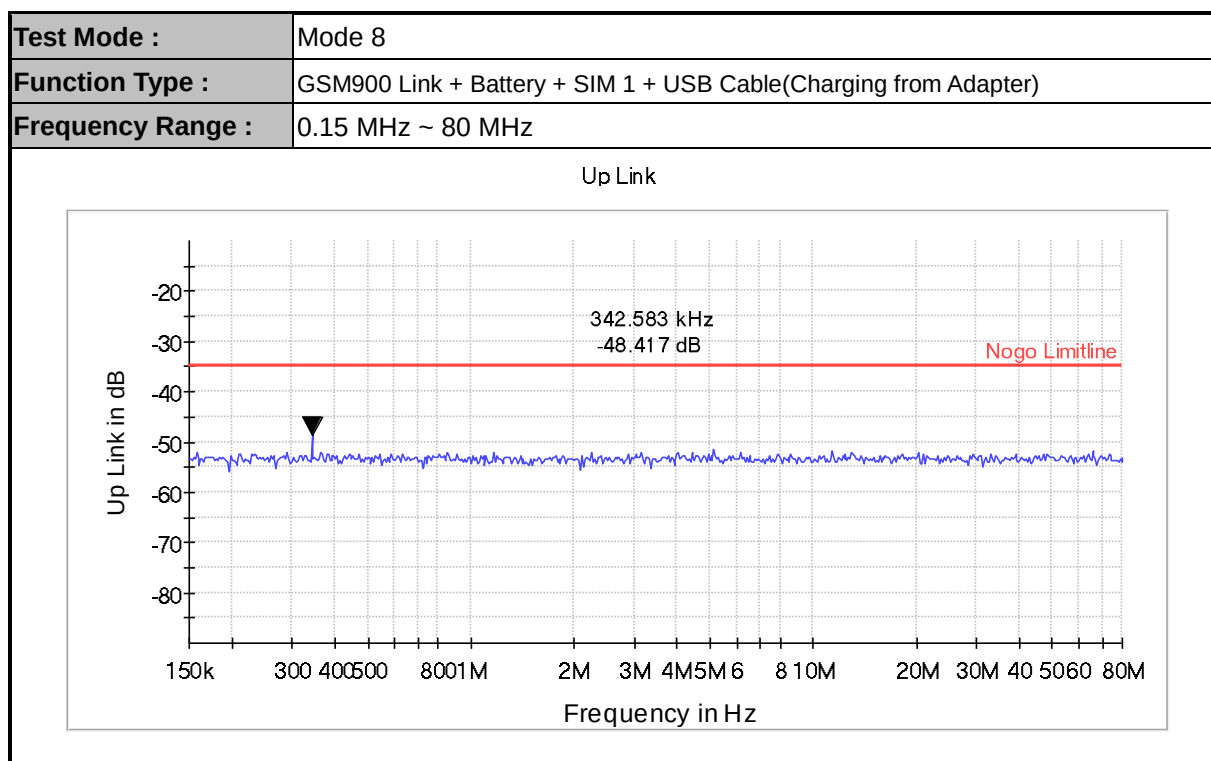
5.5.3 Test Procedures

- The EUT shall be operated within its intended climatic conditions. The temperature and relative humidity should be recorded.
- Cables selected for testing for which a CDN is suitable are to have the CDN in place during the test. All CDNs that are not being used to inject the test signal are to be terminated with a 50 ohm load.

5.5.4 Test Result

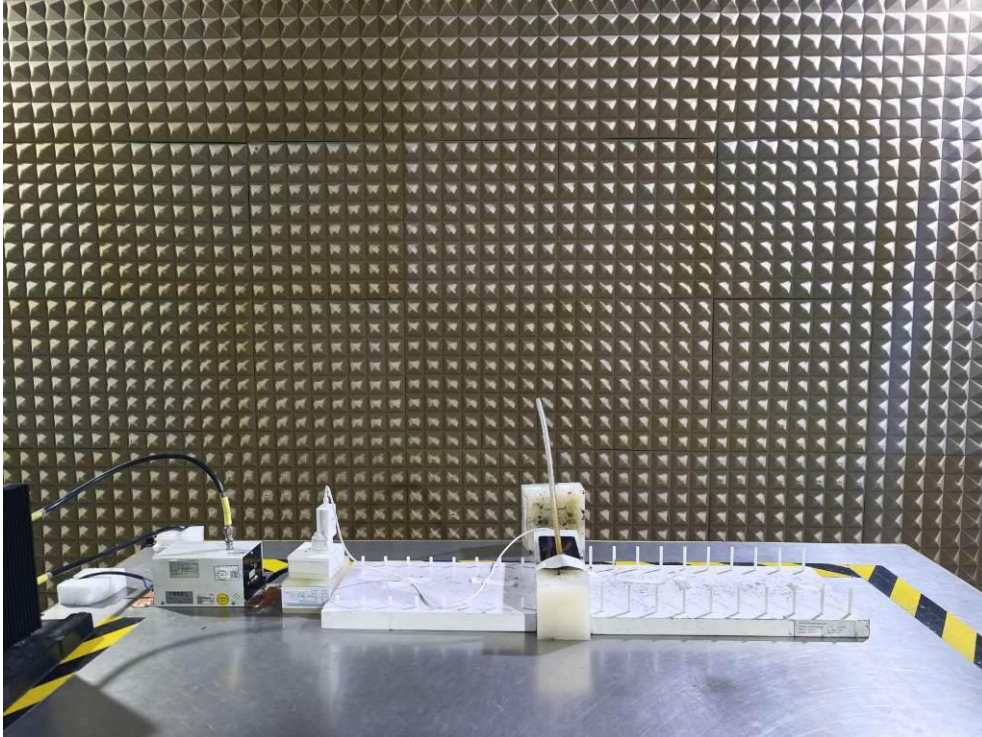
Test Standard	EN 61000-4-6:2014
Product Standard	EN 301 489-1, EN 301 489-3, EN 301 489-17, EN 301 489-19, EN 301 489-52
EUT operated voltage during test	230Vac, 50Hz
Test Frequency Range	0.15 MHz ~ 80 MHz
Test Level	3 Vrms
Test Method	CDN for Power port
Required Performance Criteria	CT/CR
Performance Criteria of EUT	CT/CR
Test Result	PASS

<Data>

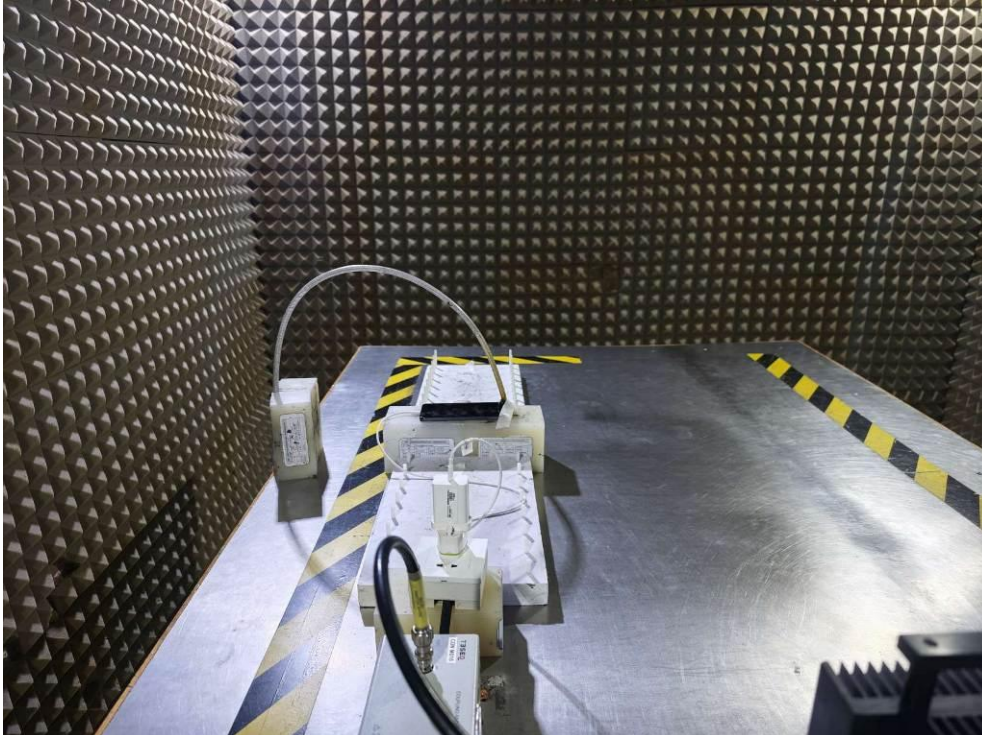


5.5.5 Setup Photographs

**Mode 8
Front View**

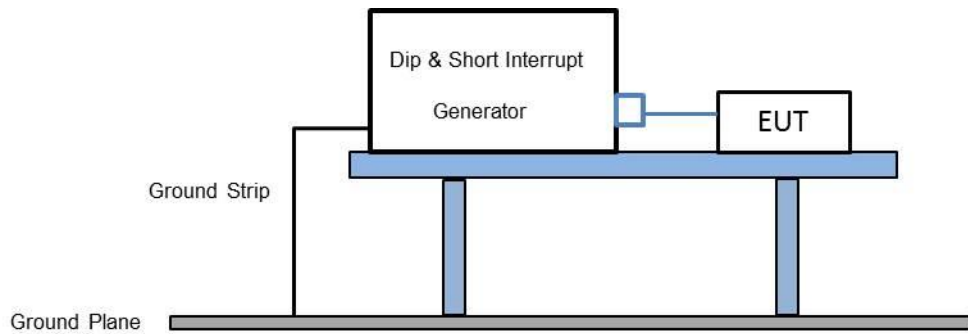


Side View



5.6 Voltage Dips and Interruptions (Refer to EN301 489-1 Section 9.7)

5.6.1 Test Setup



5.6.2 Test Instrument Setting

Test of interval	10 sec
Waveform Rise (and Fall) Time	1 ~ 5 μ s.
Duration	Sequence of 3 dips/interrupts.

5.6.3 Test Procedures

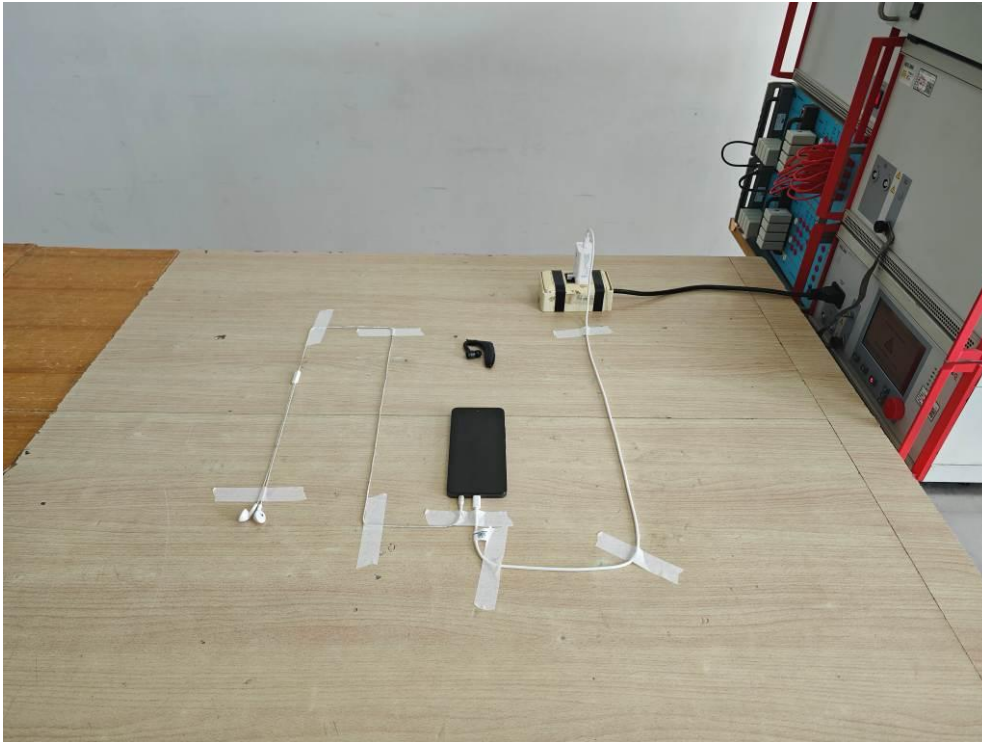
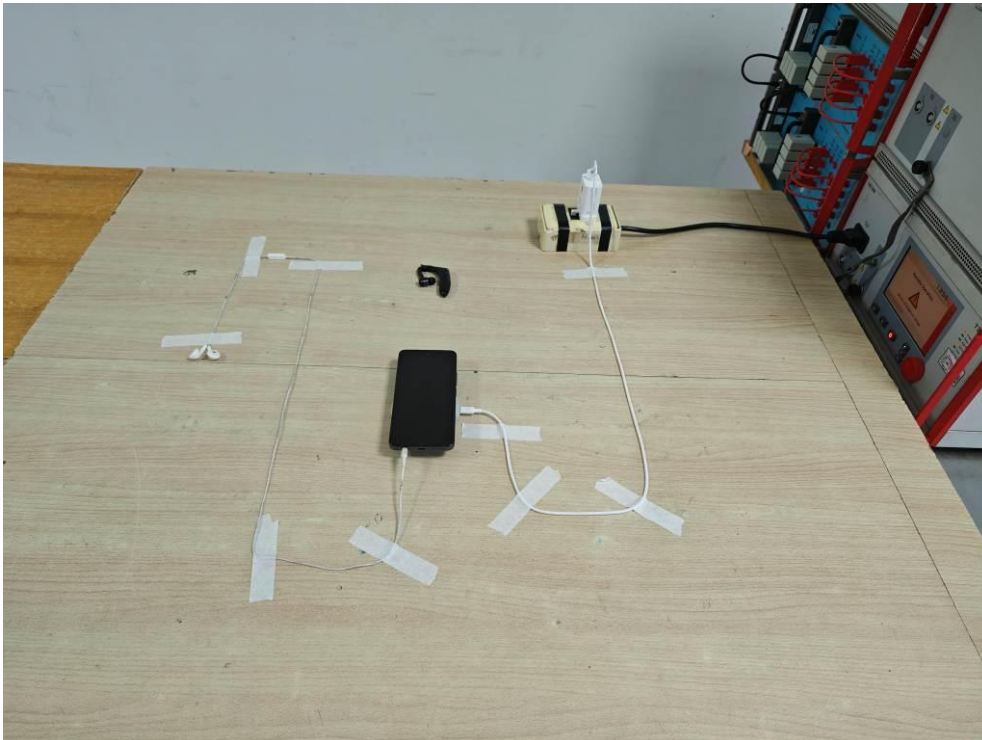
Where the equipment has a rated voltage the following shall apply:

- If the voltage range does not exceed 20% of the lower voltage specified for the rated voltage range, a single voltage within that range may be specified as a basis for test level specification.

5.6.4 Test Result

Test Standard	EN 61000-4-11:2004
Product Standard	EN 301 489-1, EN 301 489-3, EN 301 489-17, EN 301 489-19, EN 301 489-52
EUT input voltage	240 Vac, 50Hz 100 Vac, 50Hz
Requirements among Phase, Residual and Duration	As following table
Required Performance Criteria	TT/TR for Voltage Interruption TT/TR for Voltage Dips
Performance Criteria of EUT	CT/CR for Voltage Interruption CT/CR for Voltage Dips
Test Result	PASS

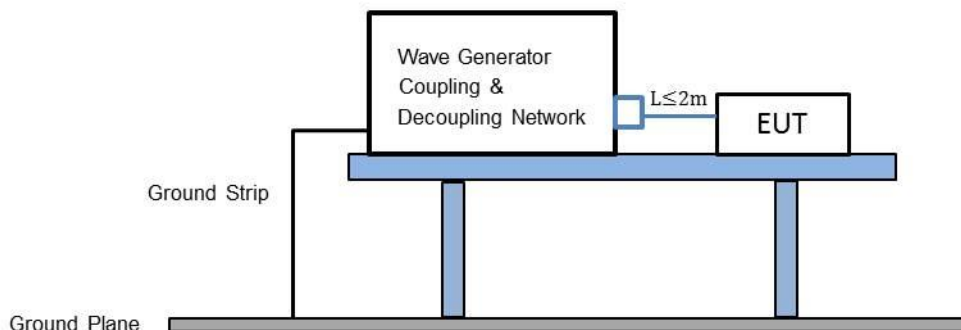
Test	Phase Angle	Residual (%)	Duration (ms)
Voltage Dips	0 °, 45 °, 90 °, 135 °, 180 °, 225 °, 270 °, 315 °	0%	10
		0%	20
		70%	500
Voltage Interruption	0 °	0%	5000

5.6.5 Setup Photographs**Mode 1~5****Mode 6**

5.7 Surges Test (Refer to EN301 489-1 Section 9.8)

5.7.1 Test Setup

<Testing for power port >



5.7.2 Test Instrument Setting

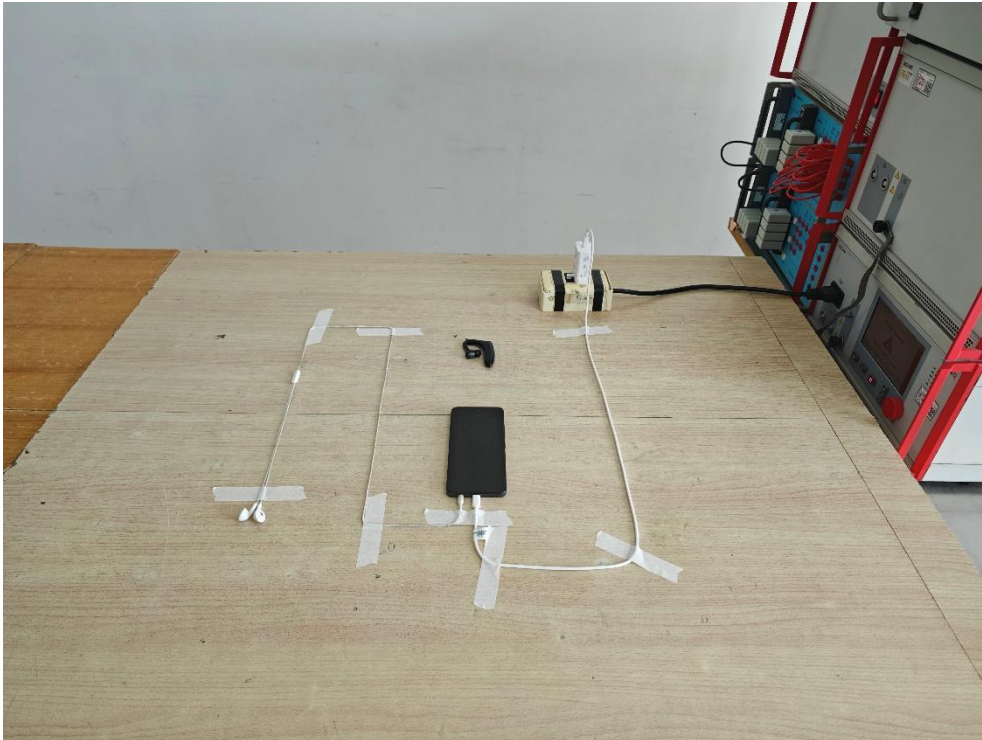
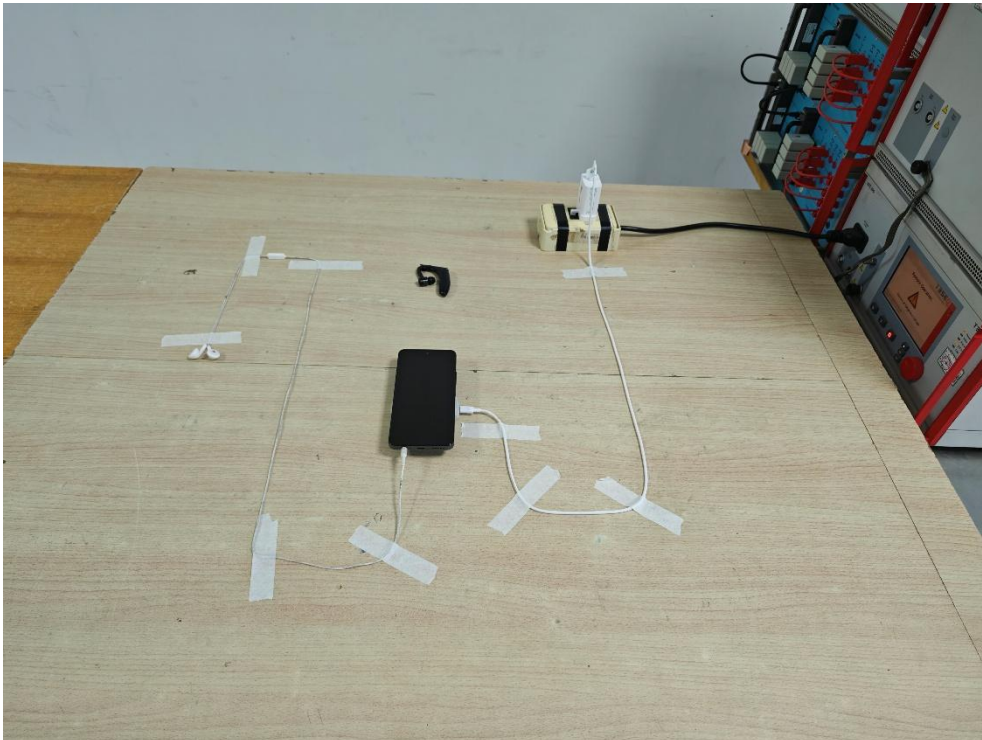
Surge wave form (Tr/Th)	For power port (if EUT supported) : 1,2/50 μs For signal port (if EUT supported) (1) outdoor cable symmetrically operated : 10/700 μs non- symmetrically operated : 1,2/50 μs (2) indoor cable (When cable length >30m) : symmetrically operated : 1,2/50 μs non- symmetrically operated : 1,2/50 μs
Number of impulses	For d.c. power ports and interconnection lines five positive and five negative surge impulses For a.c. power ports five positive and five negative impulses each at 0°, 90°, 180° and 270°
Time between successive impulses	1 minute

5.7.3 Test Procedure

- Electromagnetic conditions
The electromagnetic environment of the laboratory shall not influence the test results.
- The test procedure shall also consider the non-linear current-voltage characteristics of the equipment under test. Therefore the test voltage has to be increased by steps up to the test level specified in the product standard or test plan. All lower levels including the selected test level shall be satisfied.

5.7.4 Test Result

Test Standard	EN 61000-4-5:2014 + A1:2017
Product Standard	EN 301 489-1, EN 301 489-3, EN 301 489-17, EN 301 489-19, EN 301 489-52
EUT operated voltage during test	230 Vac, 50Hz
Test Level	on AC Power Port : ± 0.5 kV, ± 1 kV
Required Performance Criteria	TT/TR for Power Port
Performance Criteria of EUT	CT/CR for Power Port
Test Result	PASS

5.7.5 Setup Photographs**Mode 1~5****Mode 6**

6. Measurement Uncertainty

Uncertainty of Conducted Emission Measurement (150 KHz ~ 30 MHz)

Measuring Uncertainty for a Level of Confidence of 95% ($U=2U_c(y)$)	3.50dB
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Uncertainty of Radiated Emission Measurement (1000 MHz ~ 6000 MHz) - 03CH07-KS

Measuring Uncertainty for a Level of Confidence of 95% ($U=2U_c(y)$)	4.86dB
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Uncertainty of Radiated Emission Measurement (30 MHz ~ 1000 MHz) - 10CH01-KS(Horizontal)

Measuring Uncertainty for a Level of Confidence of 95% ($U=2U_c(y)$)	5.04dB
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Uncertainty of Radiated Emission Measurement (30 MHz ~ 1000 MHz) - 10CH01-KS(Vertical)

Measuring Uncertainty for a Level of Confidence of 95% ($U=2U_c(y)$)	5.04dB
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Uncertainty of Radiated Susceptibility Measurement (80 MHz ~ 6000 MHz)

Measuring Uncertainty for a Level of Confidence of 95% ($U=2U_c(y)$)	1.92dB
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Uncertainty of Electrostatic Discharge Measurement

	Measuring Uncertainty for a Level of Confidence of 95% ($U = 2U_c(y)$) %
Electrostatic Discharge ~ Rise Time	9.7%
Electrostatic Discharge ~ Peak Current	4.6%
Electrostatic Discharge ~ 30ns Current	4.6%
Electrostatic Discharge ~ 60ns Current	4.6%

Uncertainty of Conducted Susceptibility Measurement (150kHz ~ 80 MHz)

Measuring Uncertainty for a Level of Confidence of 95% ($U=2U_c(y)$)	2.60dB
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<For Electrical Fast Transients / BURST test>

Measuring Uncertainty for a Level of Confidence of 95% ($U=2U_c(y)$)- budget for voltage rise time (tr)	5.3%
Measuring Uncertainty for a Level of Confidence of 95% ($U=2U_c(y)$)- EFT/B peak voltage value (VP)	8.3%
Measuring Uncertainty for a Level of Confidence of 95% ($U=2U_c(y)$)- EFT/B voltage pulse width (tw)	2.7%

<For Surge test>

Measuring Uncertainty for a Level of Confidence of 95% ($U=2U_c(y)$)- Surge open-circuit voltage front time (TfV)	13.33%
Measuring Uncertainty for a Level of Confidence of 95% ($U=2U_c(y)$)- Surge open-circuit voltage peak value (VP)	8.20%
Measuring Uncertainty for a Level of Confidence of 95% ($U=2U_c(y)$)- surge open-circuit voltage duration (Td)	1.10%

<For Dips test>

Measuring Uncertainty for a Level of Confidence of 95% ($U=2U_c(y)$)- Surge open-circuit voltage front time (TfV)	2.66%
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7. Testing Facility

Sporton International Inc. (Kunshan) is accredited to ISO/IEC 17025:2017 by American Association for Laboratory Accreditation with Certificate Number 5145.02.

Test Lab.	Sporton International Inc. (Kunshan)
Test Site Location	No. 1098, Pengxi North Road, Kunshan Economic Development Zone Jiangsu Province 215300 People's Republic of China TEL : +86-512-57900158
Test Site No.	Sporton Site No. : EMI Test : 03CH07-KS; 10CH01-KS; CO01-KS EMS Test : RS01-KS ; CS01-KS ; ES01-KS ; EX01-KS



8. List of Measuring Equipment

<EMI>

Instrument	Manufacturer	Model No.	Serial No.	Characteristics	Calibration Date	Test Date	Due Date	Remark
Bilog Antenna	TESEQ	CBL6111D	63161	30MHz~1GHz	Oct. 24, 2025	Mar. 20, 2026	Oct. 23, 2026	Radiation (10CH01-KS)
Bilog Antenna	TESEQ	CBL6111D	63755	30MHz~1GHz	Oct. 24, 2025	Mar. 20, 2026	Oct. 23, 2026	Radiation (10CH01-KS)
EMI Test Receiver	Keysight	N9038A	MY57290151	3Hz~8.5GHz; Max 30dBm	Jul. 02, 2025	Mar. 20, 2026	Jul. 01, 2026	Radiation (10CH01-KS)
EMI test receive	R&S	ESR7	102631	9KHz~7GHz	Oct. 09, 2025	Mar. 20, 2026	Oct. 08, 2026	Radiation (10CH01-KS)
Amplifier	SONOMA	310N	422544	9KHz ~1GHZ	Oct. 10, 2025	Mar. 20, 2026	Oct. 09, 2026	Radiation (10CH01-KS)
Amplifier	SONOMA	310N	422545	9KHz ~1GHZ	Oct. 10, 2025	Mar. 20, 2026	Oct. 09, 2026	Radiation (10CH01-KS)
Bore-site Antenna Mast	EM	MBD-400-1	NA	1 m~4 m	NCR	Mar. 20, 2026	NCR	Radiation (10CH01-KS)
Antenna Mast	EM	MBD-400-1	NA	1 m~4 m	NCR	Mar. 20, 2026	NCR	Radiation (10CH01-KS)
Controller	EM	3000-1	NA	NA	NCR	Mar. 20, 2026	NCR	Radiation (10CH01-KS)
Turn Table	EM	T-300-1	NA	0~360 degree	NCR	Mar. 20, 2026	NCR	Radiation (10CH01-KS)
MXE EMI Receiver	Keysight	N9038A	MY57290151	3Hz~8.4GHz	Jul. 02, 2025	Mar. 20, 2026	Jul. 01, 2026	Radiation (03CH07-KS)
Double Ridge Horn Antenna	ETS-Lindgren	3117	00240132	1GHz~18GHz	Jun. 26, 2025	Mar. 20, 2026	Jun. 25, 2026	Radiation (03CH07-KS)
Amplifier	Keysight	83017A	MY57280106	0.5GHz-26.5GHz	Apr. 09, 2025	Mar. 20, 2026	Apr. 08, 2026	Radiation (03CH07-KS)
AC Power Source	Chroma	61601	616010002473	N/A	NCR	Mar. 20, 2026	NCR	Radiation (03CH07-KS)
Turn Table	EM	EM 1000-T	N/A	0~360 degree	NCR	Mar. 20, 2026	NCR	Radiation (03CH07-KS)
Antenna Mast	EM	EM 1000-A	N/A	1 m~4 m	NCR	Mar. 20, 2026	NCR	Radiation (03CH07-KS)
EMI Receiver	R&S	ESCI7	100768	9kHz~7GHz;	Apr. 16, 2025	Feb. 28, 2026	Apr. 15, 2026	Conduction (CO01-KS)
AC LISN (for auxiliary equipment)	MessTec	AN3016	060103	9kHz~30MHz	Jul. 03, 2025	Feb. 28, 2026	Jul. 02, 2026	Conduction (CO01-KS)
AC Power Source	Ouhan	APW-130NYT	202201040060	AC 0V~300V, 45Hz~1000Hz	Apr. 16, 2025	Feb. 28, 2026	Apr. 15, 2026	Conduction (CO01-KS)
Harmonic/ Flicker Test System	TeseQ	CNN1000-1	1521A01676	N/A	Jun. 18, 2025	Feb. 12, 2026	Jun. 17, 2026	Harmonics, Flicker (EX01-KS)
	TeseQ	NSG1007-5	1521A01676	5kVA	Jun. 18, 2025	Feb. 12, 2026	Jun. 17, 2026	Harmonics, Flicker (EX01-KS)

**<EMS>**

Instrument	Manufacturer	Model No.	Serial No.	Characteristics	Calibration Date	Test Date	Due Date	Remark
Signal Generator	R&S	SMJ100A	101908	100Khz to 6Ghz	Dec. 24, 2025	Mar. 08, 2026	Dec. 23, 2026	RS (RS01-KS)
Audio Analyzer	R&S	UPV	101925	DC~250KHz	Apr. 18, 2025	Mar. 08, 2026	Apr. 17, 2026	RS (RS01-KS)
Power Sensors	R&S	NRP-Z91	101429	9K-6GHz 200PW~200MW	Jul. 07, 2025	Mar. 08, 2026	Jul. 06, 2026	RS (RS01-KS)
Power Sensors	R&S	NRP-Z91	100034	9K-6GHz 200PW~200MW	Apr 17, 2025	Mar. 08, 2026	Apr 16, 2026	RS (RS01-KS)
Power Sensors	R&S	NRP-Z92	100213	9K-6GHz 2nW~2W	Apr. 17, 2025	Mar. 08, 2026	Apr. 16, 2026	RS (RS01-KS)
System Simulator	R&S	CMU200	116456	2G/3G Full Band	Jul. 02, 2025	Mar. 08, 2026	Jul. 01, 2026	RS (RS01-KS)
Base Station	R&S	CMW500	150792	LTE Full Band	Dec. 24, 2025	Mar. 08, 2026	Dec. 23, 2026	RS (RS01-KS)
Antenna	R&S	HL046E	100100	80MHz~3GHz	NCR	Mar. 08, 2026	NCR	RS (RS01-KS)
AC Power Source	Chroma	61602	ABP000000810	N/A	Oct. 10 2025	Mar. 08, 2026	Oct. 09, 2026	RS (RS01-KS)
Antenna	SCHWARZBECK	STLP 9149	9149-010	0.7-10.5GHz	NCR	Mar. 08, 2026	NCR	RS (RS01-KS)
Power Amplifier	R&S	BBA150	103650	2.5-6GHz,100W	Apr. 17, 2025	Mar. 08, 2026	Apr. 16, 2026	RS (RS01-KS)
Power Amplifier	AR	120S1G3M1	0325544	0.8~3GHz,120W	Apr. 17, 2025	Mar. 08, 2026	Apr. 16, 2026	RS (RS01-KS)
Power Amplifier	AR	150W1000M1	0328395	80~1000MHz, 150W	Apr. 17, 2025	Mar. 08, 2026	Apr. 16, 2026	RS (RS01-KS)
Coupling/Decoupling Network	TESEQ	CDN M016	24477	150kHz~230MHz	Jul. 04, 2025	Mar. 10, 2026	Jul. 03, 2026	CS (CS01-KS)
Signal Generator	R&S	SMB 100A	105413	9K-1.1G	Dec. 24, 2025	Mar. 10, 2026	Dec. 23, 2026	CS (CS01-KS)
Power Sensors	R&S	NRP-Z91	102869	9kHz~6GHz 200PW~200MW	Dec. 25, 2025	Mar. 10, 2026	Dec. 24, 2026	CS (CS01-KS)
Preamplifier	BONN	BSA 0125-150	076577A	150W 9kHz~250MHz	Dec. 25, 2025	Mar. 10, 2026	Dec. 24, 2026	CS (CS01-KS)
AC Power Source	Chroma	61601	616010001649	N/A	Oct. 10, 2025	Mar. 10, 2026	Oct. 09, 2026	CS (CS01-KS)
Base Station	R&S	CMW500	150792	LTE Full Band	Dec. 24, 2025	Mar. 10, 2026	Dec. 23, 2026	CS (CS01-KS)
Audio Analyzer	R&S	UPV	101468	DC~250KHz	Aug. 26, 2025	Mar. 10, 2026	Aug. 25, 2026	CS (CS01-KS)
ESD Generator	EM TEST	DITO	P1251107027	±0.5kV~16.5kV	Aug. 29, 2025	Feb. 10, 2026	Aug. 28, 2026	ESD (ES01-KS)
EMC Test System	TeseQ	NSG 3040	1746	0kV-4.4kV	Jul. 02, 2025	Feb. 12, 2026	Jul. 01, 2026	EFT, SURGE, DIP (EX01-KS)
Single motor driven variable transformer	Teseq AG	VAR 3005-S16	850	0kV~4.4kV	Jul. 02, 2025	Feb. 12, 2026	Jul. 01, 2026	EFT, SURGE, DIP (EX01-KS)

NCR: No Calibration Required.

—THE END—